

FORMALIZING ANGULAR STRUCTURE OF PARTICLE JETS

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ABSTRACT OF THE DISSERTATION

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Most of all stable matter is made of protons and neutrons in the atomic nuclei. Protons and neutrons, or hadrons are also composite objects made of quarks and gluons, or partons which are as far as we know, fundamental particles of the strong nuclear force or just the strong force. The field theory Quantum Chromodynamics (QCD) formalizes properties of the strong force that makes it impossible for partons to escape hadrons under “normal” circumstances. However, when matter is heated up to extremely high temperatures or compressed to extremely high pressure, the same property of QCD will lead to the hadronic matter undergoing a phase transition into a partonic state consisting of free quarks and gluons outside the hadronic volume, called Quark-Gluon Plasma (QGP).

QGP is created in the laboratory by colliding heavy-nuclei like $^{96}_{40}\text{Zr}$, $^{96}_{44}\text{Ru}$, $^{197}_{79}\text{Au}$, etc. at near light fields. In the very early stages of these collisions, before QGP formation, partons undergo hard scattering and produce highly energetic outgoing partons. In vacuum, these hard scattered partons would undergo successive fragmentations and eventually evolve into spatially dense ensembles of hadrons called jets. However in heavy ion collisions, these partons go through the QGP, undergoing extra radiative and collisional energy losses. These

extra interactions change the spatial compositions of hadrons inside jets, compared to the vacuum case. This phenomenon is called jet quenching, a commonly used probe of QGP, with the vacuum case approximated by equivalent proton-proton ($p + p$) collisions.

Two jet analyses are presented in this thesis. The first, jet-hadron correlations in Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV, explores the dependence of jet quenching on length of the path travelled in QGP, by calculating changes in average azimuthal distribution of final state hadrons relative to axes of differently oriented jets.

The second analyses characterizes angular spread of the jet's energy through observables called generalized angularities. The analysis begins with $p + p$ collisions at $\sqrt{s} = 200$ GeV showing distributions of various generalized angularities, which will eventually become baselines for heavy-ion measurements. Next, SoftDrop grooming applied on these jets separates the effects of jets early and late evolution. We close with initial studies for ongoing measurement of heavy-ion measurements of generalized angularities.

Along the way, various computational techniques studied for these analyses are also discussed.

Acknowledgments

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Dedication

A dedication here.

Table of Contents

Abstract	ii
Acknowledgments	iv
Dedication	v
List of Tables	viii
List of Figures	x
1. Introduction	1
1.1. The Standard Model	1
1.2. Quantum Chromodynamics	3
1.3. Hard Processes	7
1.4. From partons to jets	9
2. Solenoidal Tracker At Relativistic Heavy Ion Collider	17
2.1. Relativistic Heavy-Ion Collider (RHIC)	17
2.2. The STAR detector	21
3. Relativistic collisions of nuclei	29
3.1. Finding Jets	30
3.2. Heavy-Ion Collisions	33
3.3. Jets in QGP	41

4. Data analysis techniques	46
4.1. Machine Learning Basics	47
4.2. Jet Classification	47
4.3. Unfolding	51
5. Jet-hadron correlations relative to the event plane	56
5.1. Analysis Method	57
5.2. Results	71
5.3. Conclusions	78
6. Generalized Angularities	81
6.1. Angularities of $p + p$ jets	82
6.2. Angularities of Au+Au jets	86
6.3. Conclusion and Outlook	89
7. Summary	94
7.1. Outlook	96
Bibliography	98
Appendix A. Relativistic Kinematics	104
A.1. Laboratory vs Center-of-mass frames	104
A.2. Coordinate system	105

List of Tables

1.1. Three generations of leptons (left) and quarks (right).	1
1.2. Feynman diagrams of various interaction vertices possible in the Standard Model	2
1.3. Feynman rules for QCD in a covariant gauge with gauge parameter λ : the gluon, ghost and quark propagators, followed by the three- and four-gluon, ghost-gluon and quark-gluon vertices.	4
1.4. Leading-order DGLAP splitting functions and their diagrams	11
5.1. Summary of systematic uncertainties which are independent of the angle relative to the event plane and the momentum for both $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ and $20 < p_{T,\text{jet}} < 40 \text{ GeV}/c$ in 20-50% central Au+Au collisions.	68
5.2. Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the jet energy shift (JES) correction, each varying with event-plane orientation bins. They are displayed for $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ in 20-50% central Au+Au collisions for $p_{T,\text{assoc}} \text{ range } 1.01.5$ and $p_{T,\text{assoc}} \text{ range } 3.04.0$ bins. The values are expressed as a percent of the nominal value.	69

5.3. Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the JES correction, each varying with event-plane orientation bins. They are displayed for $20 < p_{T,\text{jet}} < 40$ GeV/ c in 20-50% central Au+Au collisions for $p_{T,\text{assoc}}^{\text{range1.01.5}}$ and $p_{T,\text{assoc}}^{\text{range3.04.0}}$ bins. The values are expressed as a percent of the nominal value.	70
5.4. Results of fits to Fig.5.8 (top panel: 15-20 GeV/ c jets) to a constant c , the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).	79
5.5. Results of fits to Fig.5.8 (bottom panel: 20-40 GeV/ c jets) to a constant c , the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).	79

List of Figures

1.1.	Summary of determinations of α_s as a function of the energy scale Q compared to the running of the coupling computed at five loops taking as an input the current Particle Data Group (PDG) world average, $\alpha_s(m_Z^2) = 0.1180 \pm 0.0009$ [1]	6
1.2.	Collinear factorization of a hadronic hard scattering.	8
1.3.	Kinematics of timelike (left) vs spacelike (right) parton branching	9
1.4.	Wide-angle emission of soft gluon q , off i and j , acts as if the emission came off the parent parton l , imagined to be on shell.	15
2.1.	An aerial view of the Relativistic Heavy Ion Collider (RHIC), showing the locations of the facilities used for accelerating different ions, and the positions of the STAR and sPHENIX detectors. [2]	18
2.2.	Summary of RHIC energies, species combinations, and luminosities for runs between 2000 and 2026. [3]	18
2.3.	Some of the final O+O collisions at $\sqrt{s_{NN}} = 200$ GeV captured by the sPHENIX and STAR detectors at the RHIC. [4]	19
2.4.	Schematic view of the AGS complex, showing various stages the ions go through before going into RHIC	20
2.5.	Schematic of the STAR detector, depicting the various detector subsystems. .	21

2.6.	The (p_T, η, φ) coordinate system, drawn for a cylindrical collider detector. The beam axis defines z and the interaction point (IP) sits at the origin. A particle of momentum $\vec{p}$ is described by its transverse momentum $\vec{p}_T$, its azimuthal angle φ in the transverse plane, and its pseudo-rapidity η , fixed by the polar angle θ with respect to the beam.	23
2.7.	The Time Projection Chamber at STAR. [5].	24
2.8.	One of the twelve sectors of the TPC. [6].	25
2.9.	The ionisation energy loss for charged particles from TPC as a function of momenta. The letters represent different particles: $\mu^\pm$ for muons, $e^\pm$ for electrons, $\pi^\pm$ for pions, $K^\pm$ for Kaons, p for protons, and d for deuterons. The solid lines are the expected value of dE/dx for the corresponding particle species from the Bichsel function [7].	26
2.10.	The Barrel Electromagnetic Calorimeter at STAR. Taken from [5].	27
3.1.	Jets reconstructed using the k_T (left), Cambridge/Aachen (middle), and anti- k_T (right) algorithms. Taken from [8]	31
3.2.	Ways of arranging constituents (each represented as a circle whose radius represent $\frac{p_{T, \text{const}}}{p_{T, \text{jet}}}$) in the (η, ϕ) space within a jet with $p_{T, \text{jet}} = 30 \text{ GeV}/c$ such that R_g of the jet increases going from left to right. Green (Red) colored con- stituents survive (are removed by) the SoftDrop grooming ($z_{\text{cut}} = 0.2, \beta = 0$).	33
3.3.	A light cone diagram showing the stages of a heavy ion collision. The abbrevi- ation T_{fo} is for the thermal freeze-out temperature, T_{ch} is for the chemical freeze-out temperature, and T_c is for the critical temperature where the phase transition between a hadron gas and a QGP occurs. τ_0 is the formation time of the QGP. Taken from [9]	34
3.4.	Typical heavy-ion collision with an impact parameter of b . The participating nuclei are colored red and the spectators are colored blue.	35

3.5.	An illustration of the total inclusive charged particles multiplicity N_{ch} and number of participant (N_{part}) and impact parameter (b) in heavy-ion collisions. Taken from [10]	36
3.6.	Schematic diagrams showing the initial overlap region (left) and the spatial anisotropy generated by this anisotropic overlap region. This anisotropy can be quantified using the Fourier coefficients of the momentum anisotropy. Taken from [9]	37
3.7.	Left panel: Elliptic flow v_2 for several particle species in Au + Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV from STAR and PHENIX experiments. Predictions from hydrodynamic models are shown as lines. Taken from [11]. Right panel: Number of constituent quarks (n_q) scaled v_2 as a function of p_{T}/n_q . Taken from [12].	38
3.8.	Measurement of the angular correlation between all charged hadrons and the leading hadron in the event for p + p (black lines), d + Au (red and green markers), and Au + Au events (blue markers), performed by the STAR experiment. The vanishing of the away-side yield around $\Delta\phi \approx \pi$ in Au + Au events is evidence of particles getting quenched in the QGP medium. Taken from [13].	42
3.9.	Nuclear modification factor R_{AA} as a function of p_{T} for identified particle species in Au + Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV measured by the PHENIX experiment. Taken from [14].	43
3.10.	R_{CP} distributions for charged hadrons (open markers) and jets (solid markers) from RHIC (blue star markers) and LHC (red circle markers) experiments. Taken from [15] and references within [16–18].	44

3.11. Differential jet shapes in Pb+Pb and $p + p$ collisions for four Pb+Pb centralities. Each spectrum is normalized so that its integral is unity. This shows that there are more particles in jets in central collisions and these modifications are primarily at large angles relative to the jet axis, as expected from partonic energy loss. Taken from [9]	45
4.1. Typical quark (left) and gluon (right) jet. Gluon jets exhibit a broader spatial radiation pattern and higher particle multiplicity compared to quark jets	48
4.2. An illustration of the deep convolutional neural network architecture. The first layer is the input jet image, followed by three convolutional layers, a dense layer and an output layer. [19]	49
4.3.	50
4.4. Per-pixel feature importance	50
4.5. ROC curves for PYTHIA6 vs. JEWEL without recoils (left) and with recoils (right).	51
4.6. Schematic figure showing the unfolding procedure. The desired physics / truth / particle level is represented by the $2 \rightarrow 2$ feynman diagram in the middle. The blue (gray) tracks and tower deposits are the dijet (underlying) event as recorded by the TPC and BEMC.	52
4.7. Illustration of the OmniFold method. In Step 1, MC is corrected to match data at the detector level, producing a weighting function $w_1(\vec{x}_{\text{reco}}^{\text{MC}})$. In Step 2, a new function $w_2(\vec{x}_{\text{true}}^{\text{MC}})$ is learned based only on particle-level quantities. The method proceeds iteratively, refining the function $w_2(\vec{x}_{\text{true}}^{\text{MC}})$ such that the particle-level MC can be reweighted to give event yields and kinematics that match those observed in the data. [20]	53

5.1.	Event-plane resolution: Second-order (fourth-order) harmonic relative to the event plane, $R_2(\Psi_2)$ ($R_4(\Psi_2)$), respectively. The approach follows the Modified Reaction Plane (MRP) method [21] utilizing the charged tracks of the TPC for event-plane reconstruction and resolution calculation for tracks ranging from 0.2-1.0 GeV/c.	58
5.2.	Example of a correlation function distribution in 2 dimensions between $15 < p_{T,\text{jet}} < 20$ GeV/c $1.0 < p_{T,\text{assoc}} < 1.5$ GeV/c (left). Example of a normalized mixed event distribution in 2 dimensions between $15 < p_{T,\text{jet}} < 20$ GeV/c $1.0 < p_{T,\text{assoc}} < 1.5$ GeV/c showing the $\Delta\eta = \eta_{\text{jet}} - \eta_{\text{assoc}}$. acceptance for associated particles given $ \eta_{\text{assoc}} < 1.0$ and $ \eta_{\text{jet}} < 0.6$ (right).	60
5.3.	(Top) Signal+background region, background-dominated region, and RPF fit to the background for the event-plane dependent jet-hadron correlations of $15 < p_{T,\text{jet}}^{\text{jet}} < 20$ GeV/c jets correlated with $p_{T,\text{assoc}}$ range 1.52.0 charged hadrons from the 20-50% most central events. (Bottom) Quality of the RPF fit to the background-dominated region expressed as (data - fit) / fit.	65
5.4.	Example of a background subtracted correlation function for $15 < p_{T,\text{jet}} < 20$ GeV/c jets correlated with $p_{T,\text{assoc}}$ range 1.52.0 charged hadrons from the 20-50% most central collisions. Correlated scaled uncertainty from the application of mixed events is displayed by the red band, while the uncertainty associated with the RPF background fit is displayed as the gray band.	65
5.5.	$p_{T,\text{jet}}$ resolution for $15 < p_{T,\text{jet}}^{\text{GEN}} < 20$ GeV/c (left) and $20 < p_{T,\text{jet}}^{\text{GEN}} < 40$ GeV/c (right) $R = 0.4$ full jets with the corresponding response matrix as an inset on the top right. Jets are measured from all angles relative to the event plane in the 20-50% most central events. This comparison is for matched GEN-level to RECO-level jets.	67

5.6.	Near-side (left) and away-side (right) uncorrected (unc.) associated yield vs $p_{T,assoc}$ for 15-20 (top) and 20-40 GeV/ c (bottom) full jets of 20-50% centrality in Au+Au collisions. The grey bands describe the systematic uncertainties of the background fits and are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is the inclusive transverse momentum bin from 1.0-10.0 GeV/ c . Points are displaced for visibility. . . .	73
5.7.	Near-side (left) and away-side (right) jet width vs $p_{T,assoc}$ for 15-20 (top) and 20-40 GeV/ c (bottom) full jets of 20-50% centrality in Au+Au collisions. The widths are extracted from the gaussian fit to the jet peak. The grey bands describe the systematic uncertainties of the background fits which are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed-event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is an inclusive transverse momentum bin from 1.0-10.0 GeV/ c . Points are displaced for visibility.	74
5.8.	Near-side (left) and away-side (right) associated-yield ratios (of out-of-plane and mid-plane to in-plane) vs $p_{T,assoc}$ for 15-20 (top) and 20-40 (bottom) GeV/ c full jets in 20-50% centrality collisions. The grey bands describe the systematic uncertainties of the background fits which are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the JES correction. Points are displaced for visibility.	78
6.1.	Two-dimensional histogram showing detector response for $p_{T,jet}$, showing probability of a $p_{T,jet}^{GEN}$ particle level jet being reconstructed as $p_{T,jet}^{RECO}$. Purple stars shows expected $p_{T,jet}^{RECO}$ given a particular bin of the $p_{T,jet}^{GEN}$	83
6.2.	$p_{T,jet}$ -resolution in different $p_{T,jet}$ bins.	84

6.3. Closures for $\lambda_{\beta \in \{0.5, 1, 2\}}^{\kappa=1}$ distributions.	84
6.4. Closures for $\lambda_{\beta \in \{0.5, 1, 2\}}^{\kappa=1}$ vs R_g profile shown for jets with $20 \leq p_{T,\text{jet}} < 40$ GeV/ c	85
6.5. Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/ c	86
6.6. Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/ c	87
6.7. Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $20 \leq p_{T,\text{jet}} < 30$ GeV/ c	88
6.8. Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $20 \leq p_{T,\text{jet}} < 30$ GeV/ c	89
6.9. Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $30 \leq p_{T,\text{jet}} < 60$ GeV/ c	90
6.10. Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $30 \leq p_{T,\text{jet}} < 60$ GeV/ c	91
6.11. Closure test for g , p_T^D and LeSub showing the ratio of distributions calculated using jets from unfolded 0-20% centrality detector level and particle level events from Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV.	91
6.12. Normalized $g = \lambda_1^1$ distributions for jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/ c (left) and $p_{T,\text{jet}} \geq 20$ GeV/ c (right) from central (0-20%, red stars) and peripheral (40-80%, blue stars) centrality ranges. Systematic uncertainties are shown as shaded boxes. Central over peripheral ratios (magenta stars) are in the lower panels with their systematic uncertainties represented as magenta boxes. Sys- tematic uncertainties are assumed to be uncorrelated between the centrality ranges.	92

6.13. Normalized $p_T^D = \sqrt{\lambda_0^2}$ distributions for jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/ c (left) and $p_{T,\text{jet}} \geq 20$ GeV/ c (right) from central (0-20%, red stars) and peripheral (40-80%, blue stars) centrality ranges. Systematic uncertainties are shown as shaded boxes. Central over peripheral ratios (magenta stars) are in the lower panels with their systematic uncertainties represented as magenta boxes. Systematic uncertainties are assumed to be uncorrelated between the centrality ranges.	92
6.14. Normalized LeSub distributions for jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/ c (left) and $p_{T,\text{jet}} \geq 20$ GeV/ c (right) from central (0-20%, red stars) and peripheral (40-80%, blue stars) centrality ranges. Systematic uncertainties are shown as shaded boxes. Central over peripheral ratios (magenta stars) are in the lower panels with their systematic uncertainties represented as magenta boxes. Systematic uncertainties are assumed to be uncorrelated between the centrality ranges.	93
A.1. η vs θ	106

Chapter 1

Introduction

1.1 The Standard Model

According to the Standard Model (SM), all matter results from interactions between fundamental particles: lepton and quarks (fermions), and mediators (bosons). Table 1.1 shows fermions of the standard model, their anti-particle equivalent would have all the signs reversed for the quantum numbers. For example, an anti-electron (e^+) and anti-electron-neutrino ($\bar{\nu}_e$) lepton number would flip to $(-1, 0, 0)$ and charges (Q s) would be $+1$ and 0 respectively.

Leptons				Quarks			
	l	Q	(L_e, L_μ, L_τ)	q	Q	$(F_d, F_u, F_s, F_c, F_b, F_t)$	
First generation	e^-	-1	$(1, 0, 0)$	d	$-1/3$	$(-1, 0, 0, 0, 0, 0)$	
	ν_e	0	$(1, 0, 0)$	u	$2/3$	$(0, 1, 0, 0, 0, 0)$	
Second generation	μ^-	-1	$(0, 1, 0)$	s	$-1/3$	$(0, 0, -1, 0, 0, 0)$	
	ν_μ	0	$(0, 1, 0)$	c	$2/3$	$(0, 0, 0, 1, 0, 0)$	
Third generation	τ^-	-1	$(0, 0, 1)$	b	$-1/3$	$(0, 0, 0, 0, -1, 0)$	
	ν_τ	0	$(0, 0, 1)$	t	$2/3$	$(0, 0, 0, 0, 0, 1)$	

Table 1.1: Three generations of leptons (left) and quarks (right).

The mediators/bosons exchanged between the other fundamental particles give rise to the fundamental interactions: electromagnetism from the photon (γ), weak force from $W^\pm$ and Z bosons, the gluon (g) for the strong force, and the Higgs boson (h) for imparting mass to the fundamental particles. Various interaction vertices possible in SM, that become the building blocks of all fundamental interactions are shown in 1.2.

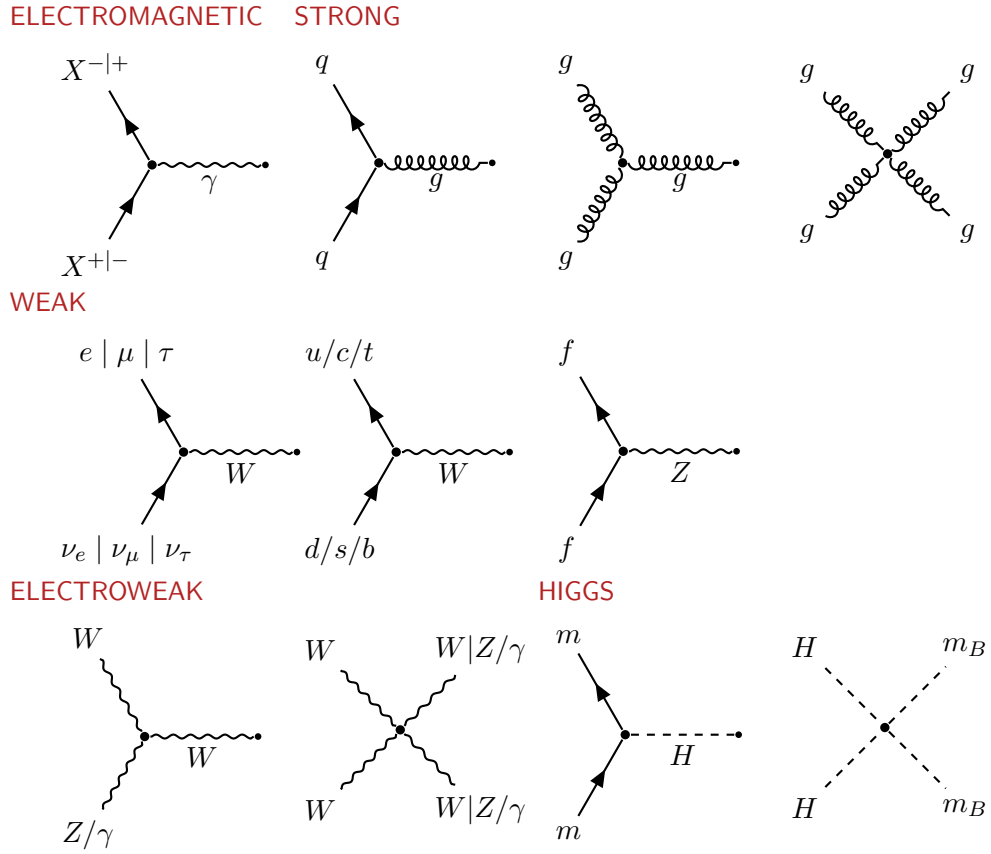


Table 1.2: Feynman diagrams of various interaction vertices possible in the Standard Model

1.2 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the component of the SM which describes the strong interactions of quarks and gluons. The QCD Lagrangian can be expected as:

$$\begin{aligned} \mathcal{L}_{\text{QCD}} = & \underbrace{\bar{q}^\alpha (i\gamma^\mu \partial_\mu \delta_{\alpha\beta} - g\gamma^\mu t_{\alpha\beta}^a A_\mu^a - m\delta_{\alpha\beta}) q^\beta - \frac{1}{4} F_{\mu\nu}^a F_a^{\mu\nu}}_{\mathcal{L}_{\text{classical}}} \\ & - \underbrace{\frac{1}{2\lambda} (\partial^\mu A_\mu^a)^2}_{\mathcal{L}_{\text{gauge-fixing}}} + \underbrace{\bar{c}^\alpha \partial^\mu (\partial_\mu \delta_{\alpha\beta} + igT_{\alpha\beta}^a A_\mu^a) c^\beta}_{\mathcal{L}_{\text{ghost}}} \end{aligned} \quad (1.1)$$

$\mathcal{L}_{\text{classical}}$ is the classical Lagrangian density shows interaction of a quark field q^α with 3 color degrees of freedom interacting with gluon fields A_μ^a with 8 color degrees of freedom. g and m are dimensionless and correspond to QCD coupling constant, and quark-mass respectively. $F_{\mu\nu}^a$ are the gluon tensors defined as,

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - gf_{abc} A_\mu^b A_\nu^c. \quad (1.2)$$

To properly define gluon propagators, we need to make a choice of gauge. The $\mathcal{L}_{\text{gauge-fixing}}$ parametrizes the class of covariant gauges with the gauge parameter λ . In non-Abelian theories, this gauge fixing term leads to creation of unphysical degrees of freedom, which are canceled out by $\mathcal{L}_{\text{ghost}}$ which contains complex, scalar fermion fields c^α .

The matrices t^a and T^a are the fundamental and adjoint representations of SU(3) in form of traceless 3×3 and 8×8 hermitian matrices satisfying:

$$\begin{aligned} [t^a, t^b] &= if_{abc} t^c, \quad \text{tr}(t^a t^b) = \frac{1}{2} \delta^{ab} \\ [T^a, T^b] &= if_{abc} T^c, \quad (T^a)_{bc} = -if^{abc}. \end{aligned} \quad (1.3)$$

Comparing with quantum electrodynamics (QED), where the photon field tensor lacks the last term of Eqn. 1.2 as it is an Abelian, $U(1)$ gauge theory, we can explain the presence

of three/four-gluon vertices and absence of such vertices for the photon in Table 1.2. The lagrangian can be used to formulate *Feynman rules* for calculating amplitude contributions from various propagators and vertices, which are shown in Tab. 1.3.

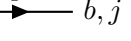
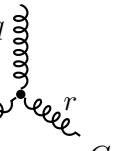
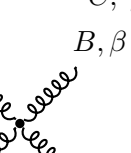
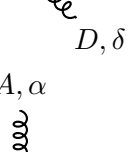
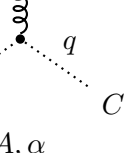
	$\delta^{AB} \left[-g^{\alpha\beta} + (1-\lambda) \frac{p^\alpha p^\beta}{p^2 + i\epsilon} \right] \frac{i}{p^2 + i\epsilon}$
	$\delta^{AB} \frac{i}{p^2 + i\epsilon}$
	$\delta^{ab} \frac{i}{(\not{p} - m + i\epsilon)_{ji}}$
	$-g f^{ABC} [(p-q)^\gamma g^{\alpha\beta} + (q-r)^\alpha g^{\beta\gamma} + (r-p)^\beta g^{\gamma\alpha}]$ (all momenta incoming, $p+q+r=0$)
	$-ig^2 f^{XAC} f^{XBD} [g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\delta} g^{\beta\gamma}]$ $-ig^2 f^{XAD} f^{XBC} [g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\gamma} g^{\beta\delta}]$ $-ig^2 f^{XAB} f^{XCD} [g^{\alpha\gamma} g^{\beta\delta} - g^{\alpha\delta} g^{\beta\gamma}]$
	$g f^{ABC} q^\alpha$
	$-ig (t^A)_{cb} (\gamma^\alpha)_{ji}$

Table 1.3: Feynman rules for QCD in a covariant gauge with gauge parameter λ : the gluon, ghost and quark propagators, followed by the three- and four-gluon, ghost-gluon and quark-gluon vertices.

1.2.1 The Running Coupling

At an energy scale Q , consider a dimensionless observable X . Lets assume that Q is sufficiently high that the quark masses become negligible in comparison. To calculate X as

a perturbation series in the QCD coupling $\alpha_s = \frac{g^2}{4\pi}$ (analogous to the QED fine structure constant), we need to remove ultraviolet divergences through renormalization. Renormalization introduces another arbitrary energy scale μ where the divergences are subtracted. In general, X would depend on the ratio Q^2/μ^2 and post-renormalization coupling $\alpha_s(\mu^2)$. However, any physical observable should be independent of μ :

$$\begin{aligned} \mu^2 \frac{dX(Q^2/\mu^2, \alpha_s(\mu^2))}{d\mu^2} &= \left[\mu^2 \frac{\partial}{\partial \mu^2} + \mu^2 \frac{\partial \alpha_s(\mu^2)}{\partial \mu^2} \frac{\partial}{\partial \alpha_s} \right] X(Q^2/\mu^2, \alpha_s(\mu^2)) = 0 \\ \implies \left[-\frac{\partial}{\partial t} + \beta(\alpha_s) \frac{\partial}{\partial \alpha_s} \right] X(e^t, \alpha_s) &= 0. \end{aligned} \quad (1.4)$$

Where, $t = \ln \frac{Q^2}{\mu^2}$, $\beta(\alpha_s) = \mu^2 \frac{\partial \alpha_s}{\partial \mu^2}$, $\alpha_s \equiv \alpha_s(\mu^2(t))$

$$\begin{aligned} \beta(\alpha_s) = -\frac{\partial \alpha_s}{\partial t} &\implies t = \int_{\alpha_s}^{\alpha_s(Q^2)} \frac{d\alpha_s}{\beta(\alpha_s)} \\ \implies \frac{\partial \alpha_s(Q^2)}{\partial t} &= Q^2 \frac{\partial \alpha_s}{\partial Q^2} = \beta(\alpha_s(Q^2)), \quad \frac{\partial \alpha_s(Q^2)}{\partial \alpha_s} = \frac{\beta(\alpha_s(Q^2))}{\beta(\alpha_s)} \end{aligned} \quad (1.5)$$

Substituting in Eq. 1.4, we see that $X(1, \alpha_s(Q^2))$ is a valid solution. Thus the scale dependence in X comes from the *running coupling* $\alpha_s(Q^2)$. Perturbatively expanding $\beta(\alpha_s)$, we can write the *renormalization group equation* as,

$$Q^2 \frac{\partial \alpha_s}{\partial Q^2} = -\alpha_s^2(Q^2) (\beta_0 + \beta_1 \alpha_s(Q^2) + \mathcal{O}(\alpha_s^2(Q^2))) \quad (1.6)$$

Where,

$$\beta_0 = \frac{1}{(4\pi)^2} \left(11 - \frac{2}{3} N_f \right), \quad \beta_1 = \frac{1}{(4\pi)^4} \left(102 - \frac{38}{3} N_f \right) \quad (1.7)$$

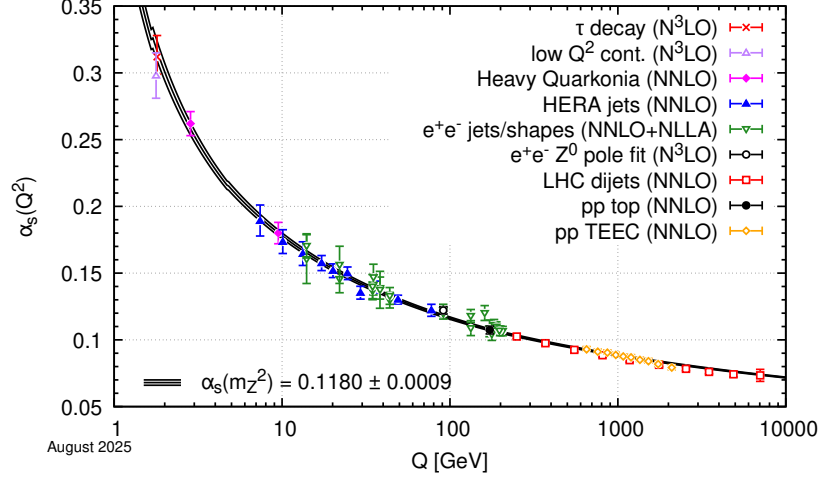


Figure 1.1: Summary of determinations of α_s as a function of the energy scale Q compared to the running of the coupling computed at five loops taking as an input the current Particle Data Group (PDG) world average, $\alpha_s(m_Z^2) = 0.1180 \pm 0.0009$ [1]

Integrating Eq. 1.6

$$\ln\left(\frac{Q^2}{Q_0^2}\right) = \int_{\alpha_s(Q_0^2)}^{\alpha_s(Q^2)} \frac{d\alpha_s}{\beta(\alpha_s)}$$

$$\Rightarrow \alpha_s(Q^2) = \frac{\alpha_s(Q_0^2)}{1 + \beta_0(\alpha_s(Q_0^2)) \ln(Q^2/Q_0^2) + \mathcal{O}(\alpha_s^2)} \quad (1.8)$$

Usually, the reference scale Q_0^2 is set to a convenient energy scale in the perturbative regime. α_s at all other scales can be inferred accordingly. Usually, Q_0 is set to mass of the Z -boson, m_Z as shown in Fig. 1.1. Another approach is to choose a scale parameter, Λ such that $\alpha_s(\Lambda) \rightarrow \infty$.

$$\ln\left(\frac{Q^2}{\Lambda^2}\right) = - \int_{\alpha_s(Q^2)}^{\infty} \frac{d\alpha_s}{\beta(\alpha_s)}$$

$$\Rightarrow \alpha_s(Q^2) = \frac{1}{\beta_0 \ln(Q^2/\Lambda^2)} \left[1 - \frac{\beta_1}{\beta_0} \frac{\ln(\ln(Q^2/\Lambda^2))}{\ln(Q^2/\Lambda^2)} + \mathcal{O}\left(\frac{1}{\ln(Q^2/\Lambda^2)^2}\right) \right] \quad (1.9)$$

Λ is called the *confinement scale*, around 200 MeV. Eqs. 1.8 and 1.9 show a monotonically decreasing relationship between α_s and Q^2 . At high momentum transfers ($Q \gg \Lambda$), the

coupling strength vanishes ($\alpha_s \rightarrow 0$). In this regime, quarks and gluons behave as quasi-free particles, making perturbative QCD (pQCD) calculations highly reliable (*Asymptotic Freedom*). At low energy scales ($Q \ll \Lambda$), the coupling diverges. In this non-perturbative QCD (npQCD) limit, isolated color charges cannot exist and partons are permanently bound into color-neutral hadrons (*Color Confinement*).

1.2.2 Quark Gluon Plasma (QGP)

Asymptotic freedom of QCD means that matter at extremely high temperatures and densities (high Q^2) undergo a *deconfinement* phase transition, with partonic degrees of freedom replacing hadronic degrees of freedom. This state of deconfined quarks and gluons is called QCD/quark-gluon plasma due to analogous phenomena observed in atomic physics. Thermodynamically, when temperature (T) or baryon number density (n_B) approaches the confinement scale Λ introduced in Sec. 1.2.1 ($T \sim \Lambda \sim \mathcal{O}(10^{12})K$, and $n_B \sim \Lambda^3 \sim 1 \text{ fm}^{-3}$) the boundaries between hadrons become ill defined resulting in hadrons “melt” into QGP. For reference, the core of our sun “only” has $T \sim \mathcal{O}(10^7)K$ and $n_B \sim \mathcal{O}(10^{-14})\text{fm}^{-3}$.

These orders of T and n_B are the theorized conditions prevalent in the universe a few μs after Big-Bang. The only natural phenomena in the present-day, cold universe that can achieve these conditions are the cores of neutron stars. Artificially, the required T and n_B can be achieved at facilities like the Large Hadron Collider (LHC) and Relativistic Heavy Ion Collider (RHIC) through collisions of heavy-ions (like Au, Pb, etc.) at near-light speeds, the aptly named *little bangs*. More on this will be discussed in Sec. 3.2.

1.3 Hard Processes

To consider any nucleus as a singular coherent object, the fluctuations inside it would involve momentum transfers below the confinement scale (Λ) introduced in 1.2.1, and would occur over timescales $\sim 1/\Lambda$. We define *hard-scattering processes* or just *hard processes* as interactions of a hard perturbative probe with high momentum transfer $Q \gg \Lambda$. Since

this interaction occurs over a time-scale $\sim 1/Q \ll 1/\Lambda$, the perturbative probe effectively interacts with a near-static “snapshot” of the nucleus with a characteristic resolution $\sim 1/Q$.

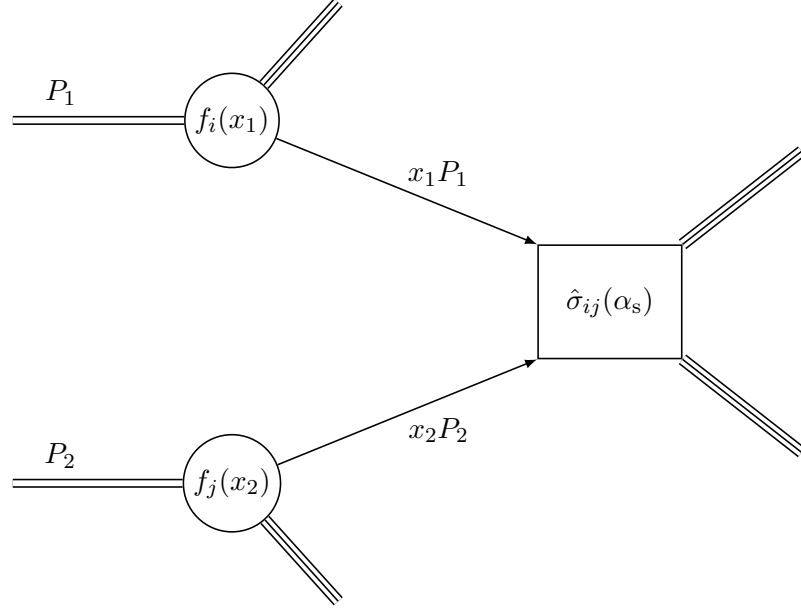


Figure 1.2: Collinear factorization of a hadronic hard scattering.

Thus, we can *factorize* the cross section of a collision process of nuclei with momentum P_1 and P_2 depicted by Fig. 1.2 as,

$$\sigma(P_1, P_2) = \sum_{i,j} \int dx_1 dx_2 f_i(x_1, \mu^2) f_j(x_2, \mu^2) \hat{\sigma}_{ij}(p_1, p_2, \alpha_s, Q^2/\mu^2) \quad (1.10)$$

Where, the interactions occur between partons i from nucleus 1 and j from nucleus 2 carrying momentum $p_1 = x_1 P_1$ and $p_2 = x_2 P_2$ respectively. μ is an arbitrary *factorization scale*, often just set to Q . The $f_k(x, \mu^2)$ are the *parton densities* giving likelihood of a parton k to carry x fraction of it's parent nucleus's momentum at a particular factorization scale μ . $\hat{\sigma}_{ij}$ is the *parton cross-section* for scattering of partons i and j . We can approximate it to the $(n+k)$ order in α_s as,

$$\hat{\sigma}_{ij}(p_1, p_2, \alpha_s, Q^2/\mu^2) = \alpha_s^k \sum_{m=0}^n \hat{\sigma}_{ij}^{(m)}(p_1, p_2, Q^2/\mu^2) \alpha_s^m \quad (1.11)$$

Different hard processes will have different leading powers k . W production will have $k = 0$ whereas perturbative dijet production with two outgoing quarks will have $k = 2$.

1.4 From partons to jets

Outgoing partons evolve from higher to lower momentum transfer scales (say, t). Initially their evolution are modeled using the pQCD mechanism called *parton shower*. Eventually, say at an infra-red scale cut-off t_0 , pQCD calculations stop being valid and npQCD processes like hadronization dominate. QCD Monte-Carlo programs combining parton shower at $t < t_0$ and hadronization process at $t > t_0$ are called *QCD event generators*

1.4.1 Parton branching

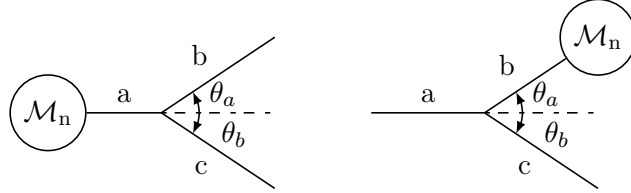


Figure 1.3: Kinematics of timelike (left) vs spacelike (right) parton branching

Consider timelike splitting of a parton, $a \rightarrow b + c$ shown in Fig. 1.4.1. Let's assume the partons are massless, and define the evolution scale (t) as,

$$t \equiv p_a^2 \gg p_b^2, p_c^2$$

$$\implies t \approx 2E_b E_c (1 - \cos(\theta)) \approx z(1-z)E_a^2 \theta^2 \quad (1.12)$$

Where, $z = E_b/E_a = 1 - E_c/E_a$ and $\theta = \theta_a + \theta_b$ is the opening angle. The cross-section of a n -parton final state branching processes can be calculated by considering n -parton matrix element.

$$d\sigma_n = \mathcal{F} |\mathcal{M}_n|^2 d\Phi_n \quad (1.13)$$

Where $\mathcal{F}$ is the initial-state flux factor and Φ_n is phase space of the n -parton final state.

In the branching process the n -parton pre-split phase space volume element:

$$d\Phi_n = \dots \frac{d^3\mathbf{p}_a}{2(2\pi)^3 E_a} \quad (1.14)$$

will be replaced by the $n + 1$ -parton phase space and the $n + 1$ -parton matrix element can be calculated from the n -parton matrix element.

$$\begin{aligned} d\Phi_{n+1} &= \dots \frac{d^3\mathbf{p}_b}{2(2\pi)^3 E_b} \frac{d^3\mathbf{p}_c}{2(2\pi)^3 E_c} = d\Phi_n \frac{1}{4(2\pi)^3} dt dz d\phi \\ |\mathcal{M}_{n+1}|^2 &\sim \frac{8\pi\alpha_s}{t} CF |\mathcal{M}_n|^2 \end{aligned} \quad (1.15)$$

Substituting in Eq. 1.13 and integrating away the $d\phi$, we get,

$$\begin{aligned} d\sigma_{n+1} &= d\sigma_n \frac{dt}{t} dz \frac{\alpha_s}{2\pi} \int \frac{d\phi}{2\pi} CF \\ &= d\sigma_n \frac{dt}{t} dz \frac{\alpha_s}{2\pi} \hat{P}_{ba} \end{aligned} \quad (1.16)$$

Same equation would hold for the space-like case. To obtain the evolution equations, we can frame the evolution in terms of the change in parton distribution function $f(x, t)$ when the evolution scale t goes from t to $t + \delta t$,

$$\begin{aligned} \delta f_i(x, t) &= \delta f_{\text{in}} - \delta f_{\text{out}} \\ &= \frac{\delta t}{t} \sum_j \int_0^1 dz \frac{\alpha_s}{2\pi} \left[\frac{\hat{P}_{ij}(z)}{z} f_j(x/z, t) - \hat{P}_{ji} f_i(x, t) \right] \end{aligned} \quad (1.17)$$

Introducing the *Sudakov form factor*, which gives the probability of a parton i evolving from scale t to a softer scale t_0 without a resolvable emission,

$$\Delta_i(t) \equiv \exp \left(- \sum_j \int_{t_0}^t \frac{dt'}{t'} \int dz \frac{\alpha_s}{2\pi} \hat{P}_{ji}(z) \right) \quad (1.18)$$

We get the familiar DGLAP-like evolution equations,

$$t \frac{\partial}{\partial t} \left(\frac{f_i}{\Delta_i} \right) = \frac{1}{\Delta_i} \sum_j \int \frac{dz}{z} \frac{\alpha_s}{2\pi} \hat{P}_{ij}(z) f_j(x/z, t) \quad (1.19)$$

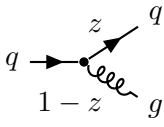
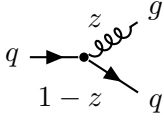
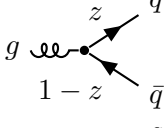
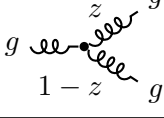
Kernel (ij)	Diagram	$\hat{P}_{ij}(z)$
qq		$C_F \left[\frac{1+z^2}{1-z} \right]$
gq		$C_F \frac{1+(1-z)^2}{z}$
qg		$T_R [z^2 + (1-z)^2]$
gg		$C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z) \right]$

Table 1.4: The four leading-order collinear splittings entering the DGLAP evolution of Eq. ?? . The daughter parton i carrying momentum fraction z is drawn on the upper leg, the recoiling parton $(1-z)$ on the lower. The plus prescription (Eq. ??) and $\delta(1-z)$ terms regularize the soft limit and enforce the momentum and flavor sum rules [22].

To deal with singularities corresponding to soft gluons at $z \rightarrow 1$, we introduce an explicit infrared cut-off $z < 1 - \epsilon(t)$. Requiring the transverse momentum in timelike branching $a \rightarrow bc$ to be positive,

$$\begin{aligned}
p_T^2 &= z(1-z)p_a^2 - (1-z)p_b^2 - zp_c^2 > 0 \\
\implies z(1-z) &> t_0/t \\
\implies z, 1-z &> \epsilon(t) = \frac{1 - \sqrt{1 - 4t/t_0}}{2} \simeq t_0/t \\
\implies t_0/t &< z < 1 - t_0/t
\end{aligned} \quad (1.20)$$

$$\Delta_i(t) \equiv \exp \left(- \sum_j \int_{2t_0}^t \frac{dt'}{t'} \int_{t_0/t'}^{1-t_0/t'} dz \frac{\alpha_s}{2\pi} \hat{P}_{ji}(z) \right) \quad (1.21)$$

1.4.2 Angular Ordering

Sec. 1.4.1 took collinear enhancements into account to all orders in α_s . However it does not take the enhancement in soft gluon emission at wide emission angle. This is demonstrated in Eq. 1.22 by considering an on-mass-shell external line of momentum p , mass m emitting a gluon with momentum q at an angle θ

$$\frac{1}{(p \pm q)^2 - m^2} = \frac{\pm 1}{2p \cdot q} = \frac{\pm 1}{2\omega E(1 - v \cos \theta)}, \quad (1.22)$$

where ω is the gluon energy, E and v the energy and velocity of the emitter and θ the emission angle. For light partons $v \rightarrow 1$ and the collinear enhancement appears as $\theta \rightarrow 0$; the enhancement as $\omega \rightarrow 0$ holds for any velocity and angle. Internal lines are off-mass-shell and carry no such enhancement, since $(p + q)^2 - m^2 \rightarrow p^2 - m^2 \neq 0$. The cross section carries a sum over all pairs of external lines $\{i, j\}$,

$$d\sigma_{n+1} = d\sigma_n \frac{d\omega}{\omega} \frac{d\Omega}{2\pi} \frac{\alpha_s}{2\pi} \sum_{i,j} C_{ij} W_{ij} \quad (1.23)$$

The weighted sum $\sum_{i,j} C_{ij} W_{ij}$ is called the *antenna pattern* of the process. The *color factor* C_{ij} and *radiation function* W_{ij} are defined as,

$$C_{ij} = -\mathbf{Q}_i \cdot \mathbf{Q}_j$$

$$W_{ij} = \frac{\omega^2 p_i \cdot p_j}{p_i \cdot q \, p_j \cdot q} = \frac{1 - v_i v_j \cos \theta_{ij}}{(1 - v_i \cos \theta_{iq})(1 - v_j \cos \theta_{jq})} \quad (1.24)$$

The vectors $\mathbf{Q}_i$ represents color charge of parton i .

$$\mathbf{Q}_i^2 = \begin{cases} C_F = 4/3 & i \in \text{quarks} \\ C_A = 3 & i \in \text{gluons} \\ 0 & i \in \text{color singlet} \end{cases} \quad (1.25)$$

W_{ij} comes from interference between lines i and j . W_{ij} can be separated into two parts $W_{ij}^{[i/j]}$ each with an *incoherent* and a *coherent* contribution from line i/j . Assuming massless emitters ($v_i = v_j = 1$) for simplicity,

$$\begin{aligned} W_{ij} &= W_{ij}^{[i]} + W_{ij}^{[j]} \\ W_{ij}^{[i]} &= \frac{1}{2} \left(\overbrace{\frac{1}{1 - \cos \theta_{iq}}}^{\text{incoherent}} + \overbrace{\frac{\cos \theta_{iq} - \cos \theta_{ij}}{(1 - \cos \theta_{jq})(1 - \cos \theta_{iq})}}^{\text{coherent}} \right) \\ &= \frac{1}{2(1 - \cos \theta_{iq})} \left(1 + \frac{\cos \theta_{iq} - \cos \theta_{ij}}{1 - \cos \theta_{ij} \cos \theta_{iq} - \sin \theta_{ij} \sin \theta_{iq} \cos \phi_{iq}} \right) \end{aligned} \quad (1.26)$$

Using the direction of line i as the z-axis, $d\Omega$ from Eq. 1.23 becomes $d \cos \theta_{iq} d\phi_{iq}$. We can carry out the ϕ_{iq} integration,

$$\begin{aligned} \int_0^{2\pi} \frac{d\phi_{iq}}{2\pi} W_{ij}^{[i]} &= \frac{1}{1 - \cos \theta_{iq}} \left(\frac{1 + \frac{\cos \theta_{iq} - \cos \theta_{ij}}{|\cos \theta_{iq} - \cos \theta_{ij}|}}{2} \right) \\ &= \begin{cases} \frac{1}{1 - \cos \theta_{iq}} & \cos \theta_{iq} > \cos \theta_{ij} \implies \theta_{iq} < \theta_{ij} \\ 0 & \text{otherwise,} \end{cases} \end{aligned} \quad (1.27)$$

Eq. 1.27 shows that azimuth-averaged soft gluon emission from a line i is confined to a cone, centered on i 's direction, extending up to the direction of line j , i 's sibling from the preceding split. The total soft gluon emission cross-section can be then easily obtained for two external lines forming a color singlet ($e^+e^- \rightarrow q\bar{q}$) have $\mathbf{Q}_i + \mathbf{Q}_j = 0$ and hence $C_{q\bar{q}} = \mathbf{Q}_q^2 = \mathbf{Q}_{\bar{q}}^2 = C_F$, and angular ordering suppresses radiation outside the cones extending from q to $\bar{q}$ and vice versa.

$$d\sigma_{q\bar{q}g} = d\sigma_{q\bar{q}} \frac{\Theta(\cos(\theta_{qg}) - \cos(\theta_{q\bar{q}}))}{1 - \cos\theta_{qg}} d(\cos\theta_{qg}) \frac{\alpha_s}{2\pi} \frac{d\omega}{\omega} C_F \quad (1.28)$$

We will come back to Eq. 1.28

1.4.3 Interference

Lets take the case of three partons $\{i, j, k\}$ forming a color singlet ($\mathbf{Q}_i + \mathbf{Q}_j + \mathbf{Q}_k = 0$) e.g., $e^+e^- \rightarrow q\bar{q}g$. First, partons l and k come from a color singlet state ($\mathbf{Q}_l + \mathbf{Q}_k = 0$). Then l splits into i and j ($\mathbf{Q}_l = \mathbf{Q}_i + \mathbf{Q}_j = -\mathbf{Q}_k$). The antennae function for this state is:

$$\begin{aligned} \sum_{a,b} C_{ab} W_{ab} &= -\mathbf{Q}_i \cdot \mathbf{Q}_j W_{ij} - \mathbf{Q}_j \cdot \mathbf{Q}_k W_{jk} - \mathbf{Q}_i \cdot \mathbf{Q}_k W_{ik} \\ &= \frac{1}{2} [\mathbf{Q}_i^2 (W_{ij} + W_{ik} - W_{jk}) + \mathbf{Q}_j^2 (W_{ij} + W_{jk} - W_{ik}) + \mathbf{Q}_k^2 (W_{ik} + W_{jk} - W_{ij})] \\ &= \mathbf{Q}_i^2 (W_{ij}^{[i]} + \tilde{W}_{jk}^{[i]} - \tilde{W}_{ik}^{[j]}) + \mathbf{Q}_j^2 (W_{ij}^{[j]} + \tilde{W}_{ik}^{[j]} - \tilde{W}_{jk}^{[i]}) \\ &\quad + \mathbf{Q}_k^2 \left(\tilde{W}_{jk}^{[i]} + \tilde{W}_{ik}^{[j]} + \frac{W_{ik}^{[k]} + W_{jk}^{[k]}}{2} \right) \end{aligned} \quad (1.29)$$

Where,

$$\tilde{W}_{jk}^{[i]} = \frac{W_{ik}^{[i]} - W_{ij}^{[i]}}{2}, \quad \tilde{W}_{ik}^{[j]} = \frac{W_{jk}^{[j]} - W_{ij}^{[j]}}{2} \quad (1.30)$$

are shifted radiation functions for likes i and j , with the collinear singularity canceled out. This allows us to make certain useful approximations. Taking $\theta_{ij} \rightarrow 0$, we can approximate the directions of i and j by l .

$$W_{ik}^{[k]} \simeq W_{jk}^{[k]} \simeq W_{lk}^{[k]}, \text{ and } \tilde{W}_{jk}^{[i]} \simeq \tilde{W}_{ik}^{[j]} \simeq W_{lk}^{[ij]}/2 \quad (1.31)$$

Where,

$$\tilde{W}_{lk}^{[ij]} = \begin{cases} W_{lk}^{[l]} & \theta_{lq} > \theta_{ij} \\ 0 & \theta_{lq} < \theta_{ij}, \end{cases} \quad (1.32)$$

$$W \simeq \mathbf{Q}_i^2 W_{ij}^{[i]} + \mathbf{Q}_j^2 W_{ij}^{[j]} + \mathbf{Q}_k^2 W_{lk}^{[k]} + \mathbf{Q}_l^2 \tilde{W}_{lk}^{[ij]}, \quad (1.33)$$

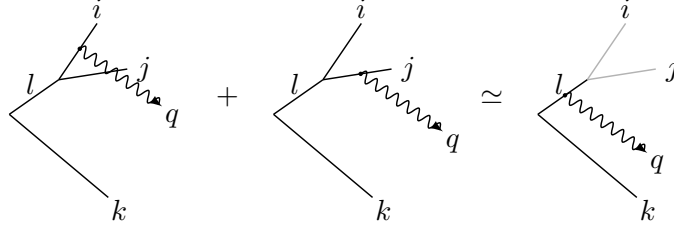


Figure 1.4: Wide-angle emission of soft gluon q , off i and j , acts as if the emission came off the parent parton l , imagined to be on shell.

Each parton radiates in proportion to its color charge squared. When i and j close in angle their incoherent contributions are limited to cones of half-angle θ_{ij} , while at larger angles, out to the direction of k , they radiate coherently in proportion to their combined charge $\mathbf{Q}_l^2$, as if from an internal line of momentum $p_l = p_i + p_j$ as shown in Fig. 1.4.

1.4.4 Coherent Parton Branching

Extended to higher orders this gives a *coherent parton branching* formalism. In place of the virtual mass-squared t the evolution variable is the opening angle, or more precisely

$$\zeta = \frac{p_b \cdot p_c}{E_b E_c} \simeq 1 - \cos \theta \quad (1.34)$$

for $a \rightarrow bc$, with angular ordering $\zeta' < \zeta$ imposed on successive branchings and the propagator factor dt/t of Sec. 1.4.1 replaced by $d\zeta/\zeta$. Using the splitting function $\hat{P}_{ba}(z)dz$ rather than the soft approximation $\mathbf{Q}_a^2 d\omega/\omega$ treats both soft and collinear enhancements correctly, so that

$$d\sigma_{n+1} = d\sigma_n \frac{d\zeta}{\zeta} dz \frac{\alpha_s}{2\pi} \hat{P}_{ba}(z). \quad (1.35)$$

The virtual mass-squared cut-off is replaced by an angular one, $\zeta_0 = t_0/E^2$ for a parton of energy E , and the convenient evolution variable is

$$\tilde{t} = E^2 \zeta \geq t_0, \quad (1.36)$$

in terms of which the ordering condition $\zeta_b, \zeta_c < \zeta_a$ and the resulting cut-off on z read

$$\tilde{t}_b < z^2 \tilde{t}, \quad \tilde{t}_c < (1-z)^2 \tilde{t}, \quad \sqrt{t_0/\tilde{t}} < z < 1 - \sqrt{t_0/\tilde{t}}, \quad (1.37)$$

with $\tilde{t} = \tilde{t}_a$ and $z = E_b/E_a$. Neglecting the masses of b and c ,

$$t = z(1-z)\tilde{t}, \quad p_t^2 = z^2(1-z)^2\tilde{t}, \quad (1.38)$$

and the Sudakov form factor of Sec. 1.4.1 becomes

$$\tilde{\Delta}_i(\tilde{t}) \equiv \exp \left(- \sum_j \int_{4t_0}^{\tilde{t}} \frac{d\tilde{t}'}{\tilde{t}'} \int_{\sqrt{t_0/\tilde{t}'}}^{1-\sqrt{t_0/\tilde{t}'}} dz \frac{\alpha_s(z^2(1-z)^2\tilde{t}')}{2\pi} \hat{P}_{ji}(z) \right) \quad (1.39)$$

At large $\tilde{t}$ this falls more slowly than its collinear counterpart, though still faster than any negative power of $\tilde{t}$. The slower fall implies less branching, which is the suppression of soft gluon emission by angular ordering.

Chapter 2

Solenoidal Tracker At Relativistic Heavy Ion Collider

2.1 Relativistic Heavy-Ion Collider (RHIC)

The Relativistic Heavy Ion Collider (RHIC), at Brookhaven National Laboratory (BNL), is the only collider with dedicated running for heavy ion research (the CERN Large Hadron Collider runs heavy ions for only about one month per year), and the only polarized proton collider ever built. Two concentric accelerator rings 2.4 miles in circumference containing a total of 1,740 superconducting magnets afford RHIC the capability to independently accelerate and collide ion beam species from ${}^1_1\text{H}(p)$ to ${}^{238}_{92}\text{U}$ at energies between 3.85 GeV and 254.9 GeV per nucleon and different spin polarizations for protons (p), as shown in Fig. 2.1. These capabilities make RHIC the most flexible and capable collider in the world for transformative studies of extreme states of nuclear matter and the origin of the proton spin.

On February 6th 2026 RHIC ended operations with collisions of ${}^{16}_8\text{O}$ nuclei at $\sqrt{s_{\text{NN}}} = 200$ GeV recorded by the *Solenoidal Tracker At RHIC (STAR)* and *sPHENIX* detectors. Over its 25 years of active existence, it has performed ~ 300 trillion collisions of 49 unique pairs of nuclei, collected ~ 450 petabytes of data, whose analysis has led to over 650 publications. Eventually, one of RHIC's ion storage rings will be removed and replaced with a new ring for storing accelerated electrons inside the existing accelerator tunnel, and the other major components of the RHIC accelerator complex will live on as parts of the upcoming Electron-Ion Collider (EIC).

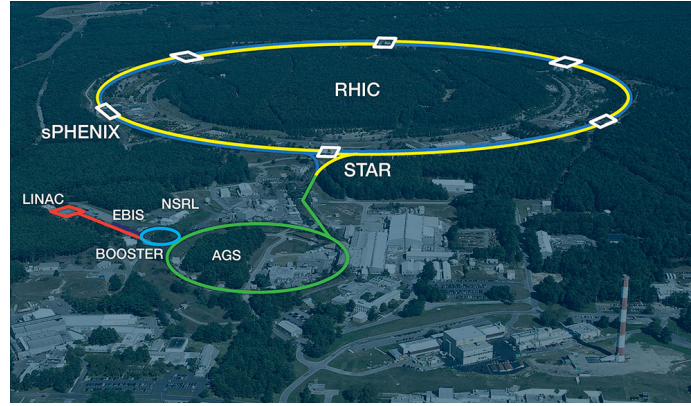


Figure 2.1: An aerial view of the Relativistic Heavy Ion Collider (RHIC), showing the locations of the facilities used for accelerating different ions, and the positions of the STAR and sPHENIX detectors. [2]

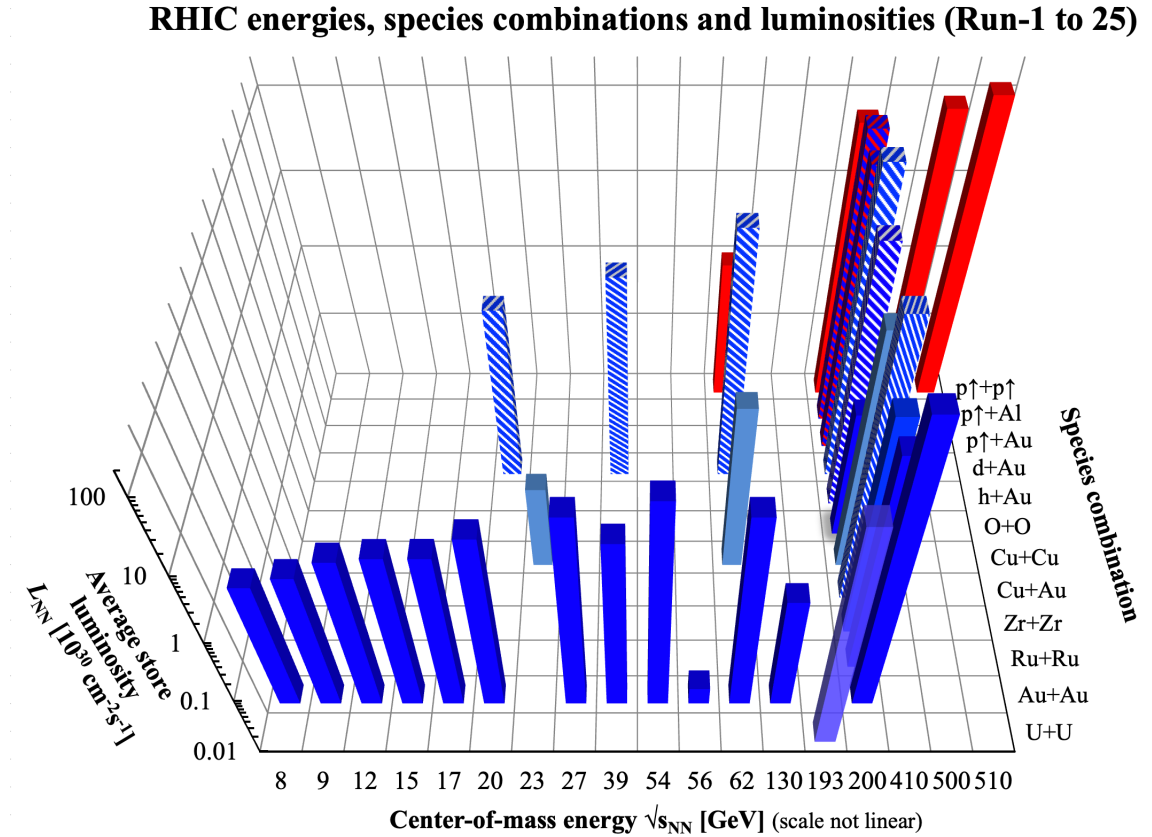


Figure 2.2: Summary of RHIC energies, species combinations, and luminosities for runs between 2000 and 2026. [3]

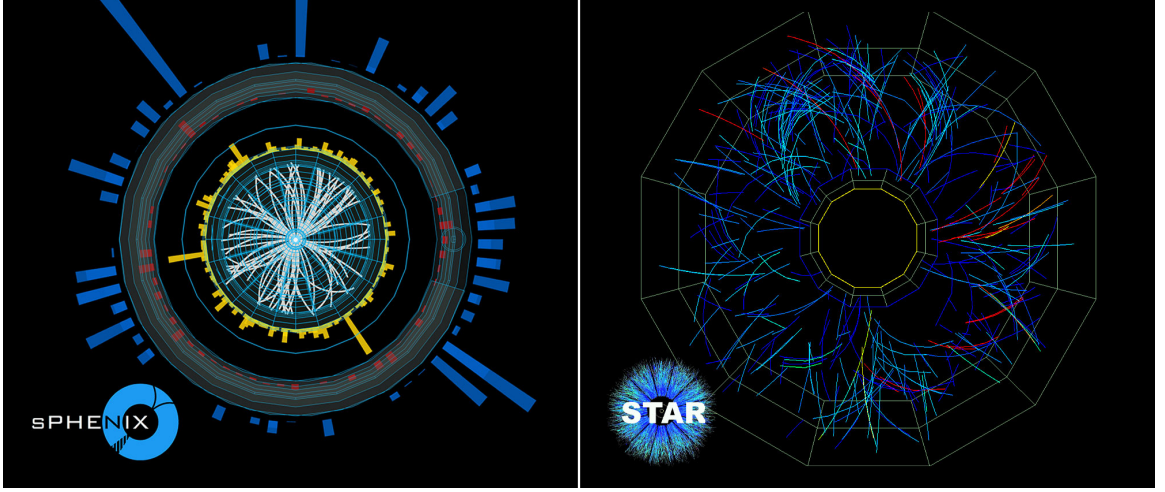


Figure 2.3: Some of the final O+O collisions at $\sqrt{s_{NN}} = 200$ GeV captured by the sPHENIX and STAR detectors at the RHIC. [4]

2.1.1 From stable matter to little bangs

Before collisions, the nuclei have to be created first by taking the corresponding atoms and stripping off all their electrons. That process looks different for different nuclei.

- **For protons** it is *Optically Pumped Polarized H^- Ion Source (OPPIS)* fed into the *Linear Accelerator (LINAC)*. The LINAC accelerates them to ~ 200 MeV and injects into the *Booster Synchrotron* at which point, the two electrons are stripped away by a carbon foil. They are accelerated to upto 2.5 GeV and fed into the *Alternating Gradient Synchrotron (AGS)* which accelerates the proton beam to about 25 GeV.
- **For heavy-ions** like $^{197}_{79}\text{Au}$, a *Laser Ion Source (LION)* extracts Au^+ ions using laser ablation, which go into *Electron Beam Ion Source (EBIS)* which ionizes them to Au^{32+} and accelerates them to 2 MeV per nucleon using electron beams. Then they enter the booster synchrotron which accelerates them to about 95 MeV per nucleon. While exiting from the synchrotron, the atoms are stripped of more electrons by passing them through a stripping foil, and atoms in the Au^{77+} state are fed to the AGS. The beam's kinetic energy up to 8.86 GeV per nucleon for these Au atoms and the final two K shell electrons are stripped.

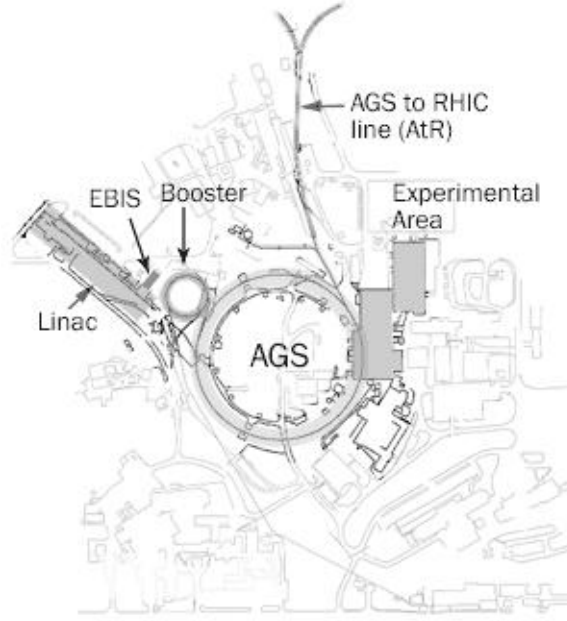


Figure 2.4: Schematic view of the AGS complex, showing various stages the ions go through before going into RHIC

when the ion beam reached top speed in the AGS, it was taken down another beamline called the AGS-to-RHIC (AtR) transfer line. At the end of this line, a switching magnet sent the ion bunches down each of two beamlines to fill RHIC's two ion storage rings. Bunches directed left filled the clockwise RHIC ring (blue) while those directed right traveled counterclockwise in the second RHIC ring (yellow). In the rings, superconducting radio-frequency (RF) cavities accelerate the ion bunches to energies of 100 GeV per nucleon for heavy ions (255 GeV for protons), with four cavities operating at 28.15 MHz to generate standing waves of electric fields for the acceleration and ten 197 MHz cavities maintaining beam stability during storage. Superconducting dipole magnets with 3.5 T fields keep the beams on a circular path, while quadrupole magnets focus the ion beams. A total of 1,740 superconducting magnets are used in the two rings at RHIC, cooled by liquid Helium at < 4.6 K. An aerial view of the RHIC complex with the different subsystems is shown in Fig 2.1.

There are 6 interaction regions (IR) around the RHIC complex, where the two beams cross and allow for detectors to be placed. 4 such detectors have existed during RHIC's

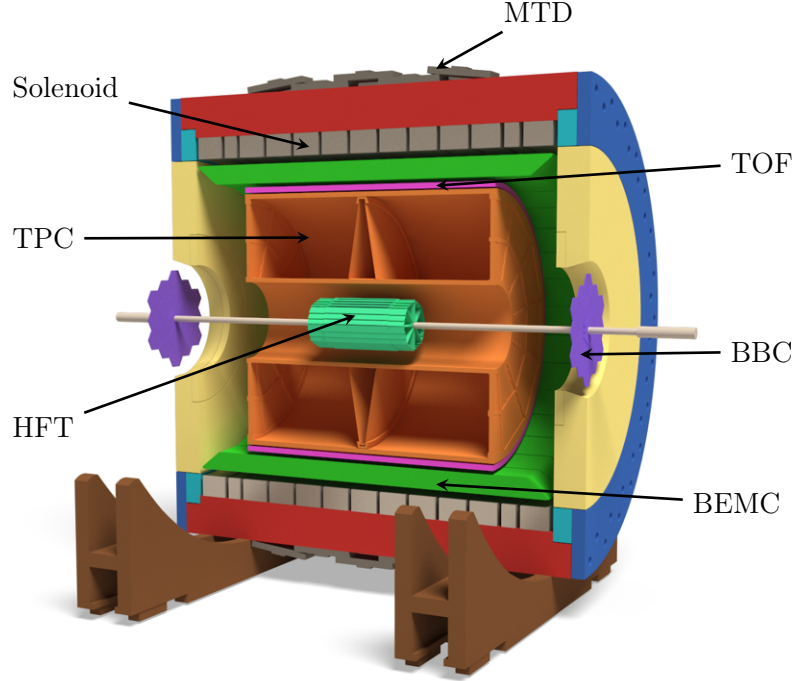


Figure 2.5: Schematic of the STAR detector, depicting the various detector subsystems.

operation: At 10 o'clock and 2 o'clock, the PHOBOS and BRAHMS experiments operated only for the first few years of RHIC (ending in 2006). At 8 o'clock, the PHENIX experiment collected data from 2000 to 2016 and was then transformed into the sPHENIX detector which began operations in 2023. STAR sits at the 6 o'clock position, and has been the only experiment running for all 25 years of RHIC's existence.

2.2 The STAR detector

STAR [23] is designed to provide hermetic coverage around the interaction point, emphasizing precise tracking capabilities in high multiplicity environments, for its main physics mission of obtaining a fundamental understanding of the microscopic structure of QCD interactions at high energy densities.

It is a highly versatile, multipurpose detector with several subsystems used for triggering, tracking and identifying the particle species for charged tracks, and calorimetry. These subsystems are surrounded by a room-temperature solenoid magnet 6.2 m long and 5.26 m

in diameter which generates a magnetic field of 0.5 T in the detector volume. The detector is cylindrical with full coverage in the azimuthal direction in the mid-rapidity region along with wide coverage in pseudorapidity. A schematic diagram of the STAR detector with its various detector subsystems is shown in Fig. 2.5. As the only experiment running continuously since the inception of RHIC, STAR has undergone major upgrades that have augmented its tracking capabilities and increased its acceptance in the forward rapidity regions. The main detectors used for this analysis are the *Time Projection Chamber* (TPC) and *Barrel Electromagnetic Calorimeters* (BEMC).

2.2.1 Detector Coordinate System

Before we get to the components of the STAR detector system, we will introduce the coordinate system used for the calculations involved. STAR is cylindrical around the beamline, so a convenient choice for z -axis is the beamline going east-west. $z = 0$ is set at the *Interaction Point* (IP), center of the detector volume, as shown in Fig. 2.6

Kinematics of ultra-relativistic particles are described by its four-momentum, (E, p_x, p_y, p_z) . However, these coordinates are not additive under Lorentz boosts in the z -direction, making calculations involving relativistic entities somewhat unwieldy. Since p_x, p_y are already invariant under Lorentz boosts along the z -axis, we can just convert them to the convenient cylindrical coordinates

$$p_T = \sqrt{p_x^2 + p_y^2}, \quad \varphi = \arctan\left(\frac{p_y}{p_x}\right) \quad (2.1)$$

Energy (E) can be replaced with the particle's mass (m), or transverse mass $m_T = \sqrt{p_T^2 + m^2}$. We can make the z -coordinate additive under Lorentz Boosts along z -axis by representing a particle's direction along the z -axis by *rapidity*

$$y = \frac{1}{2} \ln \frac{E + p_z}{E - p_z} \quad (2.2)$$

However since y is mass-dependent and most detectors don't provide robust particle

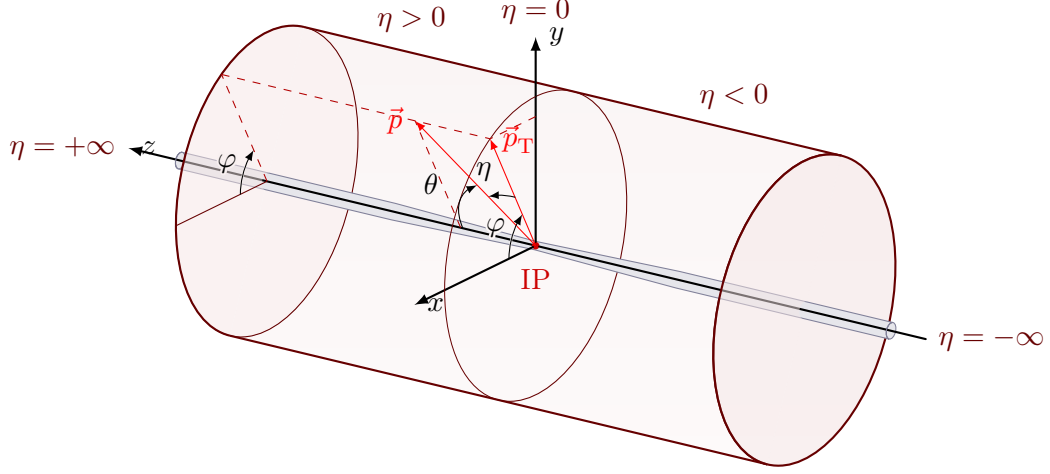


Figure 2.6: The (p_T, η, φ) coordinate system, drawn for a cylindrical collider detector. The beam axis defines z and the interaction point (IP) sits at the origin. A particle of momentum $\vec{p}$ is described by its transverse momentum $\vec{p}_T$, its azimuthal angle φ in the transverse plane, and its pseudo-rapidity η , fixed by the polar angle θ with respect to the beam.

identification at all kinematic ranges. We usually define the z -direction in a detector using *Pseudorapidity*, which is the massless limit of rapidity

$$\eta = \lim_{m \rightarrow 0} y = \frac{1}{2} \ln \frac{p + p_z}{p - p_z} = -\ln \left(\tan \frac{\theta}{2} \right) \quad (2.3)$$

Where θ is the polar angle between the particle's direction and the z -axis as shown in Fig. 2.6. Thus a particle's four-momentum is given by $(m_T, p_T, \eta, \varphi)$ or (p_T, η, φ, m) . See Appendix A.2 for a more detailed derivation of coordinates.

2.2.2 TPC

Time Projection Chamber (TPC) [6] was the the primary tracking detector for STAR, providing particle identification and momentum information for charged tracks by measuring their ionization energy loss (dE/dx) inside the TPC volume. As shown in Fig. 2.7 the TPC was a cylindrical detector 420 cm long with an outer (inner) radii of 200 (50) cm, concentric to the 0.5 T solenoid. Its acceptance covered a pseudorapidity range of $|\eta| < 1.0$, along with full azimuthal coverage. Particle momenta could be measured accurately over a range of 0.1

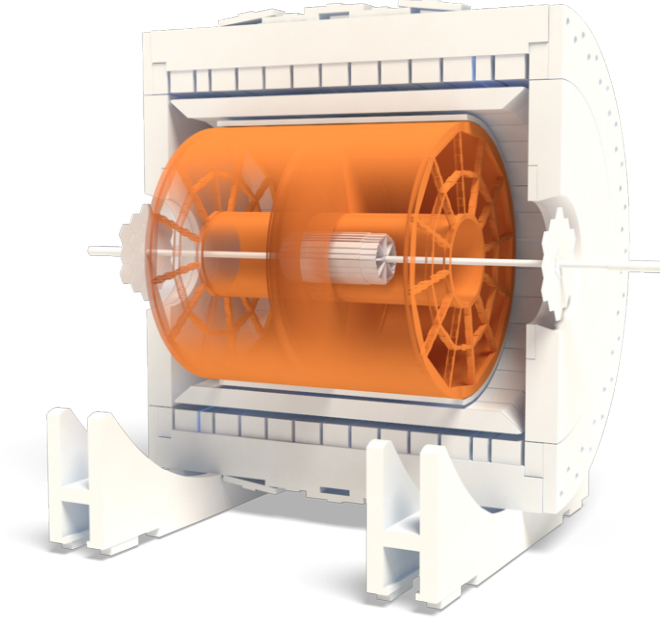


Figure 2.7: The Time Projection Chamber at STAR. [5].

to 30 GeV/ c , and particle identification was limited between 0.1 and 1 GeV/ c . The TPC volume contained an ionizable gas mixture called P10 (10 % methane, 90 % argon), chosen for its high drift velocity of 5.45 cm/ μ s in the well-defined, uniform electric field of ≈ 135 V/cm at STAR. The uniform electric field is defined between the thin conductive central membrane at $z = 0$ maintained at 28 kV acting as a cathode, and the two end caps each 210 cm away from the center at 0 V acting as anodes. The drift velocity can change over time based on atmospheric pressure fluctuations, therefore, calibrations are typically done using lasers every few hours during data-taking. To ensure minimal contamination by impurities like oxygen and water vapor maintained at 2 mbar above atmospheric pressure.

The charged particles leave ionization trails were captured by Multi-Wire Proportional Counters (MWPCs) with a segmented pad plane used in TPC readout. As particles ionize the TPC gas, freed electrons drift toward the MWPCs under the effect of the electric field. Signal contamination was prevented using a gated grid that initially restricted electron flow, opening only if a collision is deemed significant, allowing electrons to enter the MWPCs. Positive ions generated inside the MWPC are prevented from entering the TPC's drift

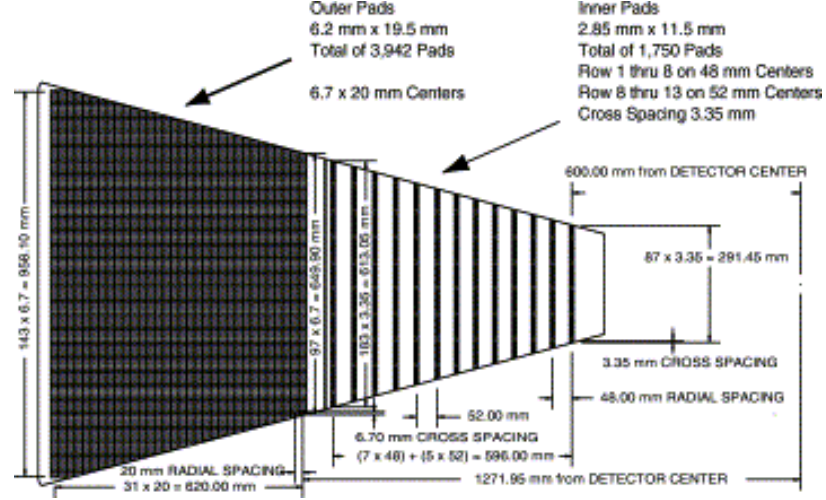


Figure 2.8: One of the twelve sectors of the TPC. [6].

volume using shield wires. The electrons passed the shield wires and were rapidly accelerated near the anode wires, creating avalanches through secondary ionization and amplifying the signal by a factor of 1,000 to 3,000. These avalanches induce signals in the surrounding pads without direct contact with the electrons. The MWPC pads detect the x-y position based on pad geometry — 3,942 pads across 32 rows in the outer sector and 1,750 pads in 13 rows in the inner sector. One such anode sector is shown in Fig. 2.8. The z information is then derived using the drift time and calibrated drift velocity of the electrons inside the detector volume. The typical resolution for position measurements from TPC is roughly 1 cm.

For a charged particle track, maximum of 45 hits could be recorded, which were fitted to a helix to calculate the particle momentum. Particles with momentum below $100 \text{ MeV}/c$ curve too tightly in the magnetic field and do not reach the TPC, while those above $30 \text{ GeV}/c$ follow nearly straight trajectories that compromise the measurement precision. For particles within these bounds, the relative momentum resolution is about 2 %. The direction of curvature gives the charge of the track. Energy loss as a function of distance travelled, dE/dx for a track was extracted using energy loss measured by a series of pads in the TPC. Figure 2.9 shows a typical plot of measured dE/dx as a function of momentum. For a given particle species with known momenta, the dE/dx curves are given by the Bichsel

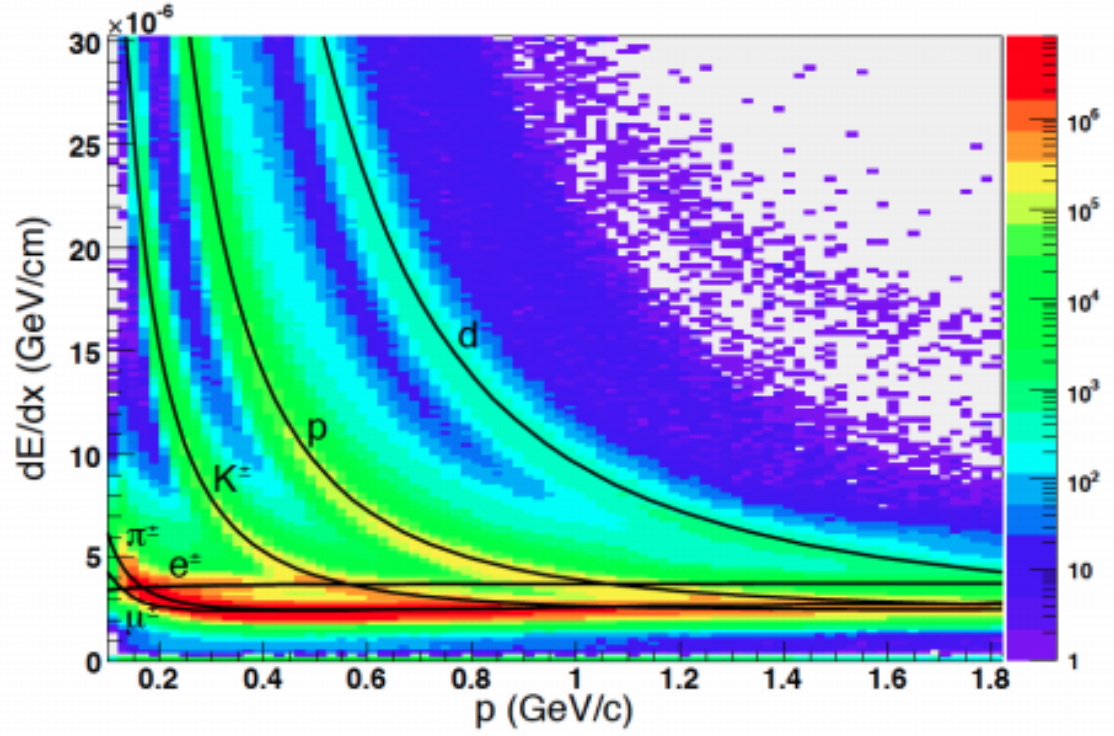


Figure 2.9: The ionisation energy loss for charged particles from TPC as a function of momenta. The letters represent different particles: $\mu^\pm$ for muons, $e^\pm$ for electrons, $\pi^\pm$ for pions, $K^\pm$ for Kaons, p for protons, and d for deuterons. The solid lines are the expected value of dE/dx for the corresponding particle species from the Bichsel function [7].

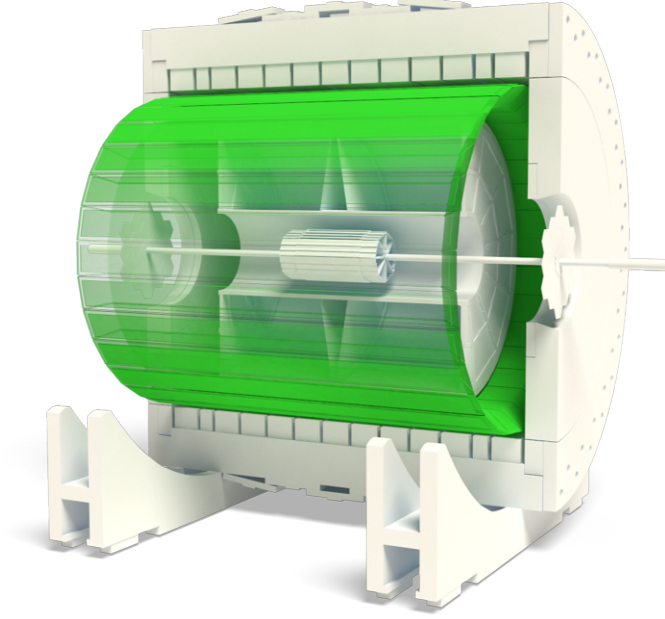


Figure 2.10: The Barrel Electromagnetic Calorimeter at STAR. Taken from [5].

function [7] which are shown as solid lines in Fig. 2.9. For each particle species, the band has non-zero thickness due to the finite momentum resolution, and selection criteria based on simulations are used to assign particle ID. The particle identification from TPC is limited to low momenta tracks (about 1 GeV/c) only beyond which the bands for different species start to overlap.

2.2.3 BEMC

The STAR Barrel Electromagnetic Calorimeter (BEMC) [24] designed to measure particle energies, particularly for electron and neutral particle identification. BEMC was a fast detector, used as a trigger to select events containing a hard process. It also provided the neutral component of the jet energy that couldn't be reconstructed by the TPC in jet-related measurements. As shown in Fig. 2.10, it was cylindrical detector located between the TPC and magnet coils. It covers the pseudorapidity range $|\eta| < 1$ and the full azimuth range ($0 \leq \phi \leq 2\pi$), matching the TPC's coverage.

The BEMC had inner and outer radii of 223.5 cm and 263 cm respectively and consisted

of 120 modules, each containing 40 towers, totaling 4800 towers. Each of the towers covers a range of $\Delta\eta \times \Delta\phi = 0.05 \times 0.05$, approximately $10 \text{ cm} \times 10 \text{ cm}$ near $\eta = 0$, growing slightly with increasing $|\eta|$ due to their projective geometry toward the center of the detector ($z = 0$). Each tower was built with 20 alternating layers of 5-mm thick lead, acting as absorbers to initiate electromagnetic showers, and 21 layers of plastic scintillators, which detect the photons produced by these showers. Together, the layers created a depth of around 20 radiation lengths (X_0), allowing the calorimeter to capture nearly all of the energy from electromagnetic showers. This depth is enough to contain 95 % of the energy from electrons and photons above 1 GeV. Within these layers, the shower's energy was most concentrated around 5-7 radiation lengths in, where the Shower Maximum Detector (SMD), a gas (90 % Argon, 10 % CO_2)-filled proportional counter improved spatial resolution to help differentiate between electromagnetic and hadronic shower profiles. The BEMC has an energy resolution that improves with particle energy, around 17% for 1.5 GeV electrons and 10% for 3 GeV electrons in head-on Au + Au collisions. This follows a typical calorimetric trend where energy resolution decreases as energy increases ($\sigma_E/E \approx 1.5\% + 15\%/\sqrt{E}$), providing complementarity with TPC where the resolution increases with increasing momentum ($\sigma_{p_T}/p_T \propto p_T$).

Chapter 3

Relativistic collisions of nuclei

Outgoing partons from hard processes undergo many successive splitting and fragmentation processes, becoming a collimated “spray” of hadrons called *jets* which are detected as dense clustering of energy in calorimeters. In principle, a *jet finding algorithm* should be able to faithfully get the kinematics of the initiating parton given the theory discussed in sec.1.4, which show partons cleanly branching into partons and eventually fragmenting into final-state hadrons that can be traced back to the shower partons. However reality is rarely this straightforward. First, the fragmentation functions that connect the partons and final state hadrons cannot be calculated in experiments. Second, it is not trivial to establish connection between final state particles and initial state partons even at the parton level, before we have the complications from hadronization and detector level reconstruction. For example, in a clean reaction like $e^+e^- \rightarrow q\bar{q}$. If one of the quark emits a gluon, then depending on the emission angle of the gluon, we might cluster 2 or 3 jets in the final state. We will further define jet finding in sec. 3.1

As discussed in sec. 1.2.2, collisions of heavy-ions at near-light speeds can create the conditions for forming QGP. Sec. 2.1 discussed RHIC and sec. 2.2 describe the STAR detector system. Before moving on to analyses of heavy-ion collisions, we need to establish a few more concepts. Nuclei like $^{197}_{79}\text{Au}$, $^{96}_{40}\text{Zr}$, $^{96}_{44}\text{Ru}$, etc. are complex arrangements of protons and neutrons that have a multitude of non-trivial initial configurations leading up to their collisions which have significant effects on the final state particle distributions. Therefore, any measurement done using heavy-ion events need to report these initial configurations of the colliding nuclei. We discuss these in sec. 3.2.

3.1 Finding Jets

To address the prevalent ambiguity of jet definitions, the Snowmass Accord [25] stated that, experimental and theoretical measurements had to use the same jet finding algorithm on final state hadrons in order to be comparable. Thus clustered jets are not-quite the proxies for partons in the same way as parton showers, but regions in the final state phase space that are most likely to have contributions from parton showers. Even after we define a jet, another ambiguity comes from the decision of what constituents in the event must be used as inputs into the jet finding algorithm. Especially, in heavy-ion collisions jet-medium interactions make that choice non-trivial. For example, is a soft particle inside the jet cone coming from soft gluon radiation from the jet or the medium? Keeping these considerations in mind, a “good” jet finding algorithm is:

- *Collinear safe*, meaning it would find the same jet with the same $p_{T,\text{jet}}$ regardless of fragmentation details
- *Infrared safe*, meaning it would find the same jets, even in the presence of a large number of soft partons

The most commonly used family of infrared and collinear safe jet algorithms, sequentially recombines constituents in an event according to the following prescription:

1. For every pair of particles $\{i, j\}$ with momenta $(\mathbf{p}_i, \mathbf{p}_j) \equiv \{(p_{T,i}, \eta_i, \varphi_i), (p_{T,j}, \eta_j, \varphi_j)\}$

$$d_{ij} = \min(p_{T,i}^{2p}, p_{T,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \text{ and } d_i = p_{T,i}^{2p} \quad (3.1)$$

where, $\Delta R_{ij}^2 = (\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2$ is the (η, ϕ) distance between particle i and j .

R is the resolution parameter of the algorithm, which decides the angular extent of clustered jets.

2. $i_{\min} = \min(\{d_{ij}\} \cup \{d_i\})$ is found. $d_{\min} \in \{d_{ij}\}$, the particles are combined into one

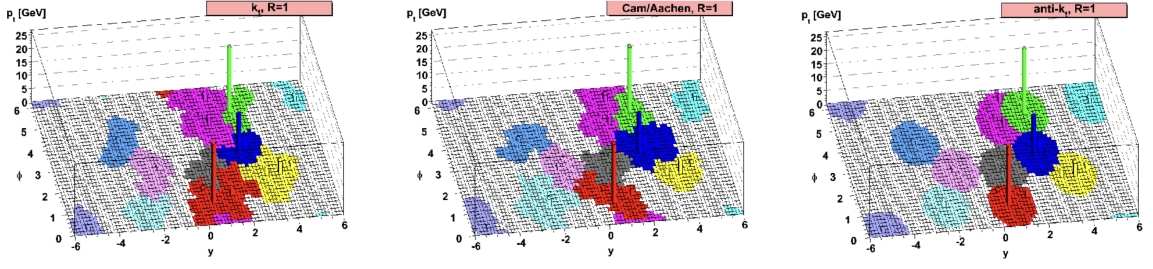


Figure 3.1: Jets reconstructed using the k_T (left), Cambridge/Aachen (middle), and anti- k_T (right) algorithms. Taken from [8]

jet candidate, adding their energies and momenta according to some *recombination scheme*, and the first step is repeated.

3. $d_{\min} \in \{d_i\}$, this is recorded as a final state jet candidate. It is removed from the list and the first step is repeated.

These steps are iterated until no particles remain. This algorithm becomes the k_T *algorithm* [26] for $p = 1$, *Cambridge-Aachen* [27] for $p = 0$ and the most widely used *anti- k_T algorithm* [8] for $p = -1$. The difference in these algorithms are shown in fig. 3.1. The *FastJet* library [28] efficiently implements these algorithms and a few more, and is the de-facto standard in the High Energy Physics (HEP) community. It also provides various recombination schemes.

- *E-scheme*, just adds the four momenta of the jet candidates. it is the default in FastJet and the most commonly used.
- *Boost-invariant* $p_t/p_t^2/E_t/E_t^2$ *schemes* involve a preprocessing stage rescaling the energy to be equal to the 3-momentum for the p_t and p_t^2 schemes or the 3-momentum to

be equal to the energy in the E_t and E_t^2 schemes. Then for all schemes the recombination p_r of p_i and p_j is a massless 4-vector satisfying

$$p_{t,r} = p_{t,i} + p_{t,j} , \quad (3.2a)$$

$$\varphi_r = \frac{w_i \varphi_i + w_j \varphi_j}{w_i + w_j} , \quad (3.2b)$$

$$y_r = \frac{w_i y_i + w_j y_j}{w_i + w_j} , \quad (3.2c)$$

where w_i is $p_{t,i}$ for the p_t and E_t schemes, and is $p_{t,i}^2$ for the p_t^2 and E_t^2 schemes.

- *Winner-takes-all (WTA) scheme* uses the four momentum of the higher momentum jet candidate

3.1.1 SoftDrop

The soft drop grooming [29] procedure can be summarized as follows. First, the observed jet is reclustered with the Cambridge/Aachen (C/A) algorithm. Second, the soft drop criterion from Eq. 3.3 is checked at each clustering step when going backward through the C/A clustering tree.

$$\frac{\min(p_{T,1}, p_{T,2})}{p_{T,1} + p_{T,2}} > z_{\text{cut}} \left(\frac{\Delta R_{12}}{R} \right)^\beta \quad (3.3)$$

Here $p_{T,\{1,2\}}$ are the momenta of the two branches that had been merged together and ΔR_{12} is their geometric distance in the (η, ϕ) plane. When the softer branch fails the criterion it is removed from the jet and the procedure continues until it is satisfied. The remaining particles in the jet constitute the soft drop groomed jet. A convenient choice can be made for the soft threshold parameter z_{cut} and the angular exponent β . For $\beta = 0$ the soft drop procedure reduces to the modified mass drop tagger (mMDT). Using SoftDrop, we can decompose angularities into pQCD and npQCD components. pQCD component is the angularity calculated from the jet after grooming has been performed and the npQCD component is calculated as the difference between the inclusive/ungroomed and groomed jet

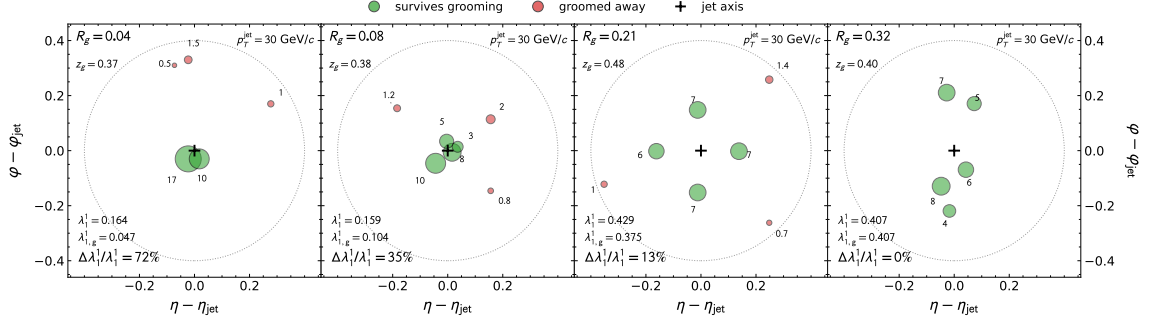


Figure 3.2: Ways of arranging constituents (each represented as a circle whose radius represent $\frac{p_{T,\text{const}}}{p_{T,\text{jet}}}$) in the (η, ϕ) space within a jet with $p_{T,\text{jet}} = 30 \text{ GeV}/c$ such that R_g of the jet increases going from left to right. Green (Red) colored constituents survive (are removed by) the SoftDrop grooming ($z_{\text{cut}} = 0.2$, $\beta = 0$).

angularities.

$$\begin{aligned}
 \mathcal{O}_\beta^\kappa &= \mathcal{O}_{\beta,\text{pQCD}}^\kappa + \mathcal{O}_{\beta,\text{npQCD}}^\kappa \\
 \mathcal{O}_{\beta,\text{pQCD}}^\kappa &\equiv \mathcal{O}_{\beta,\text{g}}^\kappa \\
 \mathcal{O}_{\beta,\text{npQCD}}^\kappa &\equiv \Delta \mathcal{O}_\beta^\kappa = \mathcal{O}_\beta^\kappa - \mathcal{O}_{\beta,\text{g}}^\kappa
 \end{aligned} \tag{3.4}$$

Using the momentum balance (z_g) and angularity of the first collinear-splitting (R_g), we get direct connection between measurement of jet shapes and calculations from field theory. Larger R_g implies that the most prominent split in the jet occurred earlier in its evolution, opening up more phase space for pQCD splittings (which survive SoftDrop grooming) because of color coherence discussed earlier. Fig. 3.1.1 shows percentage npQCD contribution to jet width $\left(\frac{\Delta \lambda_1^1}{\lambda_1^1} \times 100\right)$ decreasing with increasing R_g , going from highly collimated, pencil-like jets to jets with proper two pronged structure.

3.2 Heavy-Ion Collisions

Estimates of the energy density indicate that heavy ion collisions at RHIC and LHC reach energy densities ranging between 1-12 GeV/fm^3 [30, 31], well in excess of the 0.2-1 GeV/fm^3

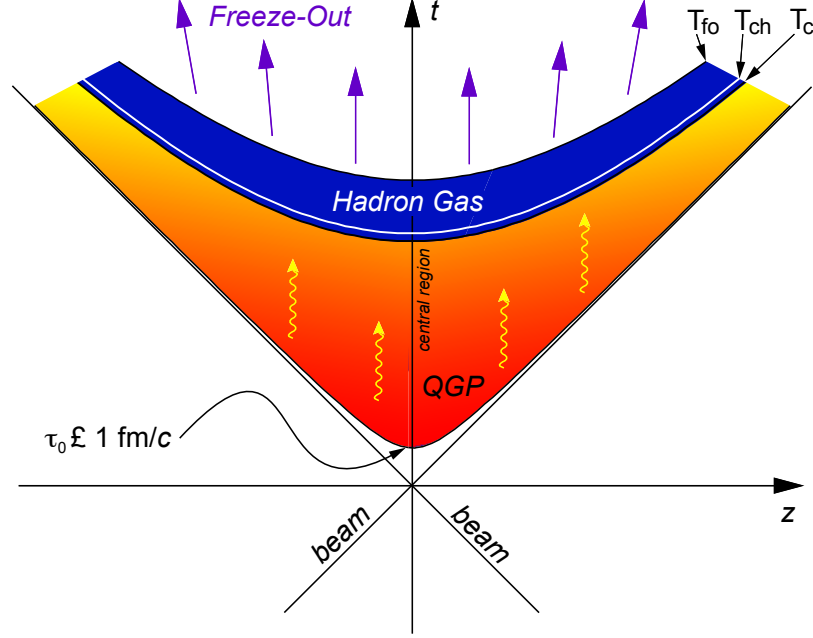


Figure 3.3: A light cone diagram showing the stages of a heavy ion collision. The abbreviation T_{fo} is for the thermal freeze-out temperature, T_{ch} is for the chemical freeze-out temperature, and T_c is for the critical temperature where the phase transition between a hadron gas and a QGP occurs. τ_0 is the formation time of the QGP. Taken from [9]

required for QGP formation [32, 33].

3.2.1 Space-time picture of QGP

The heavy ion collision and the evolution of the fireball, as depicted in Figure 3.3, has several stages, and the measurement of the final state particles can be affected by one or all of these stages depending on their production mechanism and interaction time within the medium. Immediately after the collision, the system is in a non-thermalized, pre-equilibrium state where, partons undergo hard scattering, dominated by $2 \rightarrow 2$ QCD processes with the outgoing partons back-to-back in azimuth ($\Delta\varphi \approx \pi$). At $\tau \approx 0.5 - 1$ fm/c, the system thermalizes into the QGP state which expands following hydrodynamic laws. During the expansion, the system cools down until it reaches the temperature ($T_c \approx 156$ MeV from lattice QCD studies) at which a crossover phase transition to a hadronic gas state occurs. By $\tau \approx 10$ fm/c after the collision, confinement is restored and colored partons hadronize

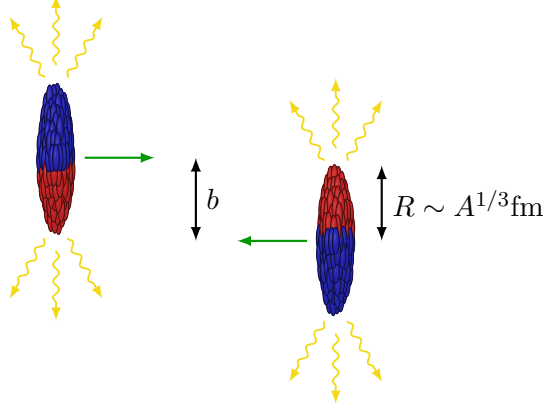


Figure 3.4: Typical heavy-ion collision with an impact parameter of b . The participating nuclei are colored red and the spectators are colored blue.

into colourless hadrons. When the chemical freeze-out temperature ($T_{ch} \approx 165$ MeV) is reached, the particle abundances of the system are frozen. Elastic interactions can still occur till the kinetic freeze-out ($T_{fo} \approx 100$ MeV), where the particle momentum spectra are established.

3.2.2 Initial State of Colliding Nuclei

The initial state of the incoming nuclei is not precisely known, but its properties impact the production of final state particles. The incoming nuclei are often modeled as either an independent collection of nucleons called a Glauber initial state [10], or a wall of coherent gluons called a Color Glass Condensate [34]. The widely used model for determining these quantities is the Glauber Monte Carlo (Glauber MC) model [35], which assumes Wood–Saxon density distribution of nucleons. The nuclei are then assigned a random impact parameter and translated to collide. If the distance between centers of nucleons is less than $\sqrt{\sigma_{\text{inel}}^{\text{NN}}/\pi}$, where $\sigma_{\text{inel}}^{\text{NN}}$ is the inelastic nucleon–nucleon cross section, the nucleons have undergone a binary collision. Nucleons that have undergone at least one binary collision are said to have participated in the reaction. This procedure is followed to obtain the number of participant nucleons (N_{part}) and the number of binary collisions (N_{coll}) event-by-event.

A schematic diagram of the geometry of heavy-ion collisions is shown in Fig. 3.4.

3.2.3 Centrality Determination

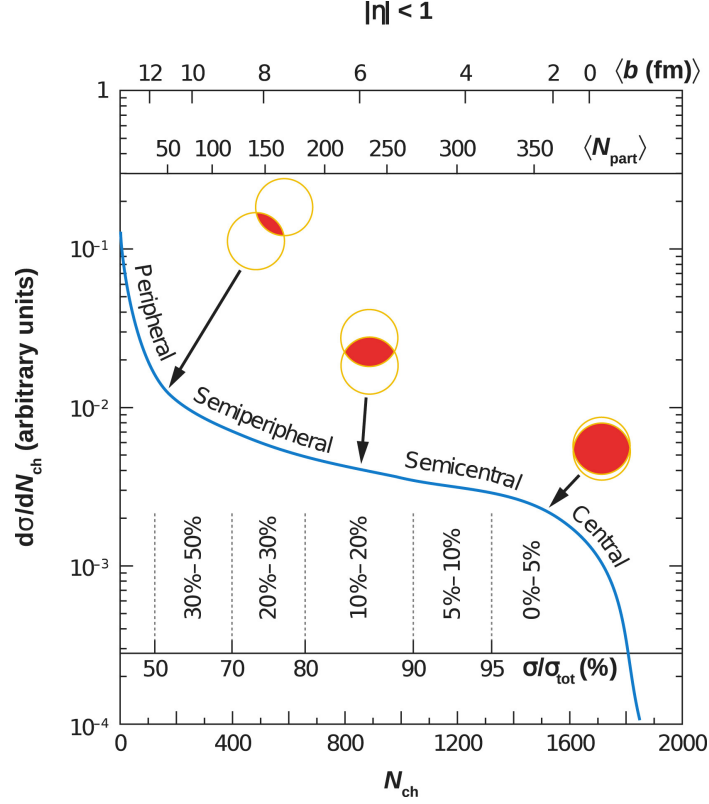


Figure 3.5: An illustration of the total inclusive charged particles multiplicity N_{ch} and number of participant (N_{part}) and impact parameter (b) in heavy-ion collisions. Taken from [10]

As the impact parameter (b) of collisions, the distance between the geometrical centers of the colliding nuclei, is not measurable in experiments, a proxy called *centrality* is defined. Centrality is characterized by the charged particle multiplicity of events, connected to the impact parameter through a simulation defining it as a function of N_{part} , N_{coll} and impact parameter b from Glauber model is mapped with the experimental data through a particle production model called the Two-Component Model [36]. Assuming that particle production is governed by contribution from hard component ($\propto N_{\text{coll}}$) and soft component (proportional to $\propto N_{\text{part}}$), it simulates the multiplicity distribution according to

$$\frac{dN_{\text{ch}}}{d\eta} = n_{pp} \left[x N_{\text{coll}} + (1 - x) \frac{N_{\text{part}}}{2} \right] \quad (3.5)$$

where the quantity n_{pp} is the multiplicity from a $p + p$ collision of the same center-of-mass energy. x is the contribution of hard component. n_{pp} is drawn from a negative binomial distribution with parameters NBD $(n_{pp}; \langle n_{pp} \rangle, k)$, where $\langle n_{pp} \rangle$ is the average multiplicity in the $p + p$ collision and k is the width of the distribution. The simulated charged multiplicity distribution is then fitted with the corresponding multiplicity distribution from the experiment. $\langle n_{pp} \rangle$, k , and x are kept as free parameters in the fitting to experimentally measured charged particle multiplicity distribution. Initial guess values for x could be set close to those determined in different experiments for faster convergence of fitting [37, 38].

An event belongs to a centrality class, say 0%–5% if its charged particle multiplicity falls in the top 5th percentile. Fig. 3.2.3 illustrates collision centrality definition in heavy-ion collisions by comparing the charge particle multiplicities with Glauber MC + Two-component model simulation. Centrality classes are classified for the simulated distribution. $\langle N_{\text{part}} \rangle$, $\langle N_{\text{coll}} \rangle$ in those centrality classes are calculated.

3.2.4 Anisotropic flow

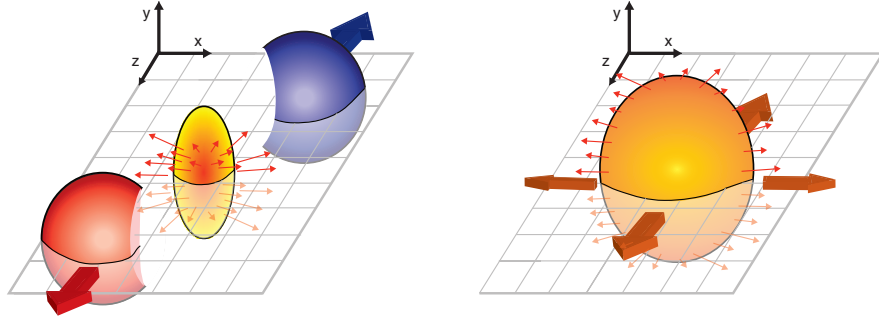


Figure 3.6: Schematic diagrams showing the initial overlap region (left) and the spatial anisotropy generated by this anisotropic overlap region. This anisotropy can be quantified using the Fourier coefficients of the momentum anisotropy. Taken from [9]

Contrary to initial naive expectations of a gas-like QGP, the QGP formed in these collisions was shown to behave like a liquid of quarks and gluons [39]. Thus, mean free path of particles in QGP fireball is much smaller than the fireball's system size. Hence, in non-central heavy ion collisions the initial state coordinate space asymmetry generates

a pressure gradient in the direction of the impact parameter b , schematically shown in fig. 3.2.4. This spatial asymmetry translates into a azimuthal momentum anisotropy as the system expands, meaning that particles are more likely to be emitted along the plane where the pressure gradients are the highest. This plane is defined by the beam axis and the impact parameter of the collision, and called the *reaction plane* (Ψ_{RP}). The anisotropy can be characterized by the Fourier decomposition of the azimuthal particle distribution with respect to Ψ_{RP} [40, 41]:

$$\frac{dN}{d(\phi - \Psi_{\text{RP}})} = \frac{N_0}{2\pi} \left(1 + 2 \sum_{n=1}^{\infty} v_n \cos[n(\phi - \Psi_{\text{RP}})] \right), \quad (3.6)$$

The *anisotropic flow* coefficients v_n , the n^{th} order term in the Fourier decomposition of the final state particle azimuthal distribution is given by,

$$v_n = \langle \cos([n(\phi - \Psi_{\text{RP}})]) \rangle \quad (3.7)$$

Due to the symmetry of the collision geometry, the most prominent contribution is from the second order coefficient v_2 , called *elliptic flow*.

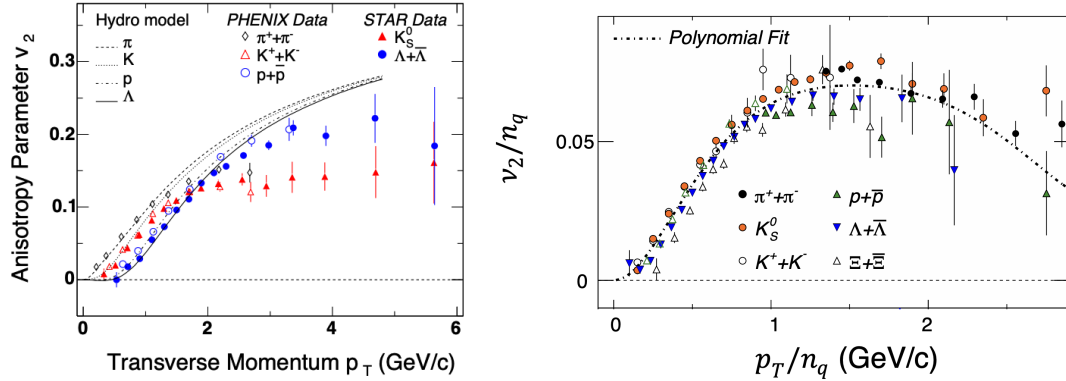


Figure 3.7: **Left panel:** Elliptic flow v_2 for several particle species in Au + Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV from STAR and PHENIX experiments. Predictions from hydrodynamic models are shown as lines. Taken from [11]. **Right panel:** Number of constituent quarks (n_q) scaled v_2 as a function of p_T/n_q . Taken from [12].

v_2 was measured by the STAR and PHENIX for several particle species and is shown in

Fig. 3.7. Large positive v_2 values indicate collective phenomena associated with a strongly coupled fluid, or a *signature of QGP*. A scaling of the v_2 values with the number of constituent quarks (2 for mesons, 3 for baryons) yielded a universal curve as a function of p_T/n_q suggesting the flow dynamics were developed in a medium with quark and gluon degrees of freedom, i.e. QGP.

Event plane reconstruction

The reaction plane, Ψ_{RP} cannot be directly measured and is approximated by the second-order symmetry plane, $\Psi_{2,EP}$, referred to as the *event plane* in this text, for simplicity. $\Psi_{\text{RP}} \rightarrow \Psi_{2,EP}$ if nucleon distributions were in their average positions and devoid of fluctuations from interactions among nucleons [42]. By measuring the charged particle azimuthal distribution, the n-th order event plane can be extracted by [43],

$$\Psi_{n,EP} = \frac{1}{n} \arctan \left(\frac{Q_{y,n}}{Q_{x,n}} \right), \quad (3.8)$$

the weighted vector $\mathbf{Q}_n = Q_{x,n}\hat{\mathbf{x}} + Q_{y,n}\hat{\mathbf{y}}$ are given by,

$$\mathbf{Q}_n = \hat{\mathbf{x}} \sum_{\text{tracks}} p_{T,\text{track}} \cos(n\phi_{\text{track}}) + \hat{\mathbf{y}} \sum_{\text{tracks}} p_{T,\text{track}} \sin(n\phi_{\text{track}}). \quad (3.9)$$

where the sum is calculated for all reconstructed charged particles (tracks) in the event, ϕ_{track} is the track's azimuthal angle, and $p_{T,\text{track}}$ the track's transverse momentum.

Due to finite acceptance and multiplicities, the calculated event plane has an underlying anisotropy that is corrected by applying two separate correction methods. First, a calibration and recentering correction procedures are applied to remove bias introduced by non-uniform acceptance of the TPC tracking system and further account for potential beam-condition effects [43–45]. This procedure involves recentering the weighted $\mathbf{Q}_2$ -vectors such that, $\langle \mathbf{Q}_2 \rangle = \mathbf{0}$. It is done by calculating a modified $\mathbf{Q}_2$, obtained by subtracting an event averaged $\langle \mathbf{Q}_2 \rangle$ from each event's nominal $\mathbf{Q}_2$.

The higher harmonics of $\Psi_{n,EP}$ still persist after recentering [43], and are removed through a second correction step, referred to as the shifting method, applied event-by-event to make the event-plane angle isotropic in the lab frame [46]. This method defines a new angle:

$$\Psi'_{2,EP} = \Psi_{2,EP} + \sum_n \frac{2}{n} (-\langle \sin(n\Psi_{2,EP}) \rangle \cos(n\Psi_{2,EP}) + \langle \cos(n\Psi_{2,EP}) \sin(n\Psi_{2,EP}) \rangle), \quad (3.10)$$

Usually 20 orders of Fourier moment are removed. Additional details of the recentering and shifting corrections can be found in Refs. [43, 44, 47]

The difference between Ψ_{RP} and $\Psi_{2,EP}$ is quantified by event-plane resolution, R_n or $R_n\{\Psi_{2,EP}\}$ given by Eqn. 3.11. The observed v_n , v_n^{obs} is corrected for this limited resolution by doing, $v_n = v_n^{\text{obs}}/R_n$ [43, 48].

$$R_n = R_n\{\Psi_{2,EP}\} = \langle \cos(n[\Psi_{RP} - \Psi_{2,EP}]) \rangle \quad n \text{ is even} \quad (3.11)$$

Furthermore, individual events are divided into two random sub-events by assigning charged tracks to sub-events, "a" and "b". The sub-events are unique, with approximately equal multiplicities. We can write the correlation of two event planes by taking the product of two sub-events, [43, 49]

$$\langle \cos(n[\Psi_{2,EP}^a - \Psi_{2,EP}^b]) \rangle = \langle \cos(n[\Psi_{2,EP}^a - \Psi_{RP}]) \rangle \langle \cos(n[\Psi_{2,EP}^b - \Psi_{RP}]) \rangle, \quad (3.12)$$

This allows calculation of the event-plane resolution directly from data. Since a and b have equal multiplicities, the resolution of each sub-event can be calculated from the correlation between the two sub-events as [43]:

$$\langle \cos(n[\Psi_{2,EP}^a - \Psi_{RP}]) \rangle = \sqrt{\langle \cos(n[\Psi_{2,EP}^a - \Psi_{2,EP}^b]) \rangle}. \quad (3.13)$$

The event-plane resolution is multiplicity dependent and calculated for separate ranges

of collision centrality using the two sub-events method. Narrower bins are calculated and then combined accordingly to match the ranges required, by averaging the results from the narrow bins weighted by the multiplicity of each bin [46].

3.3 Jets in QGP

As discussed in sec. 3.2.1, hard scatterings of partons that produce jets happens before the system thermalizes into QGP (parton producing a 10 GeV jet scatters at $\tau \sim 1/E \sim 0.2\text{fm}/c$). Thus the ensuing parton shower traverses the QGP, and resulting jet is *modified/quenched* compared to an appropriate *baseline*. The two main mechanisms for jet quenching are *collisional* and *radiative* energy losses. Collisional energy loss happens due to the elastic scattering between the parton and the medium constituents and usually dominates for lower energy partons. Radiative energy loss happens due to induced gluon radiation (*bremssstrahlung*) through inelastic scatterings with the medium particles [50–52]. For partons with $p_{T,\text{parton}} \geq 5 \text{ GeV}/c$, this is usually the dominant source of energy loss [53].

The baseline measurement is done in a collision system where either QGP is not formed or has negligible lifetime compared to central heavy-ion collisions. Two possible choices arise for the baseline, first is $p+p$ collisions from which we can calculate the *Nuclear Modification Factor* (R_{AA}) for distributions of any observable $\mathbf{O}$ calculated in $p+p$ and $A+A$ collisions, given by

$$R_{AA}(\mathbf{O}, b) = \frac{dN_{AA}/d\mathbf{O}}{\langle T_{AA}(b) \rangle \cdot d^2\sigma_{pp}/d\mathbf{O}} \quad (3.14)$$

where $T_{AA}(b) = N_{\text{coll}}(b)/\sigma_{\text{NN}}^{\text{inel}}$ is the nuclear overlap function which determines the number of participating nucleons and number of binary collisions N_{bin} for a heavy ion collision as a function of its impact parameter (b) [54]. The second common baseline used in absence of a $p+p$ baseline is peripheral collisions of heavy-ion collisions. The *Relative Nuclear*

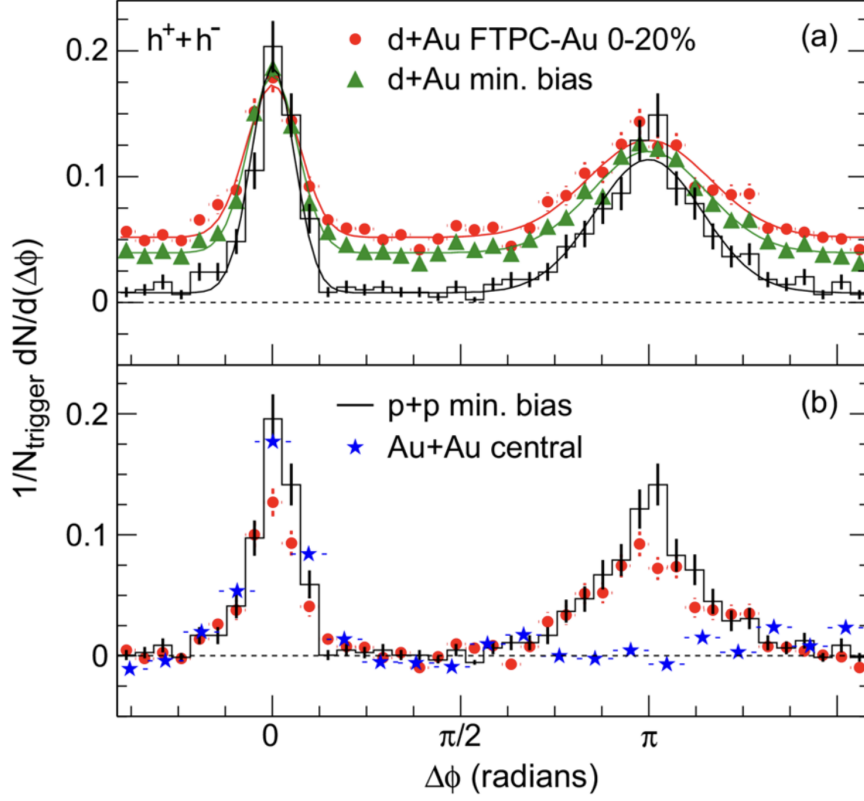


Figure 3.8: Measurement of the angular correlation between all charged hadrons and the leading hadron in the event for $p + p$ (black lines), $d + Au$ (red and green markers), and $Au + Au$ events (blue markers), performed by the STAR experiment. The vanishing of the away-side yield around $\Delta\phi \approx \pi$ in $Au + Au$ events is evidence of particles getting quenched in the QGP medium. Taken from [13].

Modification Factor R_{CP} is given by,

$$R_{CP}(\mathbf{O}, b) = \left. \frac{dN/d\mathbf{O}}{\langle N_{\text{bin}}(b) \rangle} \right|_{\text{cent}} \times \left. \frac{\langle N_{\text{bin}}(b) \rangle}{dN/d\mathbf{O}} \right|_{\text{peri}} \quad (3.15)$$

Before infrastructure became available to reliably reconstruct a sufficient number of jets in heavy-ion collisions, the effects of the medium on high energy partons were studying the *Di-hadron correlation*. Leading hadron (highest p_T in event) is used as a jet proxy and the azimuthal angular correlation ($\Delta\phi$) of all charged hadrons in the event is studied. Fig. 3.8 shows one of the first measurements performed by STAR [13], di-hadron correlations compared between $p + p$, $Au + Au$, and $d + Au$ collisions. The $p + p$ and $d + Au$ both

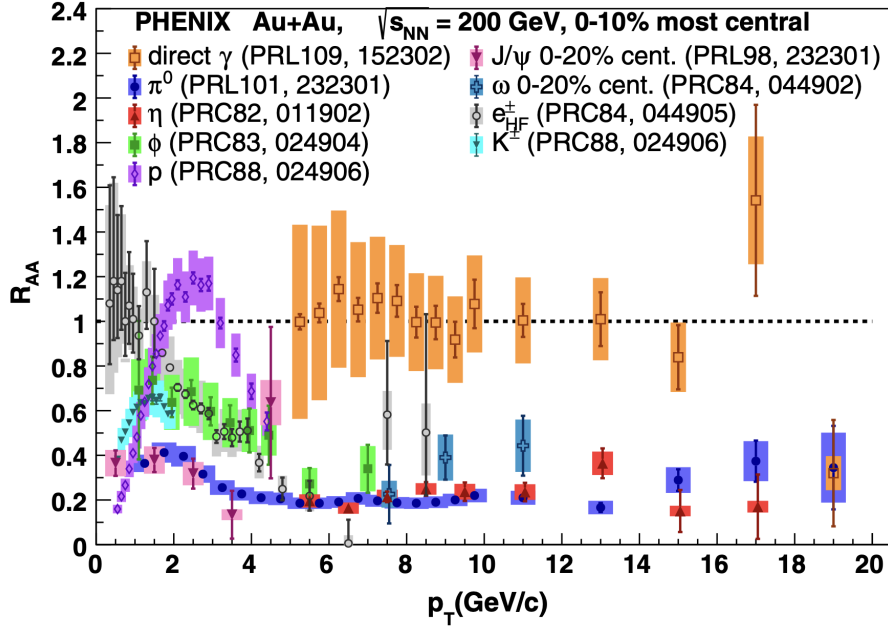


Figure 3.9: Nuclear modification factor R_{AA} as a function of p_T for identified particle species in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the PHENIX experiment. Taken from [14].

show peaks around $\Delta\varphi = 0$ and $\Delta\varphi = \pi$ as expected, given that hard scatterings are dominated by $2 \rightarrow 2$ Feynman diagrams. However, for Au + Au collisions, the peak around $\Delta\varphi = \pi$ completely vanishes. This was the expectation because depending on the distance from the center of the collision where the initial back-to-back partons were produced, one of the partons would travel a larger distance in the QGP medium and therefore lose more energy than the other one.

A closer look at jet quenching was given by the PHENIX experiment through a collection of R_{AA} measurements discussed in eq. 3.3 for identified particles spectra, shown in Fig. 3.9. Whereas, direct photons (γ) do not interact strongly, and hence have no yield modification in the medium. All strongly interacting particles show varying levels of suppression from 0.2 - 0.6, again confirming their energy loss in the presence of QGP.

Although measurements with high p_T hadrons help us understand jet quenching mechanisms in the medium, they suffer from survival bias, as these hadrons arise from a subsample of jets which usually undergo the least amount of quenching [55]. Fully reconstructed jets,

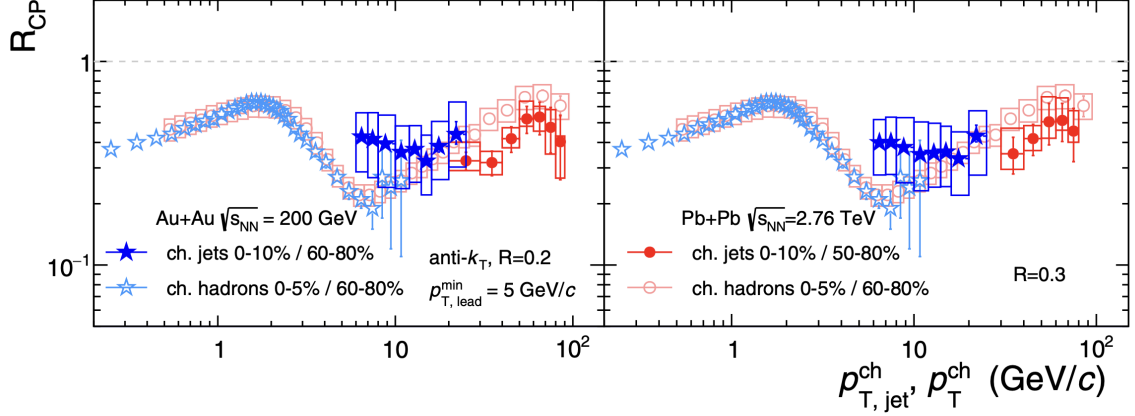


Figure 3.10: R_{CP} distributions for charged hadrons (open markers) and jets (solid markers) from RHIC (blue star markers) and LHC (red circle markers) experiments. Taken from [15] and references within [16–18].

however, are mostly unbiased probes of energy flow at high p_T , and offer a wealth of sub-structure observables which allow us to look at energy flow within the jet more differentially. This motivates the studies presented in chapters 5 and 6.

The STAR experiment measured the R_{CP} of the p_T spectra of charged particle jets (clustered charged tracks only and contain no calorimeter hits (towers) as input) from central and peripheral Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV shown in fig. 3.10 [15]. Measured for charged jets was compared to R_{CP} for charged hadrons from STAR at the same energy, and to the measured R_{CP} for charged hadrons and jets from the ALICE and ATLAS experiments at $\sqrt{s_{NN}} = 2.76$ TeV. First, the charged hadrons R_{CP} is found to be the same over the same range of p_T at RHIC and LHC energies (which are an order of magnitude apart). The charged jet R_{CP} is also found to have similar values between RHIC and LHC energies with roughly 60% of the energy lost outside the jet cone in central collisions with respect to peripheral collisions, with no significant p_T dependence in contrast to the p_T dependence observed for the charged hadrons R_{CP} . This may be because the inclusive hadron distribution usually arises from the leading hadron in the corresponding jet, and is therefore is sensitive to the fragmentation function. This fragmentation process can therefore introduce the p_T dependence of R_{CP} in hadrons.

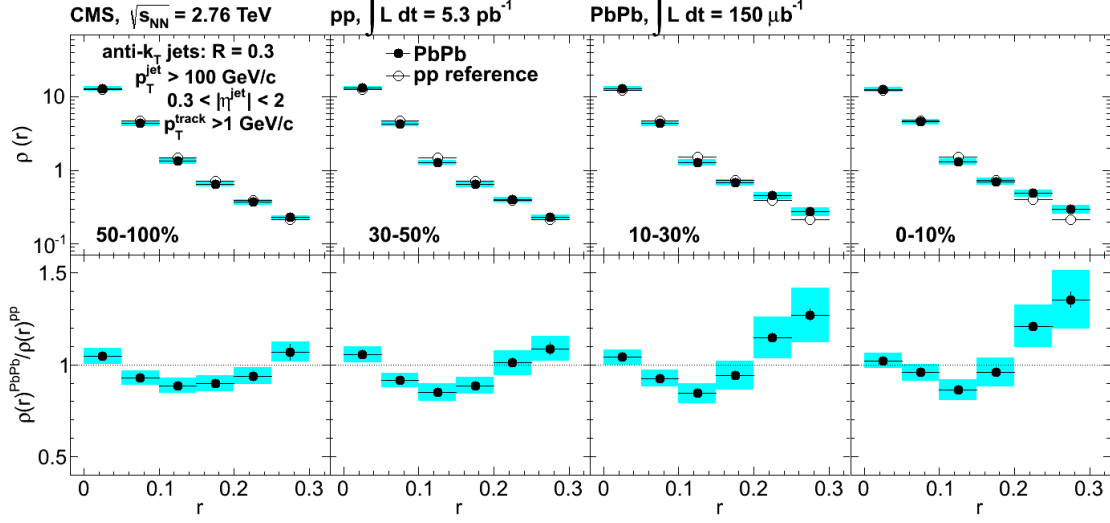


Figure 3.11: Differential jet shapes in Pb+Pb and $p+p$ collisions for four Pb+Pb centralities. Each spectrum is normalized so that its integral is unity. This shows that there are more particles in jets in central collisions and these modifications are primarily at large angles relative to the jet axis, as expected from partonic energy loss. Taken from [9]

Going a bit into the jet substructure, CMS measurement of jet's average radial momentum profile around the jet-axis, called *differential jet shape* defined as,

$$\rho(r) = \frac{1}{\delta r} \frac{1}{N_{jet}} \sum_{jets} \frac{\sum_{track \in [r_a, r_b]} p_T^{track}}{p_T^{jet}} \quad (3.16)$$

where the jet cone is divided rings of width δr which have an inner radius r_a and an outer radius r_b . From the results shown in fig. 3.11, the differential jet shapes in the most central Pb+Pb collisions are broadened in comparison to measurements done in $p+p$ collisions at the same center of mass energy [56]. These measurements demonstrate that there is an enhancement in the modification with increasing angle from the jet axis, indicating a broadening of the jet profile and a depletion near $r \approx 0.2$.

Chapter 4

Data analysis techniques

As a result of the quantum-mechanical nature of the collisions and the interaction of their products with the detectors, the observations resulting from a particular interaction are fundamentally probabilistic. Therefore, the approach to data analysis and the conclusions drawn from the data must be framed in statistical terms, which include not only low-level tasks, such as the reconstruction and identification of particles and the measurement of their energies and momenta, but also high-level tasks, such as searches for new particles and measurements. The high dimensionality and large volume of LHC data pose a problem because the statistical model over the high-dimensional space of their experimental data is not known explicitly in terms of an equation that can be evaluated. Instead, one typically has access to large samples of simulated data generated by stochastic simulation programs that model the physics of particle interactions on various scales. If the data were fairly low dimensional ($d < 5$), the problem of estimating the unknown statistical model from the simulated samples would not be difficult. Histograms or kernel-based density estimates provide reasonable estimates in low-dimensional spaces. These strategies, however, suffer from the curse of dimensionality. In a single dimension, N samples may be required to describe the source probability density function. In d dimensions, the number of samples required increases to the power of the data's dimensionality: $\mathcal{O}(N^d)$. The consequence is that any dimensionality greater than five or ten requires impractical or impossible computational resources, regardless of the speed of the sample generator.

4.1 Machine Learning Basics

Fortunately, many of the tasks encountered in HEP can be naturally reformulated as machine learning problems. Typically, the problems are formulated in terms of a search for some function $f : X \rightarrow Y$, from the space of the observed data X to a low-dimensional space of a desired target label Y , which optimizes some metric of our choosing. This metric is called a loss function and is written as $L[\mathbf{y}, f(\mathbf{x})]$.

Ideally, a learning algorithm would find the function that optimizes L over all possible values of $(\mathbf{x}, \mathbf{y})$, but this is intractable owing to the curse of dimensionality and an infinite number of functions to choose from. Instead, in supervised learning, one has labeled training data $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^N$ sampled from $p(\mathbf{x}, \mathbf{y})$. Furthermore, the function space is restricted to a model—a highly flexible family of functions $f_\phi(\mathbf{x})$ parameterized by ϕ . In this case, the algorithms minimize the loss function directly with respect to the model parameters ϕ . Neural networks, support vector machines, and decision trees are examples of types of models commonly used in machine learning. These models often have a large number of parameters, and in the case of neural networks, finding the optimal f_ϕ can be a difficult problem

An essential goal in machine learning is generalization—the ability of the model to perform well on data that were not used in training. Failure in this task is called overtraining or overfitting. A vast array of techniques exist to avoid overtraining, all of which can be considered forms of regularization. Regularization techniques such as dropout were key to advances in image recognition with deep learning

4.2 Jet Classification

For jets, classifiers can be used as a tagging mechanism. For almost every way of representing a jet, there is a class of machine learning models. We discuss two cases here

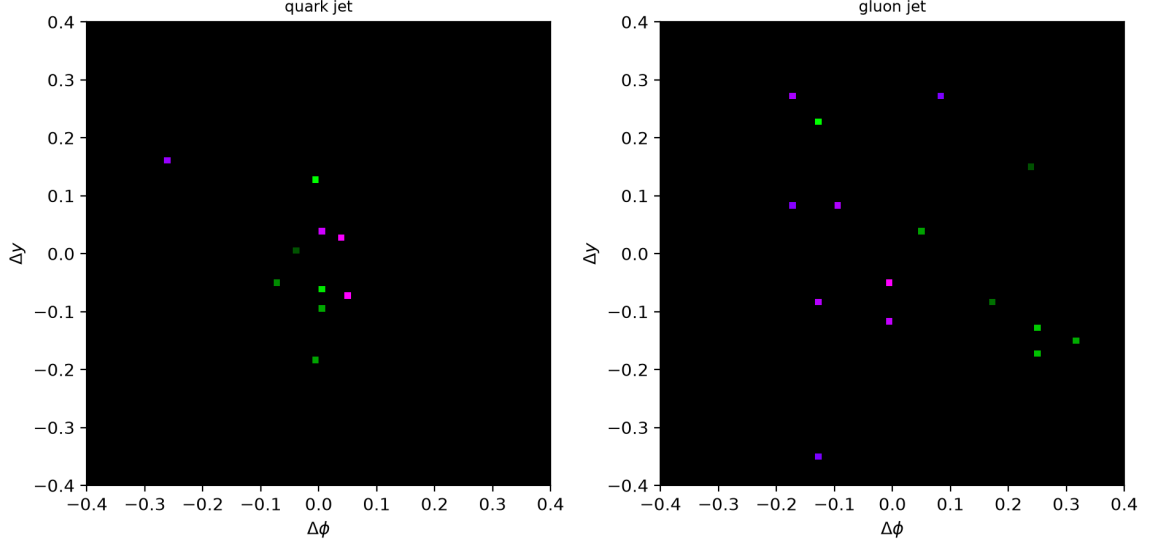


Figure 4.1: Typical quark (left) and gluon (right) jet. Gluon jets exhibit a broader spatial radiation pattern and higher particle multiplicity compared to quark jets

4.2.1 Quark-Gluon jet discrimination

Quark and gluon jet $p + p$ events are generated as $qq \rightarrow qq$ and $gg \rightarrow gg$ dijet events at $\sqrt{s_{\text{NN}}} = 200$ GeV using PYTHIA-8 (RHIC Tune). Each jet is represented as an 24×24 image following [19]. The RGB color channels of the image are made as,

- R: $\sum p_{\text{T}}$ of charged constituents,
- G: $\sum p_{\text{T}}$ of neutral constituents,
- B: number of charged constituents.

giving jets as shown in 4.2.1. Preprocessing involves centering the images around each jet's p_{T} -weighted centroid $(\bar{y}, \bar{\phi}) = \sum_i p_{\text{T},i}(y_i, \phi_i) / \sum_i p_{\text{T},i}$, followed by normalization. Data is augmented using procedures like random reflections ($\Delta y \rightarrow -\Delta y$ or $\Delta \phi \rightarrow -\Delta \phi$) and random jitter.

The images were input for a Convolutional Neural Network (CNN) implemented using PyTorch as shown in fig. 4.2. Each convolutional layer consisted of 64 filters, with filter sizes of 5×5 , 3×3 , and 3×3 , respectively (8×8 , 4×4 , and 4×4 for the 72×72 images). The

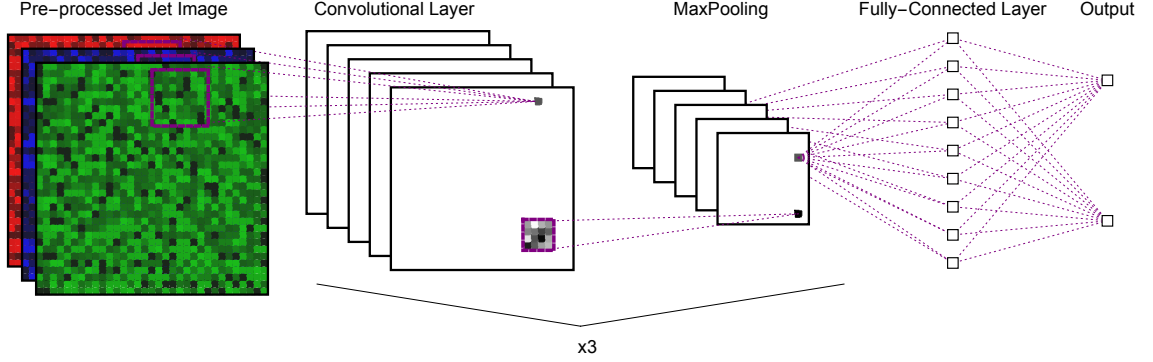


Figure 4.2: An illustration of the deep convolutional neural network architecture. The first layer is the input jet image, followed by three convolutional layers, a dense layer and an output layer. [19]

max-pooling layers performed a 2×2 down-sampling with a stride length of 2. The dense layer consisted of 128 units, with a dropout rate of 0.2 before and after it.

The network was trained with the AdamW optimizer at a learning rate of 10^{-3} and a weight decay of 0.1, using the cross-entropy loss. Training used a batch size of 128 for 20 epochs, and the model with the lowest validation loss was retained. Training images were augmented with random reflections in $\Delta\eta$ and $\Delta\phi$ and random translations of up to 0.5 pixels. The Receiver Operator Characteristic (ROC) curve is shown differential in $p_{T,\text{jet}}$ by evaluating the model on a holdout test set.

A way to gain some understanding of how the model looks at jet substructure is through *interpretability* methods. We use a simple method of just noting the average change in classifier output upon masking a pixel. As seen in fig. 4.4,

4.2.2 Characterizing jet quenching

Another way to think about jet quenching/modification, is to ask how easily can a classifier algorithm tell apart the quenched and unquenched jet. Here we take PYTHIA-6 [57] as the source of unquenched jets and JEWEL [58] generates quenched jets in different centrality bins with/without recoil centers. Jets are represented as a 9-D vectors of features: $p_{T,\text{jet}}$,

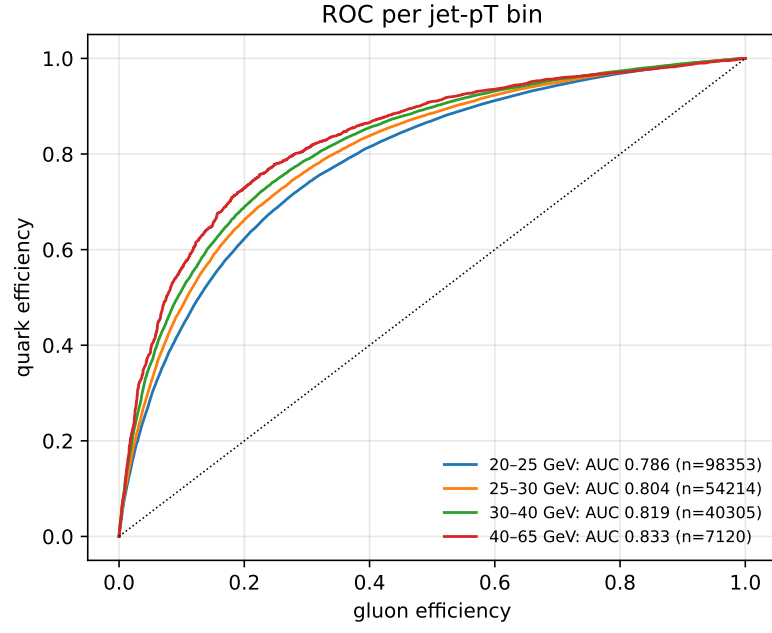


Figure 4.3:

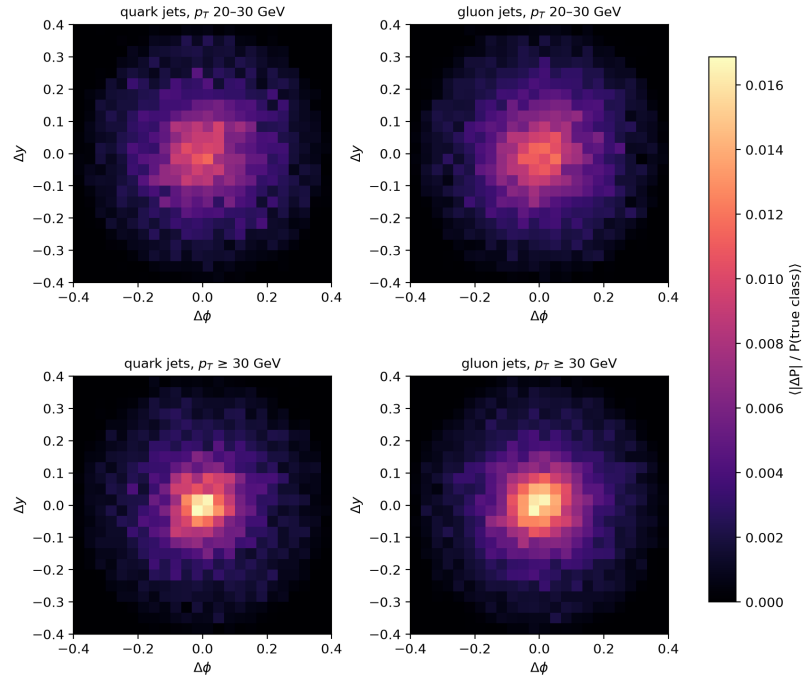


Figure 4.4: Per-pixel feature importance

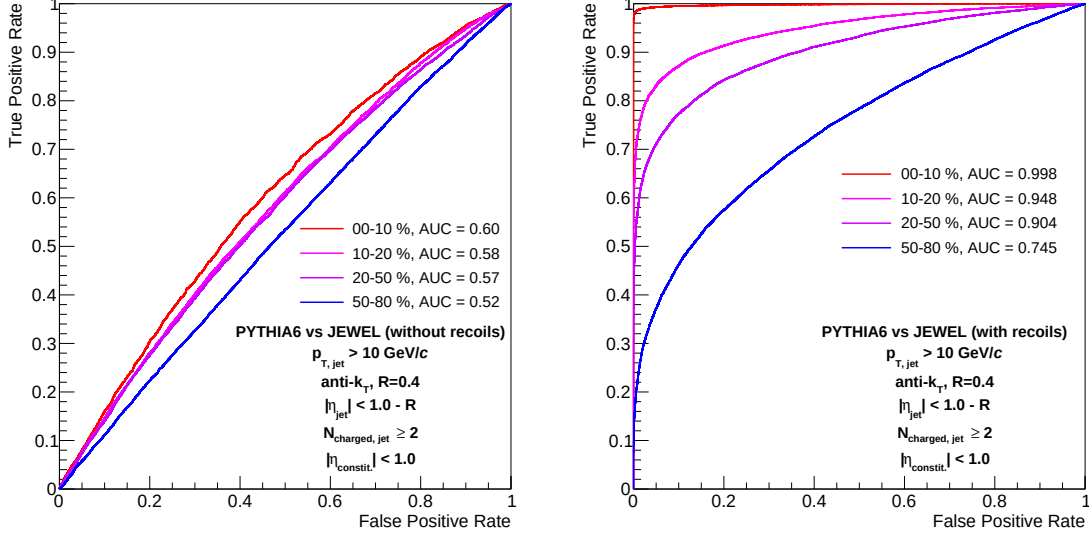


Figure 4.5: ROC curves for PYTHIA6 vs. JEWEL without recoils (left) and with recoils (right).

$N_{constit,jet}^{charged}$, $N_{constit,jet}^{neutral}$, LeSub, λ_0^2 , $\lambda_1^{0.5}$, λ_1^1 , λ_1^2 . The jets are input into Dense Neural Networks (DNNs) with three hidden layers with 100 nodes each. After training, we calculate ROC curves shown in 4.2.2 using a holdout test set. The DNN performs better than chance in all the centralities and for both with/without recoils case. The with-recoils case gives exaggerated ROC curves due to the background of recoil centers. AUC score decreases from central (0-10%) to peripheral (50-80%) events possibly implying more modification in central compared to peripheral. More information about jet quenching and machine learning can be found in [59]

4.3 Unfolding

For any experiment, there is a disconnect between the desired phenomena (*truth* level) and the measurement (*measured* level) due to limitations of the detector setup. This is portrayed in fig. 4.3 through a dijet event at the truth level, whose final state charged and neutral particles are recorded as TPC tracks and BEMC towers. Each of the detectors smear the energy and momenta and for almost all experiments, the information about particle species is lost. For example, at STAR often a π^0 mass assumption is taken for measured constituents.

Thus jets, which are composite objects of these tracks and towers are non-trivially modified by a complicated convolution of these effects. Even with the best background removal procedure, jets will contain some residual background fluctuations from the underlying event.

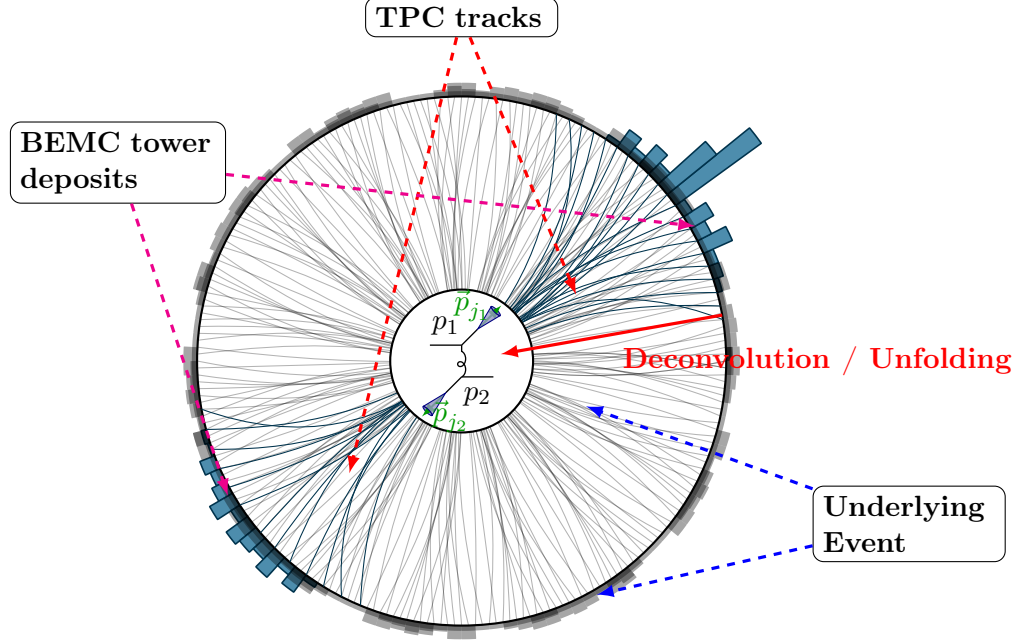


Figure 4.6: Schematic figure showing the unfolding procedure. The desired physics / truth / particle level is represented by the $2 \rightarrow 2$ feynman diagram in the middle. The blue (gray) tracks and tower deposits are the dijet (underlying) event as recorded by the TPC and BEMC.

Thus distributions of any jet observable $\mathbf{O}$ will have these *detector effects* which will need to be removed in order for the measurement to be comparable to theory calculations. In other terms, a truth value of $\mathbf{x}$ can manifest as a range of measured values $(\mathbf{x}_0, \mathbf{x}_1)$ similarly, a measured value of $\mathbf{y}$ can come from a range of truth values $(\mathbf{y}_0, \mathbf{y}_1)$. Thus, using Bayes theorem, the probability either measured value being $\mathbf{y}$ or truth value being $\mathbf{x}$ would be,

$$\begin{aligned}
 p_{\text{meas.}}(\mathbf{y}) &= \int d\mathbf{x} \, p_{\text{meas.}|\text{tr.}}(\mathbf{y}|\mathbf{x}) p_{\text{tr.}}(\mathbf{x}) \\
 p_{\text{tr.}}(\mathbf{x}) &= \int d\mathbf{y} \, p_{\text{tr.}|\text{meas.}}(\mathbf{x}|\mathbf{y}) p_{\text{meas.}}(\mathbf{y})
 \end{aligned} \tag{4.1}$$

From eq. 4.1, we measure $p_{\text{meas.}}$ through histograms, density estimates, etc. and $p_{\text{tr.}}$.

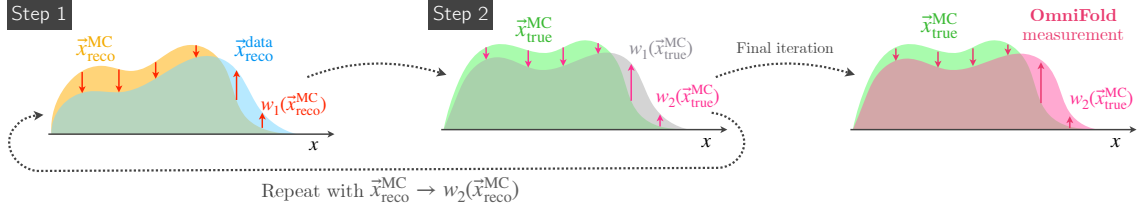


Figure 4.7: Illustration of the OmniFold method. In Step 1, MC is corrected to match data at the detector level, producing a weighting function $w_1(\vec{x}_{\text{reco}}^{\text{MC}})$. In Step 2, a new function $w_2(\vec{x}_{\text{true}}^{\text{MC}})$ is learned based only on particle-level quantities. The method proceeds iteratively, refining the function $w_2(\vec{x}_{\text{true}}^{\text{MC}})$ such that the particle-level MC can be reweighted to give event yields and kinematics that match those observed in the data. [20]

is what we want to measure. Now, $p_{\text{meas.}|\text{tr.}}$ represents the detector's *response*, which can be obtained to a high degree of accuracy from simulation programs like GEANT [60]. We can start from a *generated* truth distribution $p_{\text{gen.},0}$ and obtain the *reconstructed* measured distribution $p_{\text{reco.},0}$. Thus the assumption is,

$$p_{\text{meas.}|\text{tr.}}(\mathbf{y}|\mathbf{x}) = p_{\text{reco.}|\text{gen.},0}(\mathbf{y}|\mathbf{x}) \equiv R(\mathbf{y}, \mathbf{x}), \quad (4.2)$$

Substituting $R(\mathbf{y}, \mathbf{x})$ in eq. 4.1 and using $\omega_0(\mathbf{y}) = p_{\text{meas.}}(\mathbf{y})/p_{\text{reco.},0}(\mathbf{y})$

$$\begin{aligned} p_{\text{gen.},1}(\mathbf{x}) &\equiv \int d\mathbf{y} \, p_{\text{gen.}|\text{reco.},0}(\mathbf{x}|\mathbf{y}) p_{\text{meas.}}(\mathbf{y}) \\ &= \int d\mathbf{y} \, \frac{R(\mathbf{y}, \mathbf{x}) p_{\text{gen.},0}(\mathbf{x})}{p_{\text{reco.},0}(\mathbf{y})} p_{\text{meas.}}(\mathbf{y}) \\ &= p_{\text{gen.},0}(\mathbf{x}) \int d\mathbf{y} \, R(\mathbf{y}, \mathbf{x}) \omega_0(\mathbf{y}) \end{aligned} \quad (4.3)$$

Using $\nu_0(\mathbf{x}) = p_{\text{gen.},1}(\mathbf{x})/p_{\text{gen.},0}(\mathbf{x})$ measured distribution correspondingly becomes,

$$\begin{aligned} p_{\text{reco.},1}(\mathbf{y}) &= \int d\mathbf{x} \, (\mathbf{x}) R(\mathbf{y}, \mathbf{x}) p_{\text{gen.},1}(\mathbf{x}) \\ &= p_{\text{reco.},0}(\mathbf{y}) \int d\mathbf{x} \, \frac{R(\mathbf{y}, \mathbf{x}) p_{\text{gen.},0}(\mathbf{x})}{p_{\text{reco.},0}(\mathbf{y})} \nu_0(\mathbf{x}) \end{aligned} \quad (4.4)$$

The convolutions $\int d\mathbf{y} \, R(\mathbf{y}, \mathbf{x}) \omega_0(\mathbf{y})$ and $\int d\mathbf{x} \, R(\mathbf{y}, \mathbf{x}) \frac{p_{\text{gen.},0}(\mathbf{x})}{p_{\text{reco.},0}(\mathbf{y})} \nu_0(\mathbf{x})$ are mapping $\omega_0(\mathbf{y}) \rightarrow$

$\omega_0(\mathbf{x})$ and $\nu_0(\mathbf{x}) \rightarrow \nu_0(\mathbf{y})$ respectively. For the n^{th} iteration,

$$\begin{aligned}
\omega_{n-1}(\mathbf{y}) &= \frac{p_{\text{meas.}}(\mathbf{y})}{p_{\text{reco.},n-1}(\mathbf{y})} \\
p_{\text{gen.},n}(\mathbf{x}) &= p_{\text{gen.},n-1}(\mathbf{x}) \int d\mathbf{y} R(\mathbf{y}, \mathbf{x}) \omega_{n-1}(\mathbf{y}) \\
\nu_{n-1}(\mathbf{x}) &= \frac{p_{\text{gen.},n}(\mathbf{x})}{p_{\text{gen.},n-1}(\mathbf{x})} \\
p_{\text{reco.},n}(\mathbf{y}) &= p_{\text{reco.},n-1}(\mathbf{y}) \int d\mathbf{x} R(\mathbf{y}, \mathbf{x}) \frac{p_{\text{gen.},n-1}(\mathbf{x})}{p_{\text{reco.},n-1}(\mathbf{y})} \nu_{n-1}(\mathbf{x})
\end{aligned} \tag{4.5}$$

Given convergence, $p_{\text{reco.},n}(\mathbf{y}) \rightarrow p_{\text{gen.}}(\mathbf{y})$ and $p_{\text{tr.},n}(\mathbf{x}) \rightarrow p_{\text{unf.}}(\mathbf{x})$ is the unfolded distribution.

This algorithm can be adapted for binned data by expressing the distributions as vectors representing bin counts of histograms and $R(\mathbf{y}, \mathbf{x})$ as a matrix, giving the traditional method of *Iterative Bayesian Unfolding* [61]. Although it has been the standard for a long time, the required binning of data limits its usefulness beyond 3-dimensions. Also, not all measurements can be straightforwardly represented in the response matrix.

One of the first unbinned approach uses classifiers to calculate ratios ω and ν . For i^{th} iteration, $\omega_{i-1}(\mathbf{y})$ can be calculated using classifier $f_{\text{meas.},\text{reco}}$ which classifies a jet represented by $\mathbf{y}$ as measured or reconstructed. Thus,

$$\omega_{i-1}(\mathbf{y}) = \frac{f_{\text{meas.},\text{reco}}(\mathbf{y})}{1 - f_{\text{meas.},\text{reco}}(\mathbf{y})} \tag{4.6}$$

and $\nu_{i-1}(\mathbf{y})$ can be calculated using classifier $f_{i,i-1}$ which classifies a jet represented by $\mathbf{x}$ as being generated from i^{th} or $(i-1)^{\text{th}}$ iteration.

$$\nu_{i-1}(\mathbf{x}) = \frac{f_{i,i-1}(\mathbf{x})}{1 - f_{i,i-1}(\mathbf{x})} \tag{4.7}$$

For misses, a classifier $f_{i,i-1}^{\text{gen.}}$ separates matched generated jets weighted by $\omega_{i-1}(\mathbf{y})$ from those of the $(i-1)^{\text{th}}$ iteration; for fakes, a classifier $f_{i,i-1}^{\text{reco.}}$ separates matched reconstructed

jets weighted by $\nu_{i-1}(\mathbf{x})$ from those of the $(i-1)^{\text{th}}$ iteration,

$$\omega_{i-1}(\mathbf{x}) = \frac{f_{i,i-1}^{\text{gen.}}(\mathbf{x})}{1 - f_{i,i-1}^{\text{gen.}}(\mathbf{x})}, \quad \nu_{i-1}(\mathbf{y}) = \frac{f_{i,i-1}^{\text{reco.}}(\mathbf{y})}{1 - f_{i,i-1}^{\text{reco.}}(\mathbf{y})}, \quad (4.8)$$

evaluated at the miss's $\mathbf{x}$ and the fake's $\mathbf{y}$ respectively. Thus, it takes 2-4 classifier training per unfolding iteration.

Chapter 5

Jet-hadron correlations relative to the event plane

In this analysis, the coordinate system used to depict the distribution of associated particles is defined relative to a reconstructed jet, also called a *trigger jet*. The distribution is thus given by:

$$\frac{1}{N_{\text{trig}}} \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta}, \quad (5.1)$$

where N_{trig} is the number of trigger jets, $N_{\text{assoc,jet}}$ is the number of associated particles, $\Delta\phi$ ($= |\phi_{\text{jet}} - \phi_{\text{assoc}}|$) is the azimuthal angle of those trigger particles relative to the trigger jets, and $\Delta\eta$ ($= |\eta_{\text{jet}} - \eta_{\text{assoc}}|$) is the difference in the pseudorapidities of the trigger jet and associated particle. When studying the $\Delta\phi$ distributions, a peak is observed at $\Delta\phi = 0$ which is the triggered jet, referred to as the *near-side*. Another peak which is seen near $\Delta\phi = \pi$ is referred to as the *away-side*. The away-side consists of the hadrons associated with the trigger jet and make up the second jet which results from the early hard-scatter.

The goal of this analysis is to study the conditional yield of associated particles, the width of the near- and away-side peaks (quantified using the gaussian width) as a function of the angle between the jet axis and the event plane. The yield is estimated by:

$$\text{Yield} = \frac{1}{N_{\text{trig}}} \int_c^d \int_a^b \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} d\Delta\phi d\Delta\eta. \quad (5.2)$$

The integration limits are chosen based on practical considerations, including the detector acceptance and binning of histograms.

5.1 Analysis Method

This measurement utilizes data collected during the 2014 run from Au+Au collisions at nucleon-nucleon center-of-mass energy of $\sqrt{s_{\text{NN}}} = 200$ GeV by the STAR experiment [62] at RHIC. Events referred to as signal (same) events are required to contain a HT trigger in the BEMC [63]. Minimum-bias (MB) triggered events based on coincidence of Zero Degree Calorimeters (ZDC coincidence), Vertex Position Detectors and Beam-Beam Counters signals are used to estimate the pair-acceptance effects via a mixed-event (ME) technique [64]. For this analysis, 9.4M HT-triggered and 4.0M MB collision events are used. Events are further categorized by their centrality selection, defined in section 3.2.3. The events are required to have a reconstructed primary vertex $|v_z| < 24$ cm and centrality of 20-50%.

5.1.1 Event plane reconstruction

The event plane ($\Psi_{2,EP}$) discussed in 3.2.4 is calculated event-by-event using the Modified Reaction Plane (MRP) method [21, 65, 65, 66] using charged tracks with $0.2 < p_{\text{T,track}} < 1.0$ GeV/ c measured within the TPC. The impact of highly energetic jets on the calculation of the event-plane orientation is reduced by removing the particles within the pseudorapidity strip ($|\Delta\eta| < 0.4$) across $\Delta\phi$ surrounding the leading jet. This procedure also removes a significant portion of the away-side jet, located opposite in azimuth. The event-by-event recentering and shifting corrections for removing azimuthal anisotropy in $\Psi_{2,EP}$ are done separately for 10% centrality intervals and 4-cm v_z intervals.

The distributions of these associated tracks relative to the trigger jet are measured in three bins in the angle between the trigger jet and the event plane, in-plane ($|\Psi_{2,EP} - \phi_{\text{jet}}| < \pi/6$), mid-plane ($\pi/6 < |\Psi_{2,EP} - \phi_{\text{jet}}| < \pi/3$), and out-of-plane ($|\Psi_{2,EP} - \phi_{\text{jet}}| > \pi/3$) bins. The analysis is restricted to 20–50% central Au+Au collisions to achieve the highest event-plane resolution and therefore the analysis will be most sensitive to any path-length dependencies.

The second- (fourth-) order event-plane resolutions relative to the second-order event plane ($R_2\{\Psi_{2,EP}\}$ and $R_4\{\Psi_{2,EP}\}$ respectively) as a function of collision centrality are shown in Fig. 5.1. The resolution is peaked around the 20-30% and 30-40% centrality. The event-plane resolutions for $R_2\{\Psi_{2,EP}\}$ and $R_4\{\Psi_{2,EP}\}$ were 0.56 and 0.28, respectively. The errors on the event-plane resolution calculation were less than 1%, leading to a negligible effect on the final $\Delta\phi$ correlations. Measured values for the event-plane resolution are in good agreement with prior STAR studies [21]. These resolutions are evaluated to correct the observed flow coefficients which arise in the fits of the combinatorial background discussed in Sect. 5.1.4.

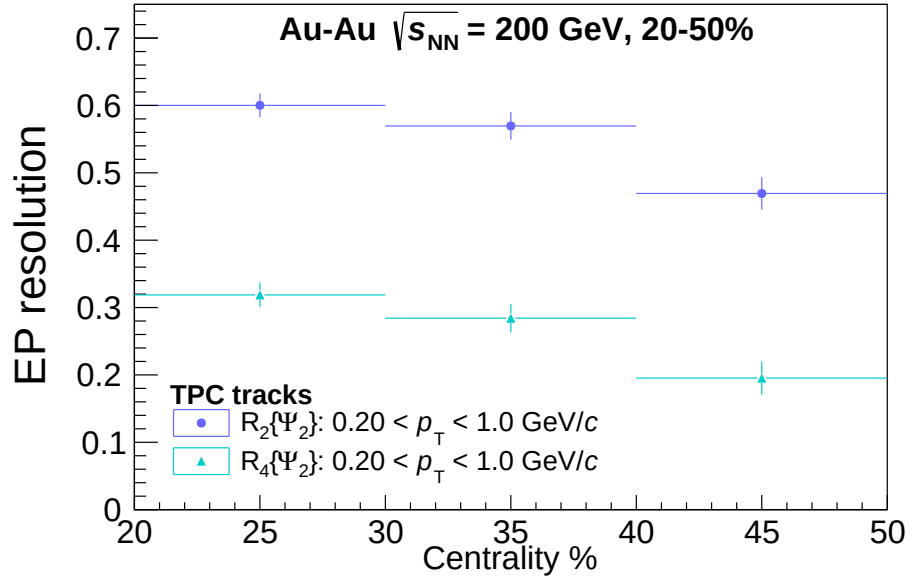


Figure 5.1: Event-plane resolution: Second-order (fourth-order) harmonic relative to the event plane, $R_2(\Psi_2)$ ($R_4(\Psi_2)$), respectively. The approach follows the Modified Reaction Plane (MRP) method [21] utilizing the charged tracks of the TPC for event-plane reconstruction and resolution calculation for tracks ranging from 0.2-1.0 GeV/c.

5.1.2 Jet reconstruction and selection

Full jets are reconstructed by measuring charged-tracks in the TPC and collecting neutral-particle information from the BEMC. The anti- k_T algorithm clusters these particles into jets by reconstructing the jet momenta as the quadratic sum of their constituent momenta using

a boost-invariant p_T^2 recombination scheme. Tracks used for reconstructing jets are assumed to be pions while the towers to have arisen from massless particles. All jets measured in this work are clustered with resolution parameter of $r = 0.4$. The area of jets, A_{jet} , is found with FastJet using active ghost particles [67]. Partially reconstructed jets at the edge of the acceptance are rejected by applying a fiducial cut, $|\eta_{\text{jet}}| < 1.0 - r$, to assure all jets fall within the acceptance of the detectors.

Jets produced in heavy-ion collisions sit on top of a large amount of underlying event. The jet signal can be found beneath tens to hundreds of particles resulting from various other processes. To reduce the influence of these background particles, this analysis requires tracks (towers) with $p_T(E_T) > 2.0 \text{ GeV}/c$ for jet reconstruction. At RHIC energies this selection reduces the median background energy density per unit A_{jet} , $\langle \rho \rangle$, down to $\approx 0.6 \text{ GeV}$. This high-constituent selection, referred to as a “hard-core” jet selection reduces fluctuations, fake jets, and background jet particles [68]. To further reduce contributions from the background and to match the trigger condition, the jets are required to contain a constituent tower that fired the HT trigger ($E_{T,\text{tower}} > 5.4 \text{ GeV}/c$). The remaining jets are studied in classes of jets with $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ and $20 < p_{T,\text{jet}} < 40 \text{ GeV}/c$.

5.1.3 Jet-hadron correlation

The measurement of the correlation function (distribution of charged hadrons relative to reconstructed jets) described in Eqn. 5.1 requires several corrections. The correlation function is measured in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) as:

$$\frac{1}{N_{\text{trig}}} \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} = \frac{1}{N_{\text{trig}}} \frac{1}{\epsilon(p_{T,\text{assoc}}, \eta_{\text{assoc}})} \frac{1}{a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)} \left(\frac{d^2 N_{\text{assoc,jet}}^{\text{meas}}}{d\Delta\phi d\Delta\eta} - \frac{d^2 N_{\text{assoc,jet}}^{\text{bkgd}}}{d\Delta\phi d\Delta\eta} \right). \quad (5.3)$$

$N_{\text{assoc,jet}}$ gives the number of pairs of trigger jets and the associated hadrons, $N_{\text{assoc,jet}}^{\text{meas}}$ and $N_{\text{assoc,jet}}^{\text{bkgd}}$ are the number of pairs measured and the pair characterized as background

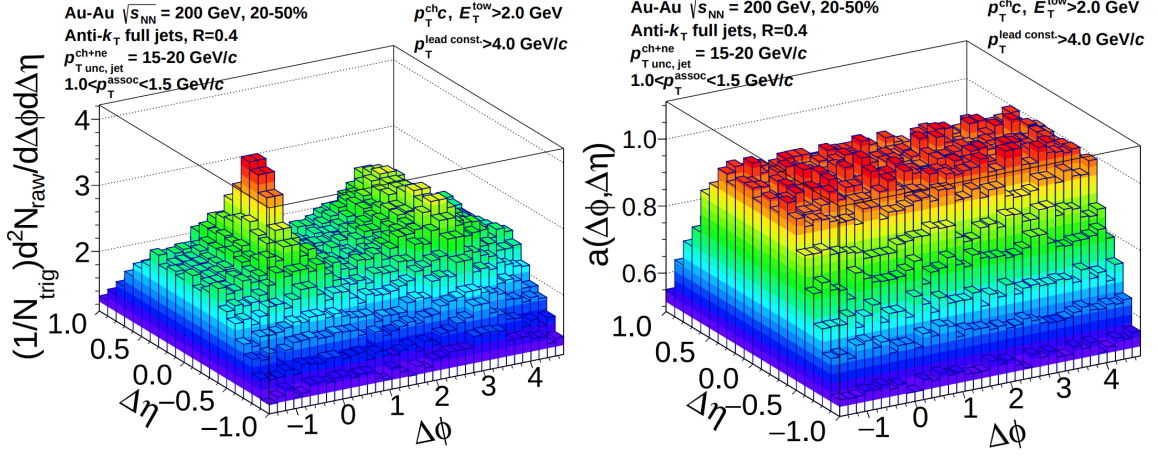


Figure 5.2: Example of a correlation function distribution in 2 dimensions between $15 < p_{T,\text{jet}} < 20$ GeV/c $1.0 < p_{T,\text{assoc}} < 1.5$ GeV/c (left). Example of a normalized mixed event distribution in 2 dimensions between $15 < p_{T,\text{jet}} < 20$ GeV/c $1.0 < p_{T,\text{assoc}} < 1.5$ GeV/c showing the $\Delta\eta = \eta_{\text{jet}} - \eta_{\text{assoc}}$. acceptance for associated particles given $|\eta_{\text{assoc}}| < 1.0$ and $|\eta_{\text{jet}}| < 0.6$ (right).

respectively. $\epsilon(p_{T,\text{assoc}}, \eta_{\text{assoc}})$ is the single-track reconstruction efficiency¹. The pair acceptance, $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$, is calculated from the raw pairs that we measure from a trigger jet associated with charged hadrons from mixed events.

The correlations are determined in bins of centrality, reconstructed trigger-jet transverse momentum ($p_{T,\text{jet}}$), associated-hadron transverse momentum ($p_{T,\text{assoc}}$), and bins of the trigger jet relative to the event plane (in, mid, out, all combined angles) defined in Sect. 5.1. The corrected correlation functions contain a large combinatorial background ($N_{\text{assoc,jet}}^{\text{bkgd}}$), which must be subtracted. This subtraction procedure is described in Sect. 5.1.4 .

The pair-acceptance correction, $1/a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$, accounts for the detector and tracking inefficiencies, and is found by correlating jets from HT-triggered events with associated hadrons from MB events of the same event class. In addition to providing the acceptance correction, the mixed-event procedure will also help remove the trivial correlation due to an η dependence in the single-particle track distributions [42]. Taking the ratio of the single-event pairs by the mixed-event pairs removes the detector effects of acceptance

¹Acceptance effects such as gaps between sector boundaries are included in this correction, however, we refer to $\epsilon(p_{T,\text{assoc}}, \eta_{\text{assoc}})$ as an efficiency correction in order to distinguish it from $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$.

and efficiency. The pair acceptance will serve as the dominant effect, given that there is little η dependence in both the tracks and jets across the acceptance range of this analysis.

The event-mixing procedure used in this analysis is well described in Ref. [64]. The mixed events used for calculating $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$ in this work are required to be within the same 10% centrality class and to have a vertex position within 4 cm along the direction of the beam (z). They are constructed separately for 20-30%, 30-40%, and 40-50% centrality classes and combined accordingly. High-momentum tracks are nearly straight, so the detector acceptance does not change significantly at high- $p_{T,\text{assoc}}$, and thus all associated momentum bins greater than 2.0 GeV/ c are combined to increase statistics. There is no difference in efficiency and acceptance within uncertainties for different orientations of the jet relative to the event plane, and therefore the same correction $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$ is applied for all angles relative to the event plane. The acceptance $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$ is normalized to 1 at its maximum, determined using the region of approximately constant acceptance ($|\Delta\eta| < 0.4$). For each $p_{T,\text{assoc}}$ bin, the projection of the flat plateau region (integrated over the region $|\Delta\eta| < 0.4$) was fit with a constant over the whole $\Delta\phi$ range. The associated uncertainties of the fits were used for the systematic uncertainty on the mixed-event normalization, which is added in quadrature and reported as the scale uncertainty. This systematic uncertainty is under 1% (1.25%) in all $p_{T,\text{assoc}}$ bins used for the reported results of jets with $15 < p_{T,\text{jet}} < 20$ GeV/ c and $20 < p_{T,\text{jet}} < 40$ GeV/ c .

A shape uncertainty in $\Delta\phi$ due to the z_{vtx} binning, similar to that in $\Delta\eta$, could lead to an additional shape uncertainty in the correlation functions. To test for this, the ratio of the 1D $\Delta\phi$ projection with the nominal z_{vtx} binning and the binning described above was calculated for each $p_{T,\text{jet}}$, $p_{T,\text{assoc}}$, and centrality bin. The variations about 1.0 are smaller than the statistical errors associated with the points. This uncertainty was therefore considered negligible.

There is an additional shape uncertainty associated with the application of the acceptance correction due to slight changes in the correlation function at large $\Delta\eta$ in the acceptance with

v_z position. The background level is determined from the level of the correlation function at large $\Delta\eta$, leading to a scale uncertainty in the background subtraction dependent on $p_{T,\text{assoc}}$. This uncertainty is from the differences between the nominal (z-integrated) and the z-vertex method (z-binned) for correcting the mixed events on the level of the background in the $0.6 < \Delta\eta < 1.2$ range, and signal plus background, in the $|\Delta\eta| < 0.6$ range. The large $\Delta\eta$ region is used to determine the background so any uncertainties in the level of the correlation function in this region lead to an uncertainty in the level of the background in the signal region. This is expressed as an additional scale band on the final results. This uncertainty is determined by varying the binning of the mixed events in v_z and is correlated for different angles relative to the event plane and for different bins in $p_{T,\text{assoc}}$. Above $p_{T,\text{assoc}} > 3 \text{ GeV}/c$, this uncertainty is negligible because the background is small.

5.1.4 Flow modulation of combinatorial background

The combinatorial background ($N_{\text{assoc,jet}}^{\text{bkgd}}$) from Eq. 5.3 can be parametrized for trigger jets restricted to the orientation $\mathfrak{R} = \text{in/mid/out-of-plane}$ relative to the event plane as [40, 69]:

$$\left(\frac{1}{\pi} \frac{dN_{\text{assoc,jet}}^{\text{bkgd}}}{d\Delta\phi} \right)_{\mathfrak{R}} = B^{\mathfrak{R}} \left(1 + \sum_{n=2}^{\infty} 2v_{n,\text{assoc}} v_{n,\text{jet}}^{\mathfrak{R}} \cos(n\Delta\phi) \right), \quad (5.4)$$

where $v_{n,\text{assoc}}$ and $v_{n,\text{jet}}^{\mathfrak{R}}$ are the Fourier coefficients of the azimuthal angle distribution (flow coefficients) of background associated particles and trigger jets restricted to the orientation $\mathfrak{R}$ respectively. $B^{\mathfrak{R}}$ gives the background level amplitude for orientation $\mathfrak{R}$. The trigger jets being restricted to orientation $\mathfrak{R}$ modifies their nominal flow coefficients $v_{n,\text{jet}}$'s into the $v_{n,\text{jet}}^{\mathfrak{R}}$ from Eq. 5.4 which are related to each other as [40]:

$$v_{n,\text{jet}}^{\mathfrak{R}} = \begin{cases} \frac{v_{n,\text{jet}}}{\beta^{\mathfrak{R}}} + \cos(n\phi_s^{\mathfrak{R}}) \frac{\sin(nc)}{nc} \frac{R_n}{\beta^{\mathfrak{R}}} + \\ \frac{1}{\beta^{\mathfrak{R}}} \sum_{k=2,4,6,..} (v_{(k+n),\text{jet}} + v_{|k-n|,\text{jet}}) \cos(k\phi_s^{\mathfrak{R}}) \frac{\sin(kc)}{kc} R_n & \text{if } n \text{ is even,} \\ v_{n,\text{jet}} & \text{if } n \text{ is odd.} \end{cases} \quad (5.5)$$

Where, $\beta^{\mathfrak{R}}$ is the $\mathfrak{R}$ dependence of $B^{\mathfrak{R}}$ given by,

$$B^{\mathfrak{R}} \propto \beta^{\mathfrak{R}} = 1 + \sum_{k=2,4,6,..} 2v_{k,\text{jet}} \cos(k\phi_s^{\mathfrak{R}}) \frac{\sin(kc)}{kc} R_n \quad (5.6)$$

$\phi_s^{\mathfrak{R}}$ and c are the center and width of the $|\Psi_{2,EP} - \phi_{\text{jet}}|$ range for jets restricted to orientation $\mathfrak{R}$. Odd $v_{n,\text{jet}}$'s mainly arise from initial state fluctuations and are therefore uncorrelated with the second-order event-plane and remain constant when the trigger jet is moved relative to the event plane, while even $v_{n,\text{jet}}$'s will change [69, 70]. The $\mathfrak{R}$ dependent background shape is dependent upon the event-plane resolution (R_n), which is fixed at the measured values. Extended details into the derivations of relevant equations can be found in Refs. [40, 71, 72].

Collective particle flow plays a major role in understanding the underlying-event background. This background consists of particles created from mechanisms unrelated to the hard process that led to a jet. Some of the jet signal is correlated with our event plane due to the path-length dependence of partonic energy loss, while soft hadrons are predominantly correlated with the event plane due to hydrodynamical flow that also contribute to the bulk-particle production.

To remove the combinatorial background comprised by contributions from the underlying event, the reaction plane fit (RPF) developed in Ref. [69] is applied in this work. The measured jet-hadron correlation signal is decomposed into a near-side and an away-side, with the former being narrow in $\Delta\phi$ and $\Delta\eta$ and the latter narrow in $\Delta\phi$, but broad in $\Delta\eta$. The narrowness of the near-side implies the signal is negligible at large $\Delta\eta$, where

the background dominates. Applying RPF, we defined our ‘signal + background’ region for $|\Delta\eta| < 0.6$ and further $0.6 \leq |\Delta\eta| < 1.2$ as a background dominated region where the signal was assumed to be negligible. The correlation function is fit with the parametrization given in Eq. 5.4, restricted to $n = 4$ for the background dominated region at large $\Delta\eta$ and small $\Delta\phi$ ($|\Delta\phi| < \pi/2$) simultaneously for in-plane , mid-plane , and out-of-plane trigger jets.

RPF improves upon prior background subtraction techniques [42, 73] by avoiding problems due to contamination from jets on both the near- and away-side by using the near-side at large $\Delta\eta$ and also using the dependence of the flow-modulated background on the angle of the trigger jet relative to the event plane to constrain the background shape and level.

For $p_{T,assoc} > 2$ GeV/ c , the combinatorial background is small, and few high momentum tracks are found at large distances in pseudorapidity from the near-side jet. So the model fit is restricted to $n = 3$ as higher order terms are no longer contributing. This occurs for $p_{T,assoc} > 4$ GeV/ c with $15 < p_{T,jet} < 20$ GeV/ c jets and $p_{T,assoc} > 3$ GeV/ c with $20 < p_{T,jet} < 40$ GeV/ c . Therefore, the RPF fits consist of six parameters ($B^{\mathfrak{R}}$, $v_{2,jet}^{\mathfrak{R}}$, $v_{2,assoc}$, $(v_{3,jet} \times v_{3,assoc})$, $v_{4,jet}^{\mathfrak{R}}$, and $v_{4,assoc}$) for low $p_{T,assoc}$, and four ($B^{\mathfrak{R}}$, $v_{2,jet}^{\mathfrak{R}}$, $v_{2,assoc}$ and $(v_{3,jet} \times v_{3,assoc})$) for high $p_{T,assoc}$.

Comparison of in-, mid-, and out-of-plane jet-hadron correlations is performed after background subtraction to explore the effects related to event-plane orientation. An example of event-plane dependent correlation function after the RPF background subtraction for jets with $15 < p_{T,jet} < 20$ GeV/ c is shown in Fig. 5.4 for in-plane (a), mid-plane (b), out-of-plane (c) and jets from all combined angles (d) for associated particles with momenta $p_{T,assoc} \text{range } 1.52.0$. The uncertainties from the RPF background subtraction are propagated using the covariance matrix from the fit and are non-trivially correlated point-to-point and between different bins relative to the event plane. These are shown in gray. The uncertainty from the acceptance correction, described in Sect. 5.1.3 is displayed by the red uncertainty band. The uncertainties on the event-plane resolution are negligible relative to that of the background subtraction and statistical uncertainties of the final results. Additional

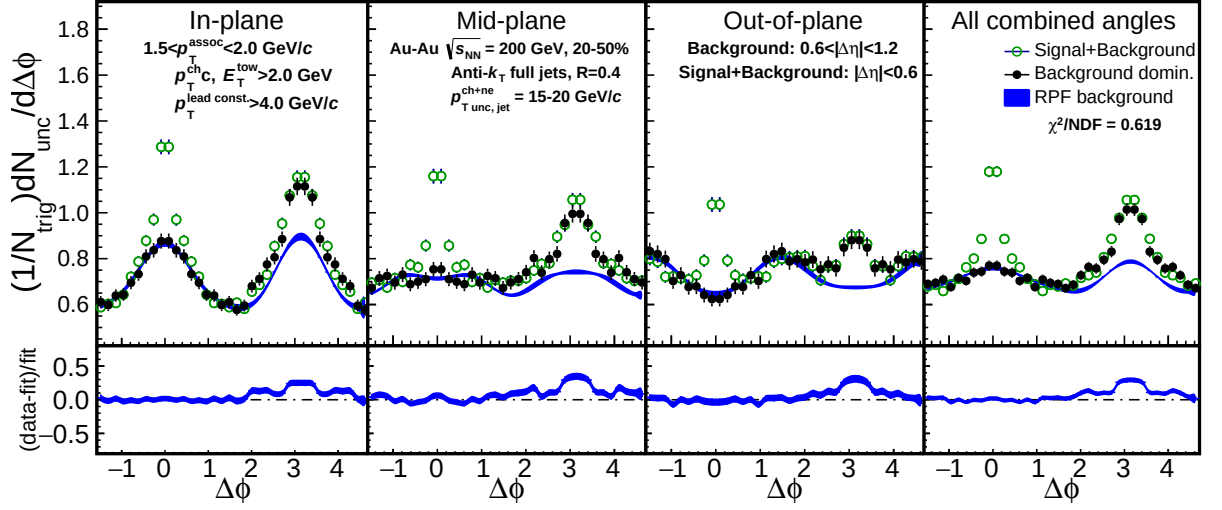


Figure 5.3: (Top) Signal+background region, background-dominated region, and RPF fit to the background for the event-plane dependent jet-hadron correlations of $15 < p_{T,jet}^{jet} < 20$ GeV/ c jets correlated with $p_{T,assoc}$ range 1.5-2.0 charged hadrons from the 20-50% most central events. (Bottom) Quality of the RPF fit to the background-dominated region expressed as $(data - fit) / fit$.

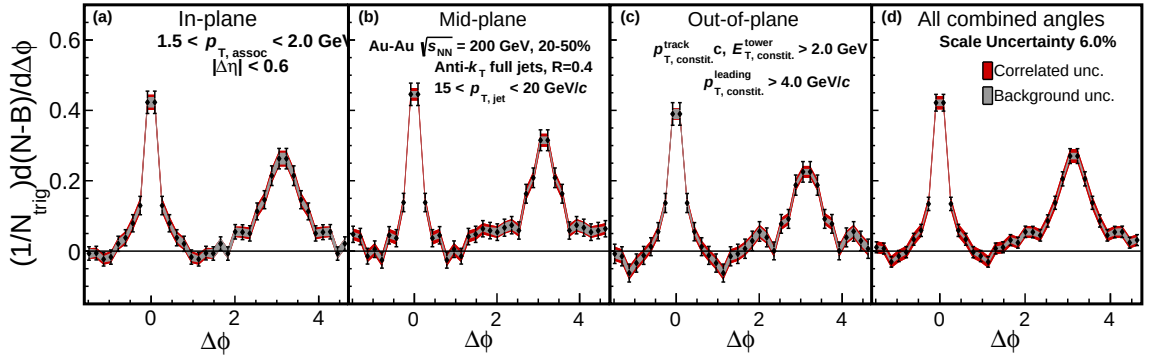


Figure 5.4: Example of a background subtracted correlation function for $15 < p_{T,jet} < 20$ GeV/ c jets correlated with $p_{T,assoc}$ range 1.5-2.0 charged hadrons from the 20-50% most central collisions. Correlated scaled uncertainty from the application of mixed events is displayed by the red band, while the uncertainty associated with the RPF background fit is displayed as the gray band.

uncertainties uncorrelated with each other, but correlated for all points in $\Delta\phi$ are given in Tab. 5.1. They are combined in quadrature and listed as the scale uncertainty on the results.

5.1.5 Systematic Uncertainties

The PYTHIA6 Perugia 2012 tune is used to create particle level dijet events embedded in MB Au+Au 200 GeV events at the detector level. This allows for further comparisons between the jets from the input PYTHIA6 tracks (generator-level jets or GEN-jets) and the jets from the embedded tracks reconstructed by GEANT (reconstructed jets or RECO-jets). RECO level events are analyzed with the same selections and parameters as used by the data analysis. All particles of GEN-level events are required to be in their final state (particles with no further daughters). GEN-level jets have $p_{T,\text{jet}}^{\text{GEN}} > 10 \text{ GeV}/c$ and the RECO-level jets have $p_{T,\text{jet}}^{\text{RECO}} > 5 \text{ GeV}/c$. Further, we require only the RECO-level jets contain a neutral component with $E_T \geq 5.4 \text{ GeV}$ to match the trigger condition applied in data. Our goal is to study the effects of the detector response on jet reconstruction and our analysis. In order to compare the same jet pre- and post-reconstruction, we apply a ‘nearness’ criteria in the $\eta - \phi$ that matches a GEN-jet to one RECO-jet satisfying:

(a) $r_{\text{GEN,RECO}} = \sqrt{(\eta_{\text{jet}}^{\text{GEN}} - \eta_{\text{jet}}^{\text{RECO}})^2 + (\phi_{\text{jet}}^{\text{GEN}} - \phi_{\text{jet}}^{\text{RECO}})^2} \leq 0.4$, $r_{\text{GEN,RECO}}$ being the separation between the gen-jet and the reco-jet in $\eta - \phi$ space, (b) the RECO-jet is the closest to the gen-jet among all reco-jets (minimize $r_{\text{GEN,RECO}}$).

We compare the resulting GEN- and RECO-level jet spectra to calculate the momentum resolution which is given by:

$$p_{T,\text{jet}} \text{ resolution}(\%) = \frac{p_{T,\text{jet}}^{\text{GEN}} - p_{T,\text{jet}}^{\text{RECO}}}{p_{T,\text{jet}}^{\text{GEN}}} \times 100\%. \quad (5.7)$$

Fig. 5.5 shows the $p_{T,\text{jet}}$ resolution for $15 < p_{T,\text{jet}}^{\text{GEN}} < 20 \text{ GeV}/c$ (left) and $20 < p_{T,\text{jet}}^{\text{GEN}} < 40 \text{ GeV}/c$ (right) $R = 0.4$ full jets. The distributions have been normalized into probability functions and are shown for the 20-50% most central events for all angles of the jet relative

to the event plane. Fig. 5.5 further shows an average energy loss of around 15% going from GEN to RECO. This net energy shift is thought to be due to counteracting effects of the tracking inefficiency at the RECO level and there being more tracks in the RECO level from the min-bias pedestal. The $p_{T,\text{jet}}$ resolution for $15 < p_{T,\text{jet}}^{\text{RECO}} < 40$ GeV/ c reco-jets is roughly 10-20%. The event-plane dependence of the $p_{T,\text{jet}}$ resolution was also studied and found to be within 1-2% of each other between different orientations of jets w.r.t event plane. There can be slight differences in the jets reconstructed at lower momenta with $15 < p_{T,\text{jet}} < 20$ GeV/ c for jets at different angles relative to the event plane, due to a low momentum embedded jet overlapping with another jet in the Au+Au data and from statistical fluctuations. As there are more jets from in-plane than out-of-plane orientations in the data, this leads to an apparent difference in the reconstructed jet spectra. Otherwise there are no significant differences between jets at different angles relative to the event plane. A map between $p_{T,\text{jet}}^{\text{GEN}}$ and $p_{T,\text{jet}}^{\text{RECO}}$, called the response matrix, is added to Fig. 5.5 as a smaller inset on the top right.

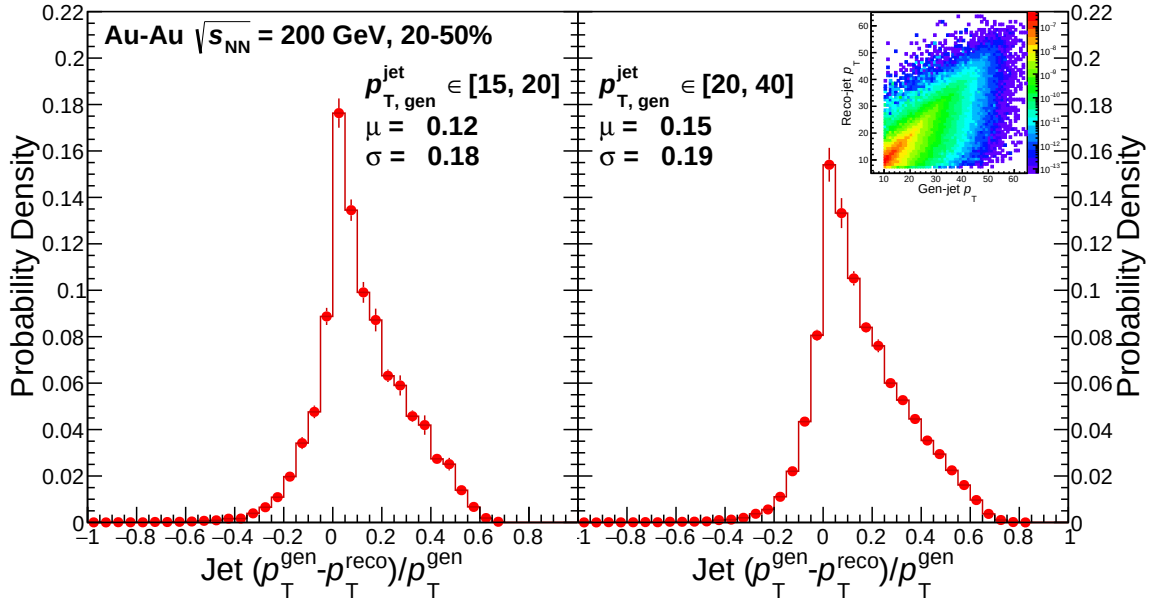


Figure 5.5: $p_{T,\text{jet}}$ resolution for $15 < p_{T,\text{jet}}^{\text{GEN}} < 20$ GeV/ c (left) and $20 < p_{T,\text{jet}}^{\text{GEN}} < 40$ GeV/ c (right) $R = 0.4$ full jets with the corresponding response matrix as an inset on the top right. Jets are measured from all angles relative to the event plane in the 20-50% most central events. This comparison is for matched GEN-level to RECO-level jets.

We summarize the systematic uncertainties in Tabs. 5.1, 5.2, and 5.3. Table 5.1 lists the sources of systematic uncertainties which are independent of the angle relative to the event plane. These sources include the single-track reconstruction efficiency (Sect. ??), uncertainties in the mixed events due to their normalization and shape in $\Delta\phi$ (Sect. 5.1.3), and uncertainties in the event-plane resolution (Sect. 3.2.3). The uncertainties are added in quadrature and lead to a 6% uncertainty in the scale of the correlation functions and yields with the single-track reconstruction efficiency being the dominate source. This uncertainty is uncorrelated for different associated-particle momenta.

Table 5.1: Summary of systematic uncertainties which are independent of the angle relative to the event plane and the momentum for both $15 < p_{T,\text{jet}} < 20$ GeV/ c and $20 < p_{T,\text{jet}} < 40$ GeV/ c in 20-50% central Au+Au collisions.

Source	Uncertainty %
Single particle reconstruction efficiency	5
Mixed event (shape $\Delta\phi$)	negligible
Mixed event normalization	< 1.25
Event-plane resolution	< 1

Additional uncertainties, highly dependent on the angle of the jet relative to the event plane and the associated particles' momentum, are summarized in Tabs. 5.2 and 5.3. These include the impact of the scale uncertainty from the mixed events (Sect. 5.1.3) and the RPF background fit (Sect. 5.1.4) on the associated yield and jet-peak width results. These uncertainties are compared for two associated particle momentum bins (1.0-1.5 and 3.0-4.0 GeV/ c) to highlight how much of an impact the background has at low momenta. The uncertainty arising from the RPF background subtraction is non-trivially correlated point-to-point in $\Delta\phi$ and for different orientations of the jet relative to the event plane, but is uncorrelated between different $p_{T,\text{assoc}}$ bins. The acceptance-correction uncertainty on the shape in $\Delta\eta$ is also correlated for different orientations of the jet relative to the event plane and uncorrelated across $p_{T,\text{assoc}}$ bins. The acceptance-shape uncertainty is the dominant source at low-momenta, while being more comparable to the background uncertainty at larger momenta.

To account for jet-energy shift (JES) due to underlying-event background contribution to the reconstructed jet energy and to provide an adequate comparison for the jet samples from different orientations relative to event plane, a random-cone study was performed on the minimum bias data of matching centrality selection. The $\langle\rho\rangle$ is found to be 0.388 GeV/ c (in-plane), 0.344 GeV/ c (mid-plane), and 0.308 GeV/ c (out-of-plane), with a corresponding RMS of 0.4 GeV/ c . Shifting the jet-momenta selection by the corresponding thresholds was utilized in the correlation analysis to remove the shift-related differences between the jet collections.

Table 5.2: Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the jet energy shift (JES) correction, each varying with event-plane orientation bins. They are displayed for $15 < p_{T,\text{jet}} < 20$ GeV/ c in 20-50% central Au+Au collisions for $p_{T,\text{assoc}}\text{range}1.01.5$ and $p_{T,\text{assoc}}\text{range}3.04.0$ bins. The values are expressed as a percent of the nominal value.

Source	Result	Orientation	Uncertainty %			
			Near-side: $p_{T,\text{assoc}}$ (GeV/ c)		Away-side: $p_{T,\text{assoc}}$ (GeV/ c)	
			1.0-1.5	3.0-4.0	1.0-1.5	3.0-4.0
Acceptance shape	Yield	in-plane	14	1.2	8.1	3.0
		mid-plane	11	1.2	7.6	3.3
		out-of-plane	11	1.1	7.3	3.0
	Width	in-plane	5.2	0.6	4.3	2.4
		mid-plane	5.4	0.5	3.9	2.2
		out-of-plane	4.3	0.4	4.9	2.1
Background fit	Yield	in-plane	11	0.8	6.1	2.1
		mid-plane	7.7	0.9	5.2	2.5
		out-of-plane	7.4	0.7	4.7	2.1
	Width	in-plane	10	0.1	8.2	0.4
		mid-plane	10	0.1	7.5	0.4
		out-of-plane	8.2	0.1	9.3	0.4
JES correction	Yield	in-plane	1.8	3.7	4.9	4.0
		mid-plane	2.8	2.5	1.4	4.2
		out-of-plane	3.7	3.3	4.8	4.2
	Width	in-plane	0.6	0.4	4.0	< 0.1
		mid-plane	0.2	< 0.1	7.9	1.5
		out-of-plane	1.6	0.7	1.9	1.7

Table 5.3: Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the JES correction, each varying with event-plane orientation bins. They are displayed for $20 < p_{T,\text{jet}} < 40$ GeV/ c in 20-50% central Au+Au collisions for $p_{T,\text{assoc}} \text{range } 1.0-1.5$ and $p_{T,\text{assoc}} \text{range } 3.0-4.0$ bins. The values are expressed as a percent of the nominal value.

Source	Result	Orientation	Uncertainty %			
			Near-side: $p_{T,\text{assoc}}$ (GeV/ c)		Away-side: $p_{T,\text{assoc}}$ (GeV/ c)	
			1.0-1.5	3.0-4.0	1.0-1.5	3.0-4.0
Acceptance shape	Yield	in-plane	35	1.2	22	2.5
		mid-plane	39	1.2	22	2.2
		out-of-plane	59	0.9	37	1.6
	Width	in-plane	30	0.7	8.4	1.9
		mid-plane	22	0.5	15	1.7
		out-of-plane	19	0.4	28	1.2
Background fit	Yield	in-plane	13	1.0	8.2	2.1
		mid-plane	14	0.7	8.0	1.2
		out-of-plane	22	0.9	14	1.5
	Width	in-plane	13	0.1	3.7	0.2
		mid-plane	10	0.1	6.8	0.2
		out-of-plane	8.6	< 0.1	13	0.1
JES correction	Yield	in-plane	0.2	0.4	< 0.1	2.6
		mid-plane	1.1	0.3	2.1	2.9
		out-of-plane	3.1	0.6	1.4	3.5
	Width	in-plane	1.2	0.5	3.3	0.3
		mid-plane	1.2	0.7	1.3	1.5
		out-of-plane	2.2	0.1	0.2	1.9

5.2 Results

Charged particle yields associated with jets are found by integrating the associated yield:

$$\text{Yield} = \frac{1}{N_{\text{trig}}} \int_c^d \int_a^b \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} d\Delta\phi d\Delta\eta. \quad (5.8)$$

The integration limits in $\Delta\phi$ for the near-side are $a = -\pi/3$ and $b = \pi/3$, while for the away-side, we have $a = 2\pi/3$ and $b = 4\pi/3$. The integration limits in $\Delta\eta$ are the same for both the near-side and away-side, $c = -0.6$ and $d = 0.6$.

Shown in Fig. 5.6 is the near-side (left) and away-side (right) associated yields vs $p_{\text{T,assoc}}$ for $15 < p_{\text{T,jet}} < 20$ GeV/ c (top) and $20 < p_{\text{T,jet}} < 40$ GeV/ c (bottom) full jets in 20-50% centrality collisions. The yields are compared for each orientation of the trigger jet reconstructed relative to the event plane (in/mid/out) for $p_{\text{T,assoc}} \in [1.0, 1.5], [1.5, 2.0], [2.0, 3.0], [3.0, 4.0], [4.0, 6.0], [6.0, 10.0]$ GeV/ c .

The main feature is the steeply falling associated yield with increasing $p_{\text{T,assoc}}$, which occurs on both the near- and away-side. Note that associated yields for $p_{\text{T,assoc}} \geq 2.0$ GeV/ c include jet constituents, which leads to the discontinuity at $p_{\text{T,assoc}} = 2$ GeV/ c . Additionally, the use of jets with ‘hard-cores’ and containment of a tower associated to the firing event trigger can lead to a surface bias of the near-side jet. This, however, maximizes the average path length the away-side recoil jets travels, increasing the likelihood of an interaction with the medium. We would expect an in-plane jet and an out-of-plane jet with the same $p_{\text{T,jet}}$ to have different distributions of hard and soft constituents. An in-plane jet with less path travelled in the medium on average would be expected to show higher yields of constituents with higher $p_{\text{T,assoc}}$ than the more quenched out-of-plane jet, which would be expected to show higher yields of constituents with lower $p_{\text{T,assoc}}$. While uncertainties are smaller for jets with $15 < p_{\text{T,jet}} < 20$ GeV/ c , both samples still lack any clear dependence on the event-plane angle. This is an indication that modifications dependent on the average path length are smaller than the experimental uncertainties. On the far right of the near- and

away-side panels is the inclusive $p_{T,\text{assoc}}$ bin with $1.0 < p_{T,\text{assoc}} < 10 \text{ GeV}/c$. The associated yields of each event-plane orientation in the inclusive $p_{T,\text{assoc}}$ selection are consistent with each other for the sample where $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$. However, in the jet sample with $20 < p_{T,\text{jet}} < 40 \text{ GeV}/c$, there are indications suggesting potential modifications. This potential modification is apparent on both the near- and away-side with the largest contributions coming from the lowest $p_{T,\text{assoc}}$ bins.

The widths are calculated by fitting a Gaussian, $Ae^{(\Delta\phi-\Delta\phi_0)^2/2\sigma^2}$, to the jet peak centered at $\Delta\phi_0 = 0$ for the near-side and $\Delta\phi_0 = \pi$ for the away-side. The gaussians are fitted separately, with a range of $|\Delta\phi| < \pi/3$ on the near-side and $|\Delta\phi - \pi| < \pi/3$ on the away-side. The Gaussian fit is repeated with different values of the background parameters and the covariance matrix is used to propagate the uncertainties. The scale uncertainties on the widths are given by $\sigma_w^{sc} = \frac{B}{\sigma_B} \times |\alpha - 1|$, where α is the p_T -dependent scale factor associated with the acceptance shape when propagating the background determined in the $0.6 \leq |\Delta\eta| < 1.2$ to the region $|\Delta\eta| < 0.6$.

From Fig. 5.7, it is clear a broadening of the jet peaks is occurring for decreasing $p_{T,\text{assoc}}$. This is expected from both collisional energy loss and gluon bremsstrahlung. With out-of-plane jets expected to traverse a longer average path length than in-plane jets, this would lead to additional interactions with the medium and more subsequent re-scatterings resulting in a larger width for jets out-of-plane relative to in-plane. Within the uncertainties there is no clear ordering. This indicates the effect of path-length dependent energy loss is not large enough to be seen by the current precision of the data. On the far right of the near- and away-side panels is the inclusive $p_{T,\text{assoc}}$ bin with $p_{T,\text{assoc}} \text{ range } 1.0-10$. The widths of each event-plane orientation in the inclusive selection for the the sample with $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ are consistent. Conversely, the $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ sample reveals indications of potential modifications. This potential modification is apparent on both the near- and away-side and primarily comes from the lowest transverse momentum bins where sample size is the largest.

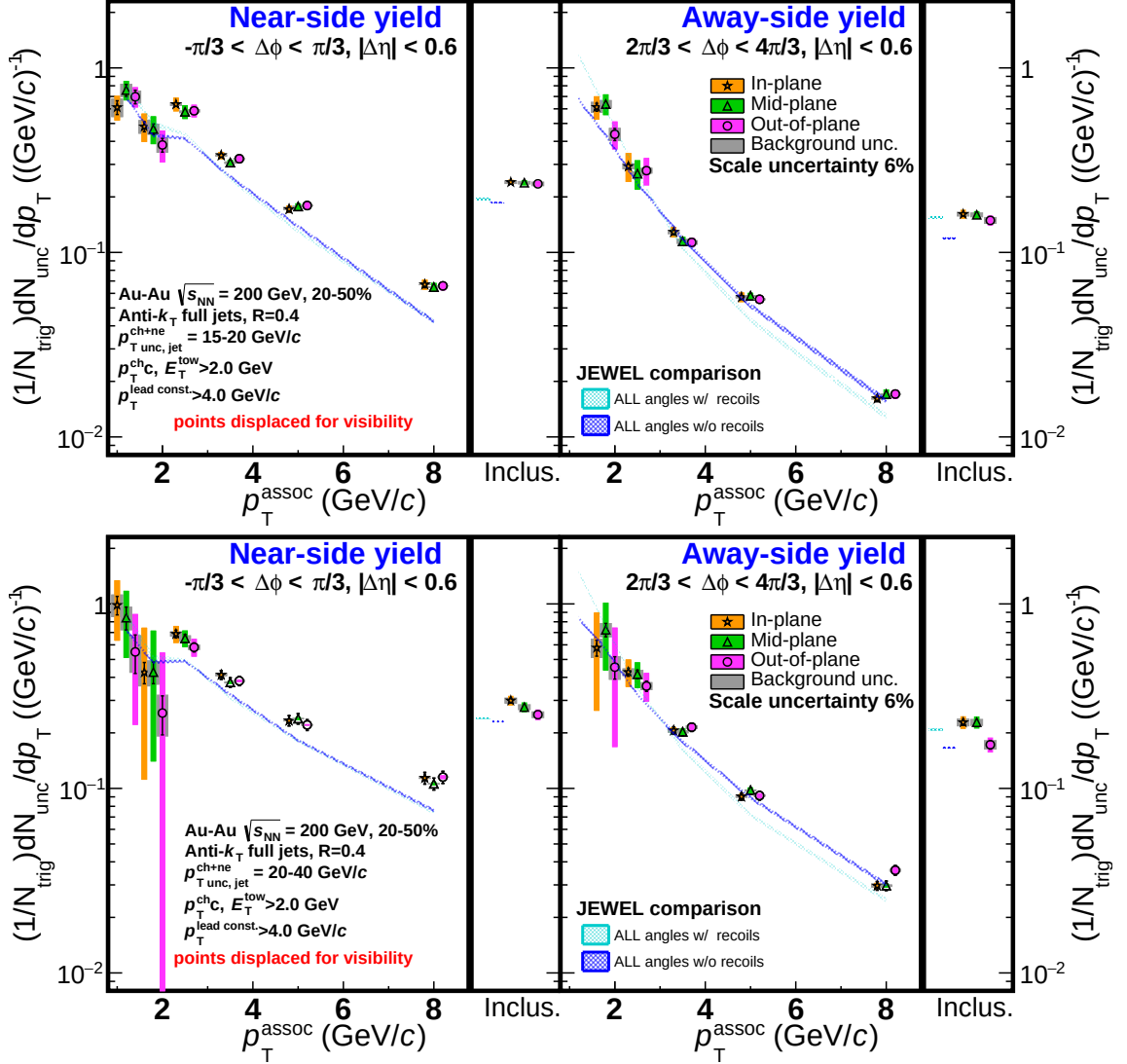


Figure 5.6: Near-side (left) and away-side (right) uncorrected (unc.) associated yield vs $p_{T,assoc}$ for 15-20 (top) and 20-40 GeV/c (bottom) full jets of 20-50% centrality in Au+Au collisions. The grey bands describe the systematic uncertainties of the background fits and are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is the inclusive transverse momentum bin from 1.0-10.0 GeV/c. Points are displaced for visibility.

The measurements presented in Figs. 5.6 and 5.7 are compared to calculations from “Jet Evolution With Energy Loss” known as JEWEL [74], a jet energy loss model based on radiative and collisional energy loss in connection with partons sampled from a longitudinally expanding medium [58]. Model calculations are provided for two regimes, for calculations

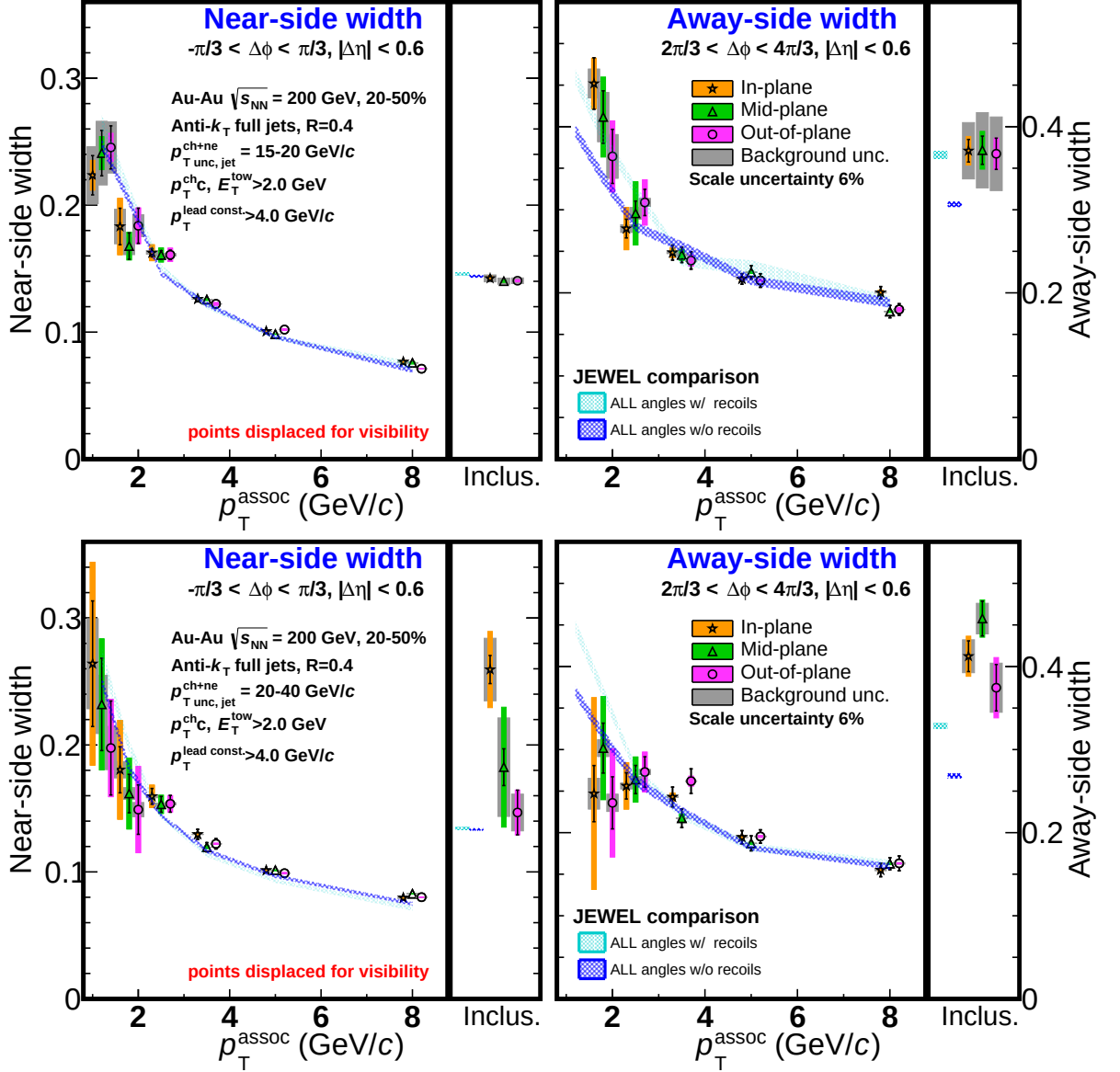


Figure 5.7: Near-side (left) and away-side (right) jet width vs $p_{T,assoc}$ for 15-20 (top) and 20-40 GeV/c (bottom) full jets of 20-50% centrality in Au+Au collisions. The widths are extracted from the gaussian fit to the jet peak. The grey bands describe the systematic uncertainties of the background fits which are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed-event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is an inclusive transverse momentum bin from 1.0-10.0 GeV/c. Points are displaced for visibility.

that a) include recoiled partons, and b) do not include recoiled partons. When no recoils are kept, the lost jet momentum is removed from the entire system. This is useful for modeling

energy loss in the hard part of the jet. When recoils are kept, the jets momentum is fully conserved, but this adds both energy and additional background particles to the di-jet. In an experimental analysis, we likely would measure some, but not all of the recoil particles as they are often indistinguishable from background.

Due to the dominant impact of jet-by-jet fluctuations on partonic energy loss over path-length dependence [75, 76], JEWEL only predicts a very slight event-plane dependence which is well below the systematic uncertainty in the measurement. Variations among event-plane orientations were not seen at the 10% level. This is therefore consistent with path-length dependence having an insignificant impact compared to jet-by-jet fluctuations in energy loss. Fluctuations in the density of the medium may also suppress observable path-length dependence and are not included in the JEWEL model. However, higher precision JEWEL calculations may be needed to discern any potential event-plane dependent effects. We thus show the JEWEL comparisons corresponding to the sample integrated over all angles relative to the event plane and compare to the results from data. Comparisons show that the away-side is well described in terms of both the associated yields and widths by including recoils at low $p_{T,assoc}$, while at high $p_{T,assoc}$, the yields are better described by not including recoils and the widths have similar results to within uncertainties for both cases. When looking at the near-side, the widths are quite similar at high $p_{T,assoc}$ with slightly larger values when including recoils at low $p_{T,assoc}$. For the associated yields at high $p_{T,assoc}$, both JEWEL cases are similar to each other, but underestimate the data and including recoils has larger yields that better match the data at low $p_{T,assoc}$.

To better quantify and examine the event-plane dependence of the yields, ratios were taken of mid-plane yields relative to in-plane yields and out-of-plane yields relative to in-plane yields. The advantage of taking ratios is a reduction in systematic uncertainties due to cancellation of uncertainties from several sources. The propagation of uncertainties was done similarly to that of Sect. 5.1.4. The yield ratio is expressed as:

$$r = \frac{Y_A}{Y_B} = \frac{Y_A^{\text{meas}} - Y_A^{\text{bkgd}}}{Y_B^{\text{meas}} - Y_B^{\text{bkgd}}}, \quad (5.9)$$

where A and B denote different event-plane orientations of the yield. The statistical errors, coming only from the terms, Y_A^{meas} and Y_B^{meas} , which were completely uncorrelated were calculated as:

$$\sigma_r^{\text{stat}} = \left| \frac{1}{r} \right| \sqrt{\left(\frac{\sigma_A}{Y_A} \right)^2 + \left(\frac{\sigma_B}{Y_B} \right)^2}. \quad (5.10)$$

The scale uncertainties, displayed as colored bands on the yield and width plots, are correlated and completely cancel in the ratio. Uncertainties from the RPF background subtraction are propagated using the covariance matrix from the background fit (σ_{ij}), but now on Eq. 5.9, which includes correlated background equations in the numerator and denominator of the ratio. The correlated background uncertainties are given by:

$$\sigma_r^{\text{bkgd}} = \sqrt{\sum_{i=0}^N \sum_{j=0}^N \frac{\partial r}{\partial p_i} \frac{\partial r}{\partial p_j} \sigma_{ij}}. \quad (5.11)$$

Where, p_i 's are the parameters of the RPF fits. Fig. 5.8 shows the near-side (left) and away-side (right) associated-yield ratios of out-of-plane/in-plane and mid-plane/in-plane for 15-20 (top) and 20-40 GeV/c (bottom) jets.

For $15 < p_{\text{T,jet}} < 20$ GeV/c, the out-of-plane to in-plane associated-yield ratio shows slight enhancement out-of-plane relative to in-plane at low- $p_{\text{T,assoc}}$, although the effect is small. This can potentially be due to additional induced gluon radiation that out-of-plane and mid-plane jets would experience relative to in-plane jets, possibly from the longer path length traversed by jets that are not in-plane. Deviations of the yield ratios from 1.0 are not statistically significant on the away-side, although a small suppression is seen in-plane relative to both mid-plane and out-of plane on both the near-side and the away-side for $p_{\text{T,assoc}} \text{range } 2.05.0$ for out/in. The suppression is expected to occur at a higher momentum

fraction (z) of the jet. On the away-side, the effects are favoring a redistribution of energy from high-momentum constituents to lower momentum constituents.

For $20 < p_{T,\text{jet}} < 40$ GeV/ c the near-side ratios are consistent with 1.0 with some movement at the two lowest $p_{T,\text{assoc}}$ bins. On the away-side, mid/in is consistent with 1.0 with a little enhancement and out/in has an enhancement at high- $p_{T,\text{assoc}}$ and suppression at low- $p_{T,\text{assoc}}$. This observation contradicts the expectations. If in-plane jets interact less, we expect ratios to be < 1.0 at high- $p_{T,\text{assoc}}$ and > 1.0 at low- $p_{T,\text{assoc}}$. This is a reminder of the competing effects in the analyzed momentum range, and it is an indication that the expected path length effects due to jet energy loss are dominated by the fluctuations in the medium.

Alternatively, the initial geometry within specific centrality bins may be fluctuating at a larger magnitude than any possible event-plane dependence [77, 78]. The possibility of this occurrence can be studied by looking at the initial configuration and selecting low and high ellipticity events by using the q_n flow vector found within a selected centrality range.

To study the impact of surface bias and event-plane resolution, a check was performed to investigate the systematic change in the ratio of yields (out/in, mid/in) with the angle of the event plane by fitting a constant to Fig. 5.8 for jets with $15 < p_{T,\text{jet}} < 20$ GeV/ c (top) and $20 < p_{T,\text{jet}} < 40$ GeV/ c (bottom). The systematic uncertainties are treated as uncorrelated point-to-point and added to the statistical uncertainties in quadrature. The results are shown in Tabs. 5.4 and 5.5 and are consistent with one. Care should be taken when interpreting the results, as various effects can be in play and medium modifications could give way to a $p_{T,\text{assoc}}$ dependence. This $p_{T,\text{assoc}}$ dependence can be seen as an enhancement of associated yields for $p_{T,\text{assoc}} \geq 2$ GeV/ c on the near-side from the geometric kinematic biases due to the “hard-core” requirement of jet reconstruction.

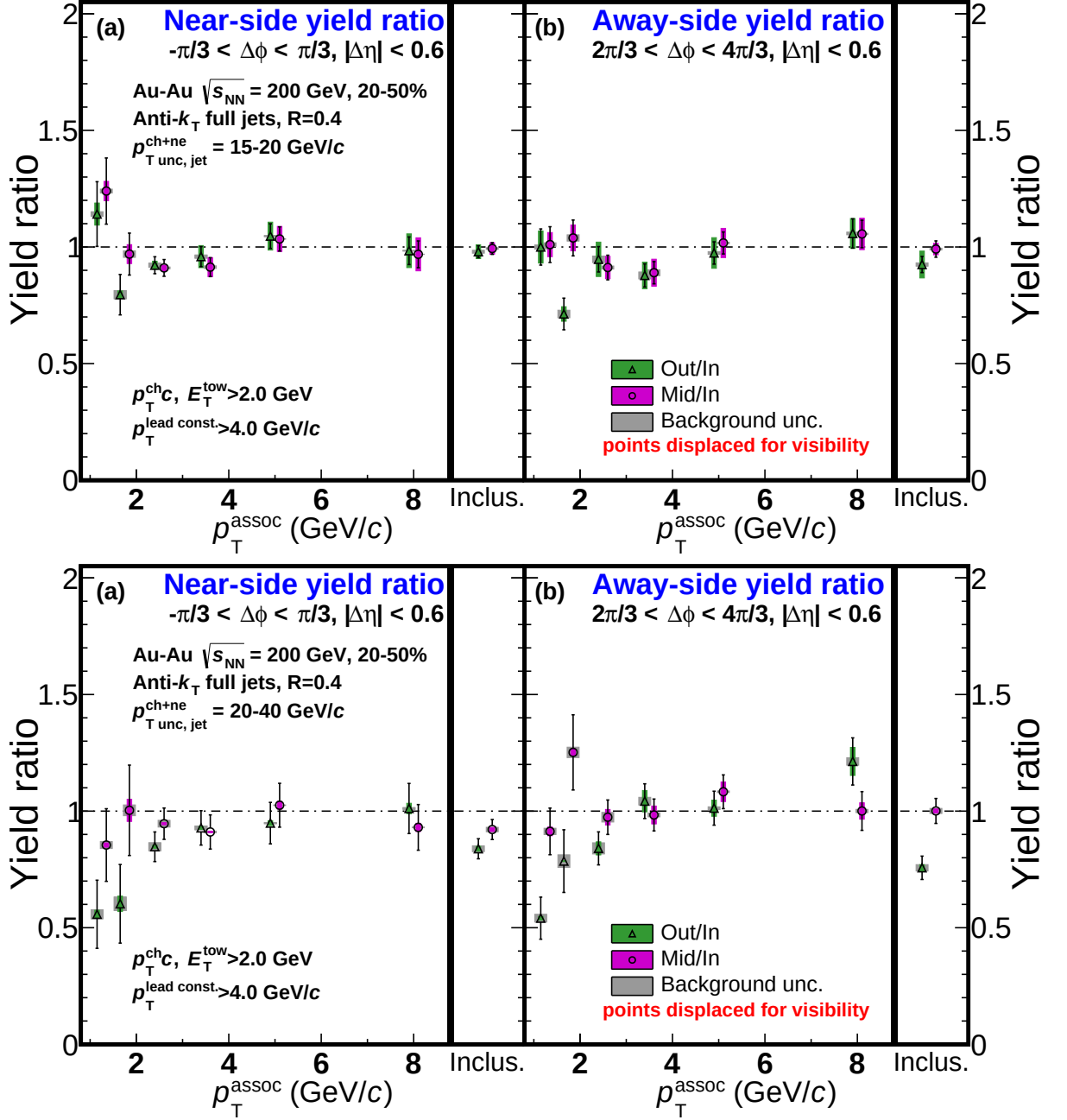


Figure 5.8: Near-side (left) and away-side (right) associated-yield ratios (of out-of-plane and mid-plane to in-plane) vs $p_{T,assoc}$ for 15-20 (top) and 20-40 (bottom) GeV/c full jets in 20-50% centrality collisions. The grey bands describe the systematic uncertainties of the background fits which are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the JES correction. Points are displaced for visibility.

5.3 Conclusions

The measurement of jet-hadron correlations relative to the event plane is reported for the 20-50% most central events in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV in STAR. Partonic

Table 5.4: Results of fits to Fig.5.8 (top panel: 15-20 GeV/ c jets) to a constant c , the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).

parameter	Near-side		Away-side	
	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$
c	0.93 ± 0.042	0.949 ± 0.038	0.89 ± 0.065	0.997 ± 0.066
χ^2/NDF	0.68	0.67	0.71	0.18
σ	-1.6	-1.3	-1.7	-0.1
90% CL	$0.86 - 1.00$	$0.89 - 1.01$	$0.78 - 1.00$	$0.94 - 1.05$

Table 5.5: Results of fits to Fig.5.8 (bottom panel: 20-40 GeV/ c jets) to a constant c , the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).

parameter	Near-side		Away-side	
	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$
c	0.874 ± 0.043	0.937 ± 0.043	0.752 ± 0.064	1.02 ± 0.075
χ^2/NDF	1.4	0.19	2.1	0.49
σ	-2.9	-1.5	-3.9	0.2
90% CL	$0.77 - 0.98$	$0.90 - 0.97$	$0.56 - 0.94$	$0.91 - 1.12$

interactions are directly related to the distance traversed in the medium, so it is expected that medium-induced jet modifications should depend on the path length. The angle of the jet, measured with respect to the event plane, is correlated on average with the jet's path length through the medium. In this analysis, the average path length of away-side jets is potentially increased due to the surface bias of the near-side trigger jet. This work utilizes the RPF background-subtraction method to remove the event-plane dependent background while reducing uncertainties and assumptions associated with previous background-subtraction techniques. Associated yields, their ratios, and jet-peak widths are extracted for each event-plane orientation and compared with different average path lengths and JEWEL model calculations. JEWEL performs better in describing the associated yields and widths at higher $p_{\text{T,assoc}}$ when recoil partons are not included. Conversely, including recoil partons leads to JEWEL providing a better description of the lower $p_{\text{T,assoc}}$ region. This study highlights the importance of conducting further tuning of Monte Carlo simulations to accurately describe the results in this analysis for jets that are biased towards hard-fragmented jets due to the HT-trigger and hard-core constituent requirements.

Within the precision of the current measurement at $\sqrt{s_{\text{NN}}} = 200$ GeV, the associated yields and jet-peak widths show no dependence on the event plane. The ratios derived from the associated yields are used to quantify the differences, but they do not deviate significantly from 1.0. For the $20 < p_{\text{T,jet}} < 40$ GeV/ c , there were indications of potential modifications observed in the inclusive bin of $1.0 < p_{\text{T,assoc}} < 10$ GeV/ c . The results presented in this study align with the findings observed in hadron-hadron and jet-hadron correlations reported in Ref. [73] for RHIC and Ref. [42] for LHC energies. The lack of clear event-plane dependence in our data indicates that any dependence of these modifications on the average path-length is less than our experimental uncertainties.

Chapter 6

Generalized Angularities

Generalized jet angularities are a family of jet observables defined as,

$$\lambda_{\beta}^{\kappa} = \sum_{\text{const} \in \text{jet}} \left(\frac{p_{\text{T,const}}}{p_{\text{T,jet}}} \right)^{\kappa} \left(\frac{\Delta R_{\text{const,jet}}}{R} \right)^{\beta} \quad (6.1)$$

where $p_{\text{T,jet/const}}$ is the jet/constituent's momentum, and $\Delta R_{\text{const,jet}}$ is the (η, ϕ) distance of a constituent from the jet-axis given by,

$$\Delta R_{\text{const,jet}} = \sqrt{(\eta_{\text{jet}} - \eta_{\text{const}})^2 + (\phi_{\text{jet}} - \phi_{\text{const}})^2} \quad (6.2)$$

Parameters κ and β tune experimental sensitivity to hard and wide-angle radiation, respectively. λ_{β}^1 's are infra-red and collinear (IRC) safe angularities [79], which probe the average angular spread of energy around the jet-axis. They are radial moments of the jet's transverse momentum profile, also known as differential jet-shape ($\pi_T(\mathbf{r})$), given by,

$$\pi_T(\mathbf{r}) = \lim_{\delta r \rightarrow 0} \left\langle \frac{1}{\delta r} \frac{\sum_{|\mathbf{r}_{\text{const}} - \mathbf{r}| < \delta r/2} p_{\text{T,const}}}{p_{\text{T,jet}}} \right\rangle_{\text{jets}}, \quad (6.3)$$

where $\mathbf{r}_{\text{const}} = (\eta_{\text{const}} - \eta_{\text{jet}})\hat{\eta} + (\phi_{\text{const}} - \phi_{\text{jet}})\hat{\phi}$, and it follows that,

$$\lambda_{\beta}^1 = \int_{\text{jet}} r^{\beta} \pi_T(\mathbf{r}) d\mathbf{r}. \quad (6.4)$$

However, as an experimental observable, it is useful to restrict the summation in Eq. 6.1 to only charged constituents / tracks and mitigate contamination from towers like pileup.

This comes at the cost of IRC safety of λ_β^1 as theoretical predictions will involve non-perturbative fragmentation functions.

6.1 Angularities of $p + p$ jets

The analysis uses data from $p + p$ collisions at $\sqrt{s} = 200$ GeV collected in 2012 using the STAR detector system. Charged-particle tracks and neutral energy depositions (towers) are measured using STAR's Time Projection Chamber (TPC) [6] and Barrel Electromagnetic Calorimeter (BEMC) [24] detectors respectively.

To enhance jet signal, only High-Tower (HT) triggered events, with at least one tower with $E_{T,\text{tower}} \geq 4$ GeV are considered. Events containing a track with $p_{T,\text{track}} \geq 30$ GeV/ c are rejected to avoid contamination from cosmic rays. Tracks(Towers) with $p_{T,\text{track}} \geq 0.2$ GeV/ c ($E_{T,\text{tower}} \geq 0.2$ GeV) are considered for jet clustering. Clustered jets completely falling within acceptance ($|\eta_{\text{jet}}| \leq 0.6$), with $p_{T,\text{jet}} \geq 10$ GeV/ c and a plurality of charged tracks ($N_{\text{ch},\text{jet}} \geq 2$) are used for further study.

The tracks and towers are clustered into jets using the anti- k_T algorithm with a jet resolution parameter $R = 0.4$, implemented using the FastJet library [80]. After obtaining the jets, the softer constituents are groomed using Recursive SoftDrop with $\beta = 0$ and $z_{\text{cut}} = 0.2$. The groomed-jet's constituents lose non-kinematic information like charge, which is regained by matching them to the original jet's constituents.

6.1.1 Detector effects and Unfolding

The embedding simulation starts from PYTHIA6 (STAR Tune) at the particle level. The particles are then reconstructed as detector hits using the STAR detector system implemented in GEANT3 and embedded into zero-bias events from 200 GeV $p + p$ collisions (2012). The jets from particle and detector levels are matched using a $p_{T,\text{jet}}$ ordered nearness criteria. The matched jets can be used to show detector responses like shown in Fig. 6.1.1 and $p_{T,\text{jet}}$ -resolution shown in fig. 6.1.1.

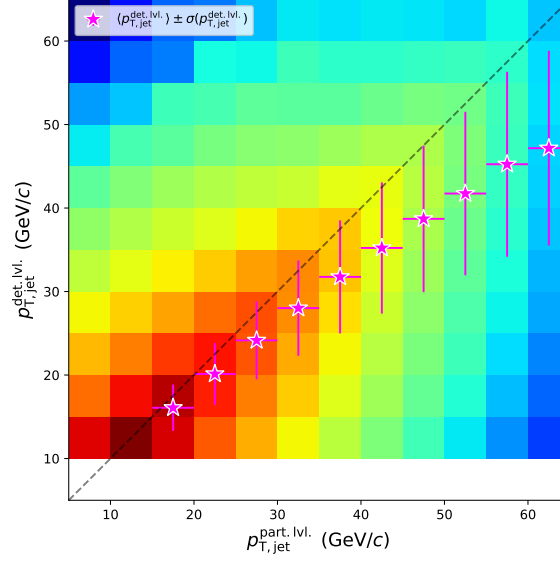
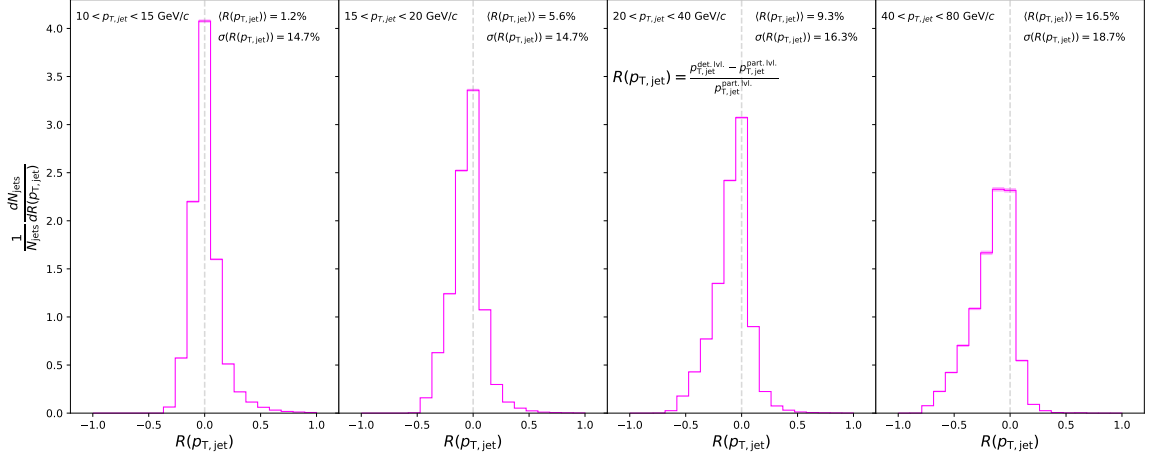
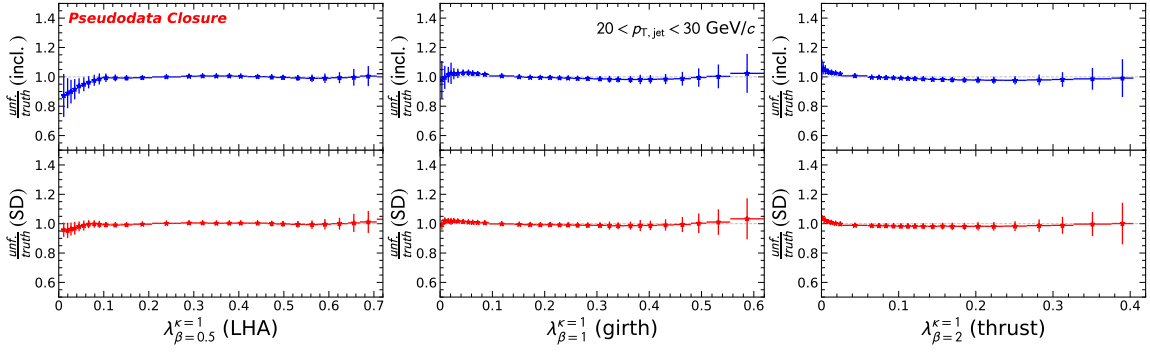


Figure 6.1: Two-dimensional histogram showing detector response for $p_{T,jet}$, showing probability of a $p_{T,jet}^{GEN}$ particle level jet being reconstructed as $p_{T,jet}^{RECO}$. Purple stars shows expected $p_{T,jet}^{RECO}$ given a particular bin of the $p_{T,jet}^{GEN}$.

Unfolding is done using the Multifold method. Every jet is represented as a vector of 23 features. The four classifiers of Multifold are parametrized as DNNs with 3 hidden layers with 256 nodes each. The model weights are trained using the AdamW optimizer with a learning rate of 10^{-3} and weight decay of 10^{-2} . Convergence is achieved in two unfolding iterations. Statistical uncertainties are calculated using bootstrap ensemble with 50 model replicas each sub-sampling 500k each of the data-like and sim-like jets, so each model replica sees 1M jets every epoch. All the required unfolded distributions and profiles are calculated by using the per-jet unfolding weights.

Closure test of the unfolding shown in Fig. 6.1.1 and 6.1.1 is done by doing the same unfolding procedure replacing measured data with pseudo-data for which the particle level is known (pseudo-truth). The pseudo-unfolded distributions are then compared to the pseudo-truth via ratios

Figure 6.2: $p_{T,jet}$ -resolution in different $p_{T,jet}$ bins.Figure 6.3: Closures for $\lambda_{\beta \in \{0.5, 1, 2\}}^{\kappa=1}$ distributions.

6.1.2 Systematic Uncertainties

Jet Reconstruction Related Uncertainties

Analysis was rerun with each of the following changes (compared to the nominal) done to the detector level of the embedding simulation. The differences in the final plots for the source was calculated. The Barlow test was performed on each source of systematic error to remove the statistical fluctuation from the variation.

- Tracking Efficiency - The uncertainty in tracking efficiency is estimated at $\pm 4\%$ for $p + p$ events.
- Variation of Hadronic Correction - For the towers, we remove the full contribution of

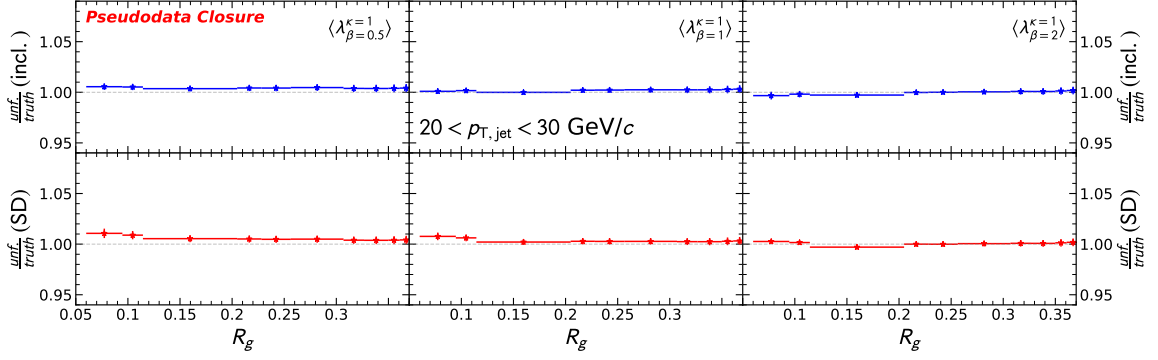


Figure 6.4: Closures for $\lambda_{\beta \in \{0.5,1,2\}}^{\kappa=1}$ vs R_g profile shown for jets with $20 \leq p_{T,\text{jet}} < 40 \text{ GeV}/c$.

the track momentum for our central values. As a variation on that, we remove half of the total track momentum contribution.

- Tower Gain Uncertainty - Tower energies increased by 3.8%
- $p_{T,\text{jet}}$ resolution - Use the $p_{T,\text{jet}}$ resolution values from fig. 6.1.1 to shift the $p_{T,\text{jet}}$ windows in which results are presented.

Unfolding related uncertainties

- Variation of the regularization parameter(number of iterations) - For our central values, we have used 2 iterations as we found the result to converge reliably by that point. We vary the iterations between 3 and 5.
- Prior variation - Varied the prior of the embedding sample from PYTHIA6 to HERWIG7
- Non Closure - Any deviations from 1 for ratios shown in fig. 6.1.1 and fig. 6.1.1

6.1.3 Results and Discussions

The $\lambda_{\beta \in \{0.5,1,2\}}^{\kappa=1}$ distributions are shown in Figs. Grooming shifts the distributions to smaller angularity values. On average higher- β angularities are more affected by grooming, as contributions from early, soft, wide-angle radiation are more significant.

The $\langle\lambda_{\beta}^{\kappa=1}\rangle$ vs R_g plots are shown in figs. $\langle\lambda_{\beta}^{\kappa=1}\rangle$ rises monotonically with R_g , steepening with β . Shows angular ordering as high- R_g jets are unaffected by grooming, since widest split already passes z_{cut} . The distributions and R_g profiles are generally well explained within statistical/systematic uncertainties by PYTHIA-6, PYTHIA-8 and HERWIG-7.

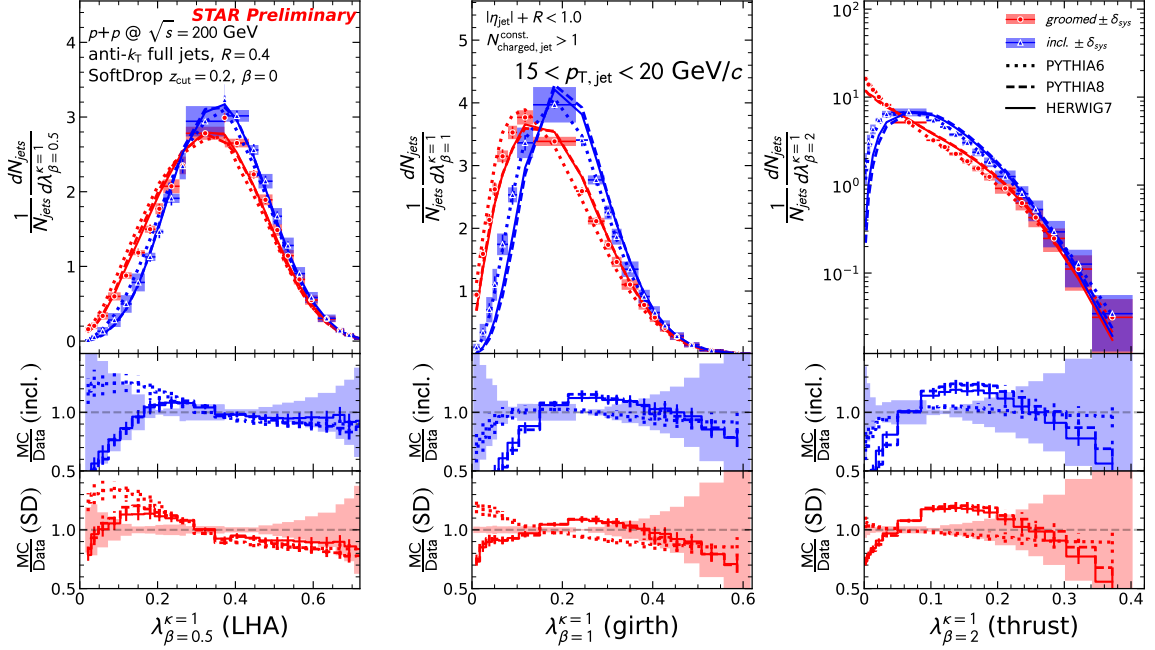


Figure 6.5: Distributions of λ_{β}^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/c.

6.2 Angularities of Au+Au jets

Charged-particle tracks and neutral energy depositions (towers) are measured using STAR’s Time Projection Chamber (TPC) [6] and Barrel Electromagnetic Calorimeter (BEMC) [24] detectors respectively. Together, they provide full azimuthal coverage with a pseudorapidity acceptance of $|\eta| \leq 1$. The tracks and towers are clustered into jets using the anti- k_T algorithm with a jet resolution parameter $R = 0.4$, implemented using the Fast-Jet library [80]. To suppress contributions of fake tracks and combinatorial background (especially in the context of the larger heavy-ion background), a “hard-core” constituent selection is applied, as was done in previous STAR analyses [81], which only allows tracks

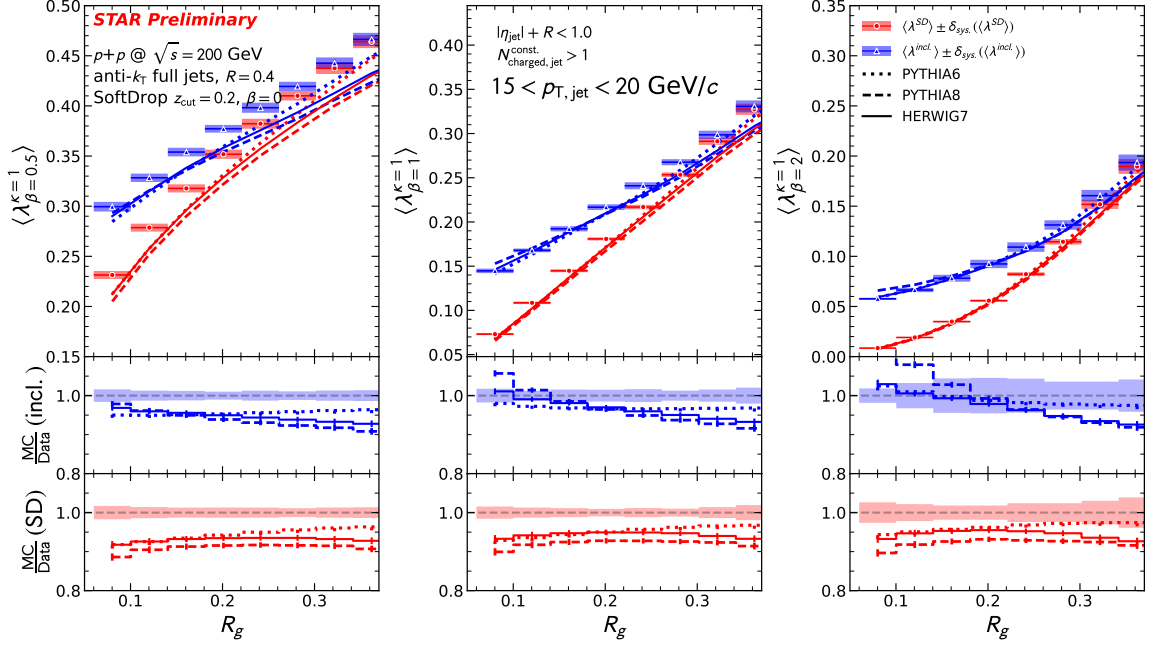


Figure 6.6: Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/c.

(towers) with $cp_{T,\text{trk}}(E_{T,\text{tower}}) \geq 2$ GeV to be clustered into jets. To enhance jet signal, only High-Tower (HT) triggered events, with at least one tower with $E_{T,\text{tower}} \geq 5.4$ GeV are considered. After clustering, only jets completely falling within acceptance ($|\eta_{\text{jet}}| \leq 0.6$) are kept. Jets with area, $A_{\text{jet}} < 0.4$ are rejected to further reduce the fake jet contribution.

Residual background and detector effects are removed by using a mapping between particle and detector level jets from an embedding simulation which involves PYTHIA-6 STAR tune [82] events processed into detector hits using GEANT3 [83] and added to real minimum-bias events from Au+Au collision environment. MultiFold [84] method was used to simultaneously unfold $p_{T,\text{jet}}$, η_{jet} , ϕ_{jet} , $N_{\text{cons,charged}}$, g , p_T^D and LeSub. MultiFold uses Dense Neural Networks (DNNs) implemented using the EnergyFlow package [85], which are trained on full embedding sample at the detector level and the generator level. Closure of the MultiFold is a measure of confidence in the algorithm determined by applying a pre-trained MultiFold on a detector level jet sample for which the particle level is known and comparing the two through ratios. Proper closure is demonstrated in Fig.6.11 for the observables studied in these

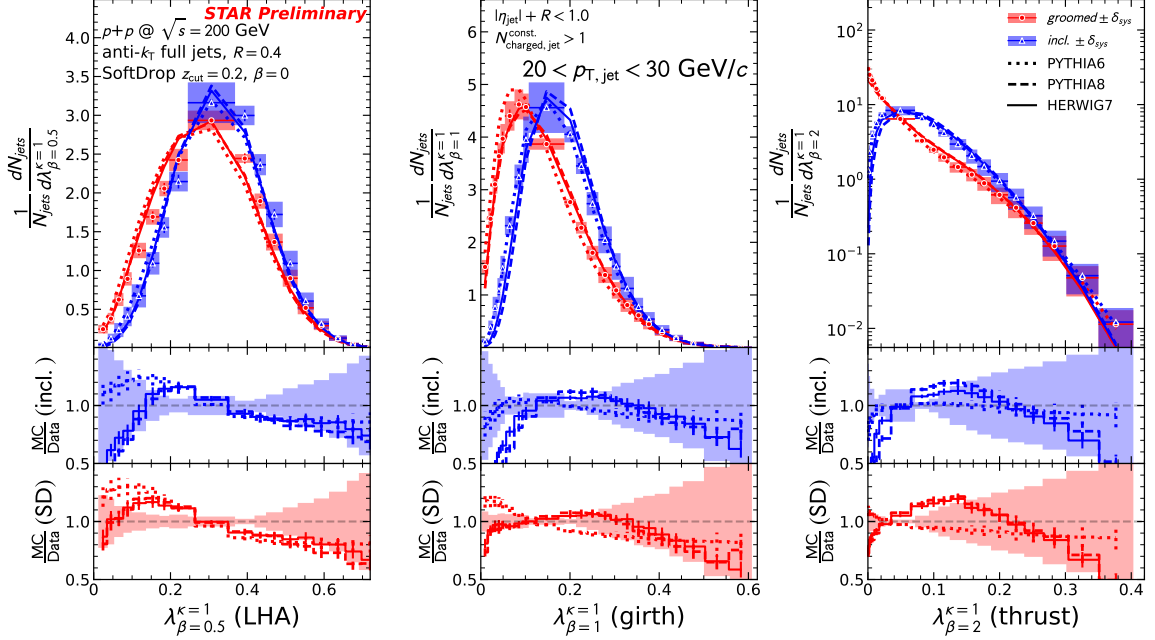


Figure 6.7: Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $20 \leq p_{T,jet} < 30$ GeV/c.

proceedings, with the ratios of distributions being consistent with unity within statistical errors.

6.2.1 Result and Discussion

Fully corrected measurements of g , p_T^D and LeSub distributions calculated from jets with $p_{T,jet} > 20$ GeV/c from events within centrality ranges of 0-20% (central) and 40-80% (peripheral), and ratio comparisons between distributions from central and peripheral events are shown in Fig. 6.12, 6.13, 6.14 respectively. The p_T^D distribution shows a discontinuity at 0.7 due to the strong dependence on number of constituents in jet. Systematic uncertainties were assumed to be uncorrelated, with uncertainties from various sources being added in quadrature. Parameters varied for calculating the uncertainties in this study are the MultiFold regularization parameters of batch size and number of iterations. Any residual non-closure from MultiFold were also included in the calculation of systematic uncertainties. A prior variation uncertainty was added from previous calculations using STAR $\sqrt{s} = 200$

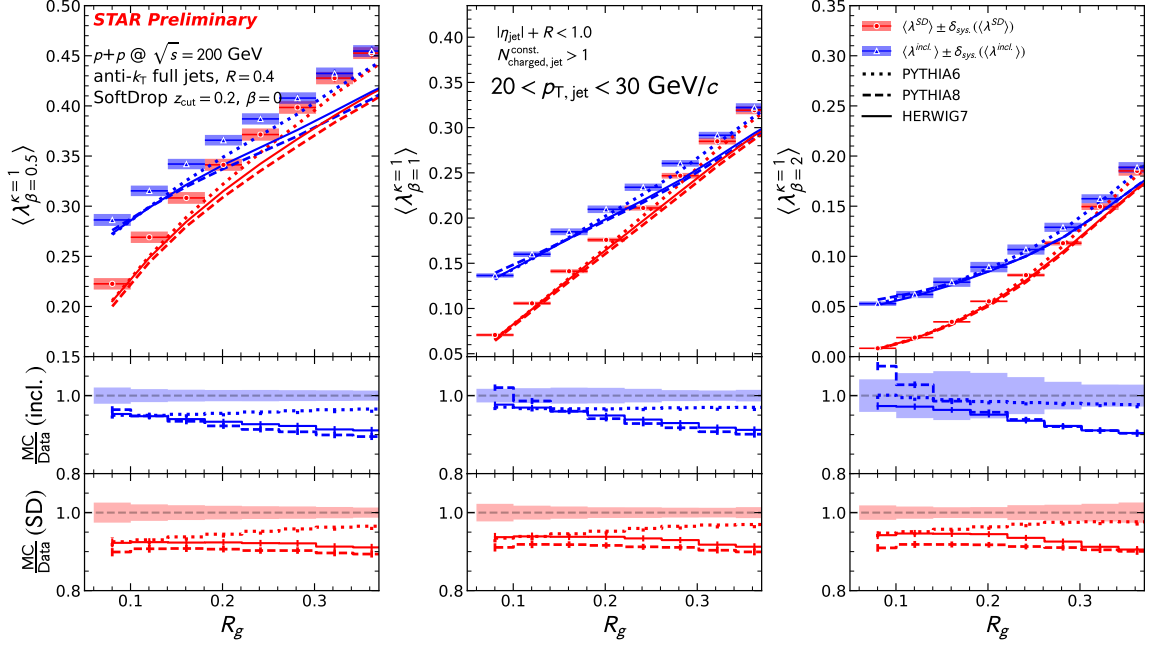


Figure 6.8: Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $20 \leq p_{T,\text{jet}} < 30$ GeV/c.

GeV p+p collision data.

Results are consistent between central and peripheral collisions within conservative systematic uncertainties. This can be attributed to the assumption of uncorrelated systematic uncertainties, and a bias in the analysis towards hard-fragmented jets due to the selection of events and constituents for jet clustering. Harder-fragmented jets, on average are narrower and have lower interaction time with QGP, which could make any relative modification in central events compared to the peripheral ones lower than what would be potentially observed in an inclusive jet analysis.

6.3 Conclusion and Outlook

Inclusive and groomed angularities have been measured for p+p collisions at $\sqrt{s} = 200$ GeV. The distributions, presented in different $p_{T,\text{jet}}$ ranges show generally show good agreement with PYTHIA-6, PYTHIA-8 and HERWIG-7 tuned to the RHIC energies. The R_g profiles conform with expectations from QCD. This measurement will be used as baselines for the

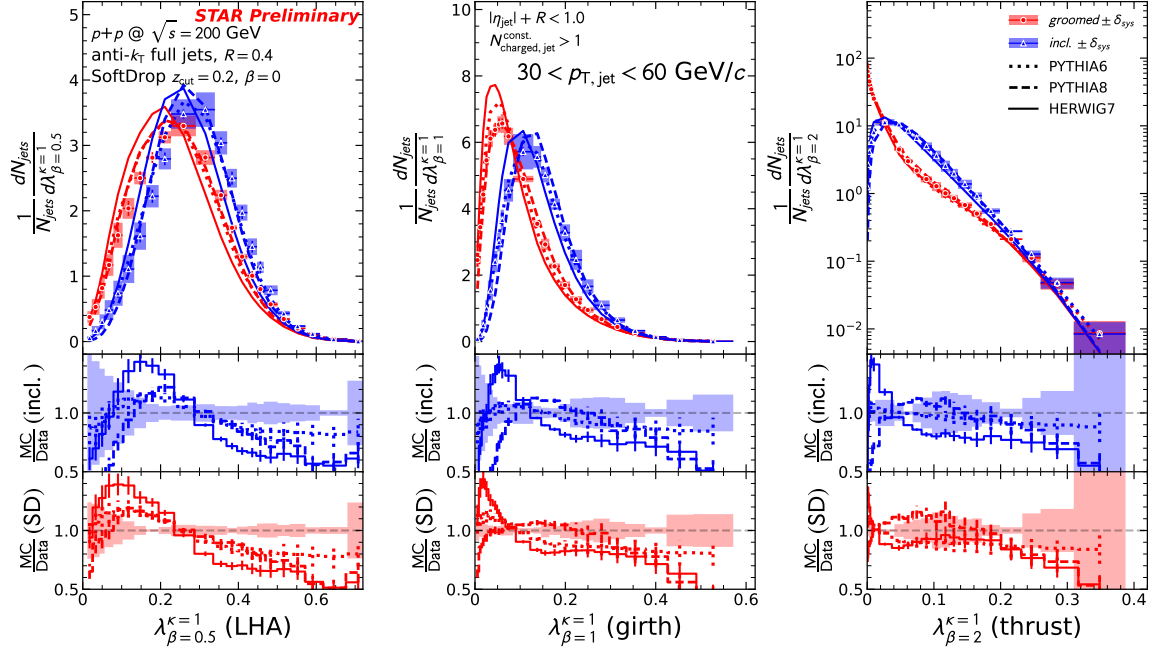


Figure 6.9: Distributions of λ_β^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $30 \leq p_{T,\text{jet}} < 60 \text{ GeV}/c$.

ongoing heavy-ion measurements.

Measurements for $g = \lambda_1^1$, $p_T^D = \sqrt{\lambda_0^2}$ and LeSub done with hard-core jets from Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ show almost no quenching from R_{CP} calculations discussed. However, that could have been due to the crude hard-core selections, introducing a survivor bias, and utilizing grooming going forward might be useful.

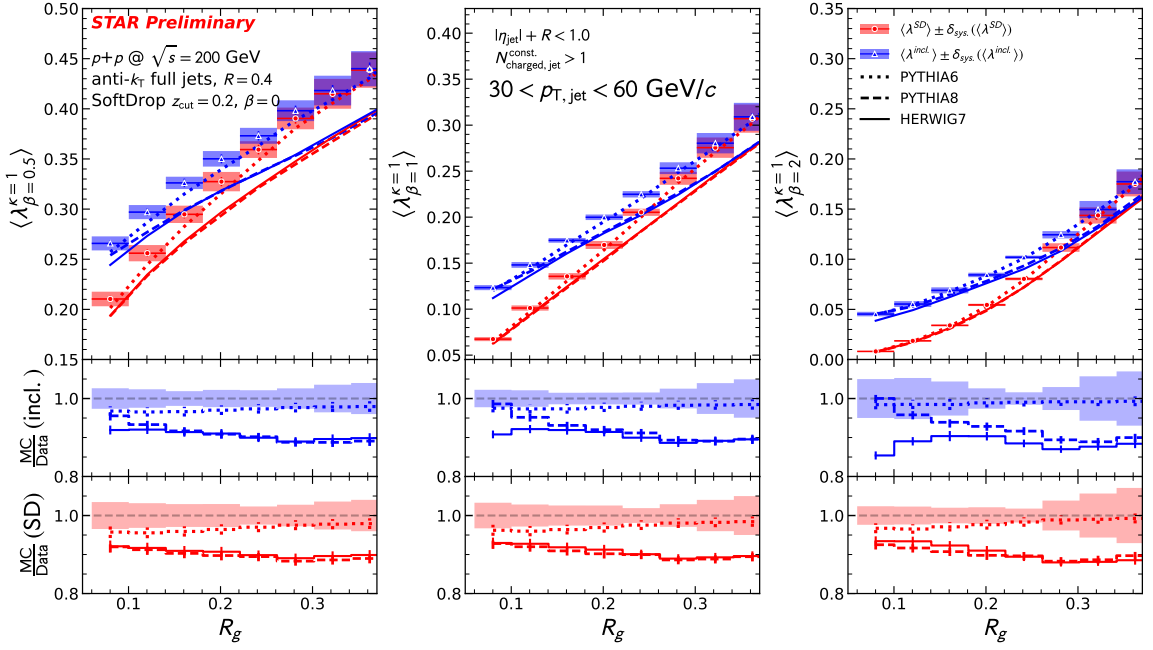


Figure 6.10: Distributions of λ_{β}^1 for $\beta = 0$ (left), $\beta = 1$ (middle) and $\beta = 2$ (right) for jets with $30 \leq p_{T,jet} < 60$ GeV/c.

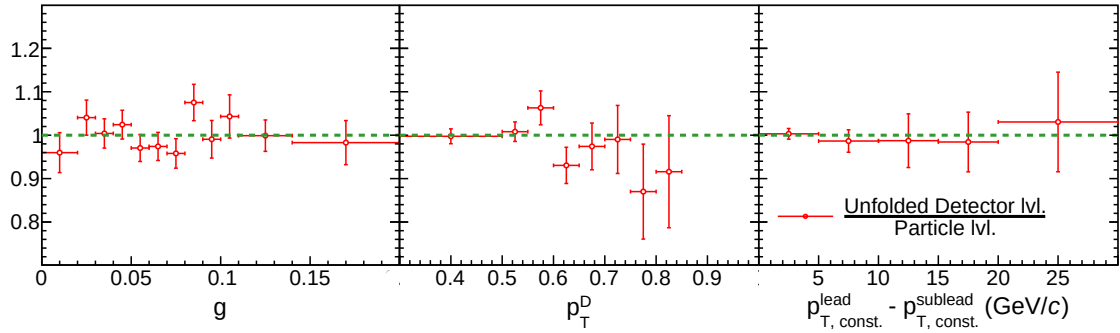


Figure 6.11: Closure test for g , p_T^D and LeSub showing the ratio of distributions calculated using jets from unfolded 0-20% centrality detector level and particle level events from Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV.

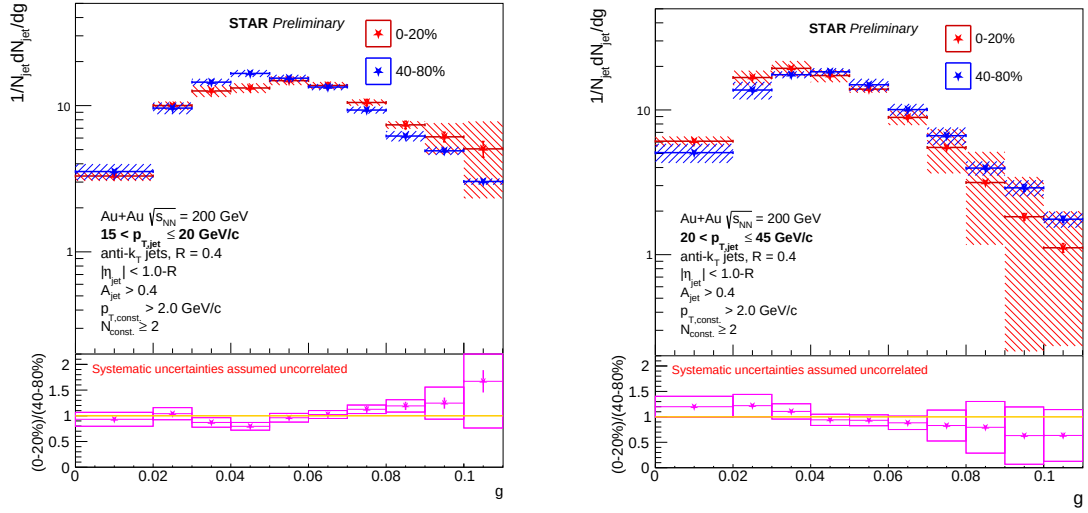


Figure 6.12: Normalized $g = \lambda_1^1$ distributions for jets with $15 \leq p_{T,jet} < 20$ GeV/c (left) and $p_{T,jet} \geq 20$ GeV/c (right) from central (0-20%, red stars) and peripheral (40-80%, blue stars) centrality ranges. Systematic uncertainties are shown as shaded boxes. Central over peripheral ratios (magenta stars) are in the lower panels with their systematic uncertainties represented as magenta boxes. Systematic uncertainties are assumed to be uncorrelated between the centrality ranges.

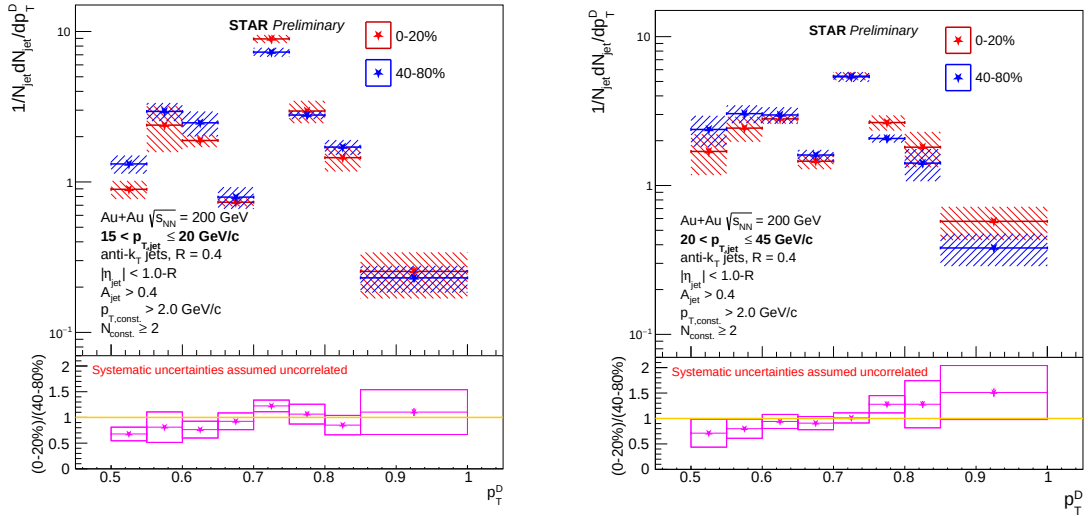


Figure 6.13: Normalized $p_T^D = \sqrt{\lambda_0^2}$ distributions for jets with $15 \leq p_{T,jet} < 20$ GeV/c (left) and $p_{T,jet} \geq 20$ GeV/c (right) from central (0-20%, red stars) and peripheral (40-80%, blue stars) centrality ranges. Systematic uncertainties are shown as shaded boxes. Central over peripheral ratios (magenta stars) are in the lower panels with their systematic uncertainties represented as magenta boxes. Systematic uncertainties are assumed to be uncorrelated between the centrality ranges.

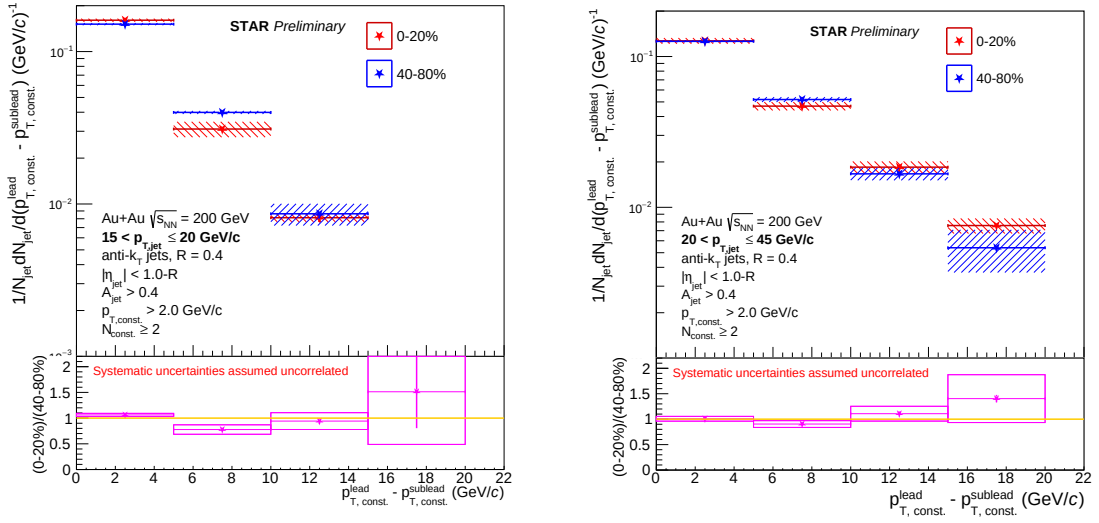


Figure 6.14: Normalized LeSub distributions for jets with $15 \leq p_{T,\text{jet}} < 20 \text{ GeV}/c$ (left) and $p_{T,\text{jet}} \geq 20 \text{ GeV}/c$ (right) from central (0-20%, red stars) and peripheral (40-80%, blue stars) centrality ranges. Systematic uncertainties are shown as shaded boxes. Central over peripheral ratios (magenta stars) are in the lower panels with their systematic uncertainties represented as magenta boxes. Systematic uncertainties are assumed to be uncorrelated between the centrality ranges.

Chapter 7

Summary

This thesis is about jets, the angular structure of the energy inside them, and what that structure can tell us about QCD in vacuum and in QGP. Ch. 1 set up the theory needed to talk about them. The running of α_s gives asymptotic freedom on one end and color confinement on the other, and the confinement scale Λ separates the perturbative hard process from the non-perturbative hadronization. The evolution of outgoing partons in between is the parton shower, whose collinear enhancements are captured by the DGLAP splitting functions and Sudakov form factors of sec. 1.4.1. The soft, wide-angle enhancement is captured by the antenna pattern, which after azimuthal averaging confines soft radiation from a parton to a cone extending up to its color partner, i.e. *angular ordering*. This makes the opening angle (rather than the virtuality) the natural evolution variable of the shower (sec. 1.4.4), and is the reason why the angular distribution of energy in a jet is an imprint of the shower's history. The observables studied in ch. 6 are radial moments of exactly this distribution.

Ch. 2 described RHIC, from the ion sources to the storage rings, and the STAR detector. All the analyses presented here use the TPC for charged tracks and the BEMC for neutral energy and triggering, together giving full azimuthal coverage for $|\eta| < 1$. Ch. 3 then established the vocabulary of the analyses: jets as defined by IRC-safe sequential recombination algorithms (sec. 3.1), SoftDrop grooming as a way of separating the pQCD and npQCD parts of a jet, the Glauber picture of the initial state, centrality as the experimental proxy for the impact parameter (sec. 3.2.3), anisotropic flow and the event plane (sec. 3.2.4), and jet quenching along with the R_{AA} and R_{CP} used to quantify it. The di-hadron and

R_{CP} measurements reviewed there are what motivate going from high- p_{T} hadrons to fully reconstructed jets and their substructure.

Ch. 4 collected the data analysis techniques common to the analyses. Two classification studies were shown as illustrations: a CNN acting on jet images separates quark and gluon jets in $p + p$ collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV, and a DNN acting on a vector of substructure observables separates PYTHIA-6 jets from JEWEL jets with an AUC that decreases going from central to peripheral collisions. The main part of the chapter is a general framework for unfolding, starting from Bayes theorem on truth and measured distributions and the assumption that the detector response $R(\mathbf{y}, \mathbf{x})$ is transferable from simulation. Each iteration in eq. 4.5 is written in terms of two ratios, ω at the measured level and ν at the particle level, and the convolutions with $R(\mathbf{y}, \mathbf{x})$ which map $\omega(\mathbf{y}) \rightarrow \omega(\mathbf{x})$ and $\nu(\mathbf{x}) \rightarrow \nu(\mathbf{y})$. Binning this gives Iterative Bayesian Unfolding and replacing the ratios with classifier outputs gives MultiFold, with misses and fakes handled by evaluating the same maps at the unmatched jets. Written this way, the binned and unbinned methods are the same algorithm, differing only in how the ratios are estimated.

Ch. 5 presented jet-hadron correlations relative to the event plane in 20-50% central Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV. Since the angle of a jet to the event plane is correlated on average with its path length through the medium, medium-induced modifications were looked for as differences between in-, mid- and out-of-plane trigger jets. The flow modulated combinatorial background was removed with the RPF method, which fits the background dominated region at large $\Delta\eta$ simultaneously for the three orientations. Associated yields, their ratios and jet-peak widths were extracted for $15 < p_{\text{T,jet}} < 20$ GeV/ c and $20 < p_{\text{T,jet}} < 40$ GeV/ c jets. Within the precision of the measurement, neither the yields nor the widths show a dependence on the event plane, and the yield ratios are consistent with 1.0, with indications of potential modification only in the inclusive $1.0 < p_{\text{T,assoc}} < 10$ GeV/ c bin of the higher $p_{\text{T,jet}}$ sample. JEWEL predicts an event-plane dependence well below the systematic uncertainties, consistent with path-length dependence of energy loss

being dominated by jet-by-jet fluctuations, and describes the low- $p_{T,\text{assoc}}$ region better with recoils and the high- $p_{T,\text{assoc}}$ region better without.

Ch. 6 presented the measurements of generalized angularities λ_β^κ . In $p + p$ collisions at $\sqrt{s} = 200$ GeV, inclusive and SoftDrop groomed $\lambda_{\beta \in \{0.5, 1, 2\}}^{\kappa=1}$ distributions and their profiles against R_g were fully corrected using MultiFold with 23 features per jet, converging in two iterations. Grooming shifts the distributions to smaller angularities, more so for higher β where early, soft, wide-angle radiation contributes more. $\langle \lambda_\beta^{\kappa=1} \rangle$ rises monotonically with R_g , steepening with β , and high- R_g jets are unaffected by grooming because their widest split already passes z_{cut} , which is angular ordering seen in data. PYTHIA-6, PYTHIA-8 and HERWIG-7 tuned to RHIC energies describe the distributions and profiles within uncertainties, so these measurements serve as the baselines for the heavy-ion measurements. In Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV, $g = \lambda_1^1$, $p_T^D = \sqrt{\lambda_0^2}$ and LeSub were unfolded with MultiFold for hard-core jets in 0-20% and 40-80% central events, the first use of the method on heavy-ion data. Central to peripheral ratios are consistent with unity within conservative systematic uncertainties. Together with the jet-hadron results, this points to the same conclusion: the HT-trigger and hard-core constituent selections bias both analyses towards hard-fragmented, narrow jets which interact the least with the medium, and a survivor bias is the likely reason why the expected path-length and centrality dependencies are not seen.

7.1 Outlook

The natural next step is to redo the Au+Au angularity measurement with the inclusive constituent selection and grooming used in the $p + p$ analysis, so that the $p + p$ results can serve as the baseline they were meant to be and the R_g profiles can separate the pQCD and npQCD parts of the modification. The unfolding framework of ch. 4 was written with this in mind, as the larger heavy-ion background makes misses and fakes far from negligible. On the jet-hadron side, selecting on the initial state ellipticity through the q_n flow vector

within a centrality class would test whether fluctuations in the initial geometry are what washes out the event-plane dependence. Finally, the quark-gluon and quenched-unquenched classifiers of ch. 4 suggest that the amount of modification can itself be quantified as a discrimination problem, which would extend naturally to the angularities once the groomed heavy-ion measurement is available.

Bibliography

- [1] F. Takahashi et al. Review of Particle Physics. *Int. J. Mod. Phys. A*, 41:2630011, 2026.
- [2] Karen McNulty Walsh. 2024.
- [3] Wolfram Fischer. 2026.
- [4] Karen McNulty Walsh. 2026.
- [5] Nuclear science and technology, star detector. 2024.
- [6] M. Anderson et al. The STAR time projection chamber: a unique tool for studying high multiplicity events at RHIC. *Nuc. Inst. and Methods in Physics Research*, 499(2-3):659–678, mar 2003.
- [7] Hans Bichsel. A method to improve tracking and particle identification in tpcs and silicon detectors. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 562(1):154 – 197, 2006.
- [8] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. The anti-kt jet clustering algorithm. *Journal of High Energy Physics*, 2008(04):063, apr 2008.
- [9] Megan Connors, Christine Nattrass, Rosi Reed, and Sevil Salur. Jet measurements in heavy ion physics. *Reviews of Modern Physics*, 90(2), 2018.
- [10] M. L. Miller et al. Glauber modeling in high energy nuclear collisions. *Ann. Rev. Nucl. Part. Sci.*, 57:205–243, 2007.
- [11] Miklos Gyulassy and Larry McLerran. New forms of qcd matter discovered at rhic. *Nuclear Physics A*, 750(1):30–63, 2005. Quark-Gluon Plasma. New Discoveries at RHIC: Case for the Strongly Interacting Quark-Gluon Plasma. Contributions from the RBRC Workshop held May 14-15, 2004.
- [12] Paul Sorensen. Evidence from identified particles for active quark and gluon degrees of freedom. *Journal of Physics G: Nuclear and Particle Physics*, 32(12):S135, nov 2006.
- [13] J. Adams et al. Evidence from d + Au measurements for final state suppression of high p_T hadrons in Au+Au collisions at RHIC . *Phys. Rev. Lett.*, 91:072304, 2003.
- [14] M J Tannenbaum. How do quarks and gluons lose energy in the qgp? *Journal of Physics: Conference Series*, 589(1):012019, mar 2015.
- [15] Measurement of inclusive charged-particle jet production in au + au collisions at 200

- gev. *Phys. Rev. C*, 102:054913, Nov 2020.
- [16] J. Adams et al. Transverse-momentum and collision-energy dependence of high- p_T hadron suppression in Au + Au collisions at ultrarelativistic energies. *Phys. Rev. Lett.*, 91:172302, Oct 2003.
 - [17] B. Abelev et al. Measurement of charged jet suppression in pb-pb collisions at $\sqrt{s_{\mathrm{NN}}} = 2.76$ tev. *Journal of High Energy Physics*, 2014(3):13, 2014.
 - [18] G. Aad et al. Measurement of charged-particle spectra in pb+pb collisions at $\sqrt{s_{\mathrm{NN}}} = 2.76$ TeV with the atlas detector at the lhc. *Journal of High Energy Physics*, 2015(9):50, 2015.
 - [19] Patrick T. Komiske, Eric M. Metodiev, and Matthew D. Schwartz. Deep learning in color: towards automated quark/gluon jet discrimination. *Journal of High Energy Physics*, 2017(1), January 2017.
 - [20] Florencia Canelli, Kyle Cormier, Andrew Cudd, Dag Gillberg, Roger G. Huang, Weijie Jin, Sookhyun Lee, Vinicius Mikuni, Laura Miller, Benjamin Nachman, Jingjing Pan, Tanmay Pani, Mariel Pettee, Youqi Song, and Fernando Torales Acosta. A practical guide to unbinned unfolding. *The European Physical Journal C*, 86(2), February 2026.
 - [21] H. Agakishiev et al. Event-plane-dependent dihadron correlations with harmonic v_n subtraction in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV. *Phys. Rev.*, C89(4):041901, 2014.
 - [22] R. K. Ellis, W. J. Stirling, and B. R. Webber. Hadroproduction of jets and photons. In *QCD and Collider Physics*, Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology, pages 237–269. Cambridge University Press, Cambridge, 1996.
 - [23] K.H. Ackermann et al. Star detector overview. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 499(2):624–632, 2003. The Relativistic Heavy Ion Collider Project: RHIC and its Detectors.
 - [24] M. Beddo et al. The STAR barrel electromagnetic calorimeter. *Nuc. Inst. and Methods in Physics Research*, 499(2):725–739, 2003. The Relativistic Heavy Ion Collider Project: RHIC and its Detectors.
 - [25] *Proceedings: 1990 DPF Summer Study on High-energy Physics (Snowmass 1990): Research directions for the decade*, Singapore, 9 1992. World Scientific.
 - [26] Stephen D. Ellis and Davison E. Soper. Successive combination jet algorithm for hadron collisions. *Phys. Rev. D*, 48:3160–3166, Oct 1993.
 - [27] M. Wobisch and T. Wengler. Hadronization corrections to jet cross sections in deep-inelastic scattering, 1999.
 - [28] M. Cacciari, G. P. Salam, and G. Soyez. FastJet User Manual. *Eur.Phys.J.*, C72:1896,

2012.

- [29] Andrew J. Larkoski, Simone Marzani, Gregory Soyez, and Jesse Thaler. Soft drop. *Journal of High Energy Physics*, 2014(5), May 2014.
- [30] A. Adare et al. Transverse energy production and charged-particle multiplicity at midrapidity in various systems from $\sqrt{s_{NN}} = 7.7$ to 200 gev. *Phys. Rev. C*, 93:024901, Feb 2016.
- [31] J. Adam et al. Measurement of transverse energy at midrapidity in pb-pb collisions at $\sqrt{s_{NN}} = 2.76$ tev. *Phys. Rev. C*, 94:034903, Sep 2016.
- [32] A. Bazavov, Tanmoy Bhattacharya, C. DeTar, H.-T. Ding, Steven Gottlieb, Rajan Gupta, P. Hegde, U. M. Heller, F. Karsch, E. Laermann, L. Levkova, Swagato Mukherjee, P. Petreczky, C. Schmidt, C. Schroeder, R. A. Soltz, W. Soeldner, R. Sugar, M. Wagner, and P. Vranas. Equation of state in $(2 + 1)$ -flavor qcd. *Phys. Rev. D*, 90:094503, Nov 2014.
- [33] Frithjof Karsch. *Lattice QCD at High Temperature and Density*, pages 209–249. Springer Berlin Heidelberg, Berlin, Heidelberg, 2002.
- [34] Edmond Iancu, Andrei Leonidov, and Larry McLerran. Nonlinear gluon evolution in the color glass condensate: I. *Nuclear Physics A*, 692(3):583–645, 2001.
- [35] C. Loizides, J. Nagle, and P. Steinberg. Improved version of the phobos glauher monte carlo. *SoftwareX*, 1-2:13–18, 2015.
- [36] Dmitri Kharzeev and Marzia Nardi. Hadron production in nuclear collisions at rhic and high-density qcd. *Physics Letters B*, 507(1):121–128, 2001.
- [37] BB Back, MD Baker, M Ballintijn, DS Barton, RR Betts, AA Bickley, R Bindel, A Budzanowski, W Busza, A Carroll, et al. Collision geometry scaling of au+ au pseudorapidity density from s nn= 19.6 to 200 gev. *Physical Review C—Nuclear Physics*, 70(2):021902, 2004.
- [38] B. Abelev et al. Centrality determination of pb-pb collisions at $\sqrt{s_{NN}} = 2.76$ tev with alice. *Phys. Rev. C*, 88:044909, Oct 2013.
- [39] Ulrich Heinz and Raimond Snellings. Collective flow and viscosity in relativistic heavy-ion collisions. *Annual Review of Nuclear and Particle Science*, 63(1):123–151, October 2013.
- [40] J. Bielcikova et al. Elliptic flow contribution to two particle correlations at different orientations to the reaction plane. *Phys.Rev.*, C69:021901, 2004.
- [41] S. Voloshin and Y. Zhang. Flow study in relativistic nuclear collisions by Fourier expansion of Azimuthal particle distributions. *Z.Phys.*, C70:665–672, 1996.
- [42] S. Acharya et al. Jet-hadron correlations measured relative to the second order event plane in Pb-Pb collisions at $\sqrt{s_{NN}} = 2.76$ TeV. *Phys. Rev. C*, 101:064901, Jun 2020.

- [43] A. M. Poskanzer and S.A. Voloshin. Methods for analyzing anisotropic flow in relativistic nuclear collisions. *Phys.Rev.*, C58:1671–1678, 1998.
- [44] J. Barrette et al. Proton and pion production relative to the reaction plane in Au + Au collisions at AGS energies. *Phys. Rev.*, C56:3254–3264, 1997.
- [45] J. Barrette et al. Energy and charged particle flow in 10.8A GeV/c Au+Au collisions . *Phys. Rev. C*, 55:1420–1430, Mar 1997.
- [46] S. A. Voloshin, A. M. Poskanzer, and R. Snellings. Collective phenomena in non-central nuclear collisions. *Landolt-Bornstein*, 23:293–333, 2010.
- [47] Q. Wang. *Charge Multiplicity Asymmetry Correlation Study Searching for Local Parity Violation at RHIC for STAR collaboration*. PhD thesis, Kansas U., New York, 2012.
- [48] E. Abbas et al. Performance of the ALICE VZERO system. *JINST*, 8:P10016, 2013.
- [49] J. Barrette et al. Energy and charged particle flow in a 10.8A GeV/c Au + Au collisions . *Phys.Rev.*, C55:1420–1430, 1997.
- [50] A. Majumder and M. van Leeuwen. The theory and phenomenology of perturbative qcd based jet quenching. *Progress in Particle and Nuclear Physics*, 66(1):41–92, 2011.
- [51] Guang-You Qin and Xin-Nian Wang. Jet quenching in high-energy heavy-ion collisions. *International Journal of Modern Physics E*, 24(11):1530014, 2024/10/14 2015.
- [52] Karen M. Burke et al. Extracting the jet transport coefficient from jet quenching in high-energy heavy-ion collisions. *Phys. Rev. C*, 90:014909, Jul 2014.
- [53] David d’Enterria. 6.4 jet quenching: Datasheet from landolt-börnstein - group i elementary particles, nuclei and atoms · volume 23: “relativistic heavy ion physics” in springermaterials (https://doi.org/10.1007/978-3-642-01539-7_16). Copyright 2010 Springer-Verlag Berlin Heidelberg.
- [54] K.J. ESKOLA, R. VOGT, and X.-N. WANG. Nuclear overlap functions. *International Journal of Modern Physics A*, 10(20n21):3087–3090, 1995.
- [55] J. Casalderrey-Solana et al. Simultaneous description of hadron and jet suppression in heavy-ion collisions. *Phys. Rev. C*, 99:051901, May 2019.
- [56] S. Chatrchyan et al. Modification of jet shapes in pbbp collisions at $\sqrt{s_{NN}}=2.76$ tev. *Physics Letters B*, 730:243–263, 2014.
- [57] T. Sjöstrand, S. Mrenna, and P. Skands. Pythia 6.4 physics and manual. *Journal of High Energy Physics*, 2006(05):026–026, May 2006.
- [58] Korinna C. Zapp. JEWEL 2.0.0: directions for use. *Eur. Phys. J.*, C74(2):2762, 2014.
- [59] Yue Shi Lai, James Mulligan, Mateusz Płoskoń, and Felix Ringer. The information content of jet quenching and machine learning assisted observable design, 2022.

- [60] R. Brun et al. *GEANT 3: user's guide Geant 3.10, Geant 3.11; rev. version.* CERN, Geneva, 1987.
- [61] Lydia Brenner et al. Comparison of unfolding methods using RooFitUnfold. *International Journal of Modern Physics A*, 35(24):2050145, 2020.
- [62] K. H. Ackermann et al. STAR detector overview. *Nucl. Instrum. Meth.*, A499:624–632, 2003.
- [63] M. Beddo et al. The STAR Barrel Electromagnetic Calorimeter. *Nucl. Instrum. Meth.*, A499:725–739, 2003.
- [64] L. Adamczyk et al. Measurements of jet quenching with semi-inclusive hadron+jet distributions in Au + Au collisions at $\sqrt{s_{NN}} = 200$ gev. *Phys. Rev. C*, 96:024905, Aug 2017.
- [65] W. M. Castilho, W. Qian, Y. Hama, and T. Kodama. Event-plane dependent di-hadron correlations with harmonic v_n subtraction in a hydrodynamic model. *Physics Letters B*, 777:369–373, 2018.
- [66] J. Adams et al. Azimuthal anisotropy in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. *Physical Review C*, 72(1):014904, Jul 2005.
- [67] M. Cacciari, G. P. Salam, and G. Soyez. The Catchment Area of Jets. *JHEP*, 04:005, 2008.
- [68] L. Adamczyk et al. Dijet imbalance measurements in Au + Au and *pp* collisions at $\sqrt{s_{NN}} = 200$ GeV at star. *Phys. Rev. Lett.*, 119:062301, Aug 2017.
- [69] N. Sharma et al. Background subtraction methods for precision measurements of di-hadron and jet-hadron correlations in heavy ion collisions. *Phys. Rev.*, C93(4):044915, 2016.
- [70] G. Aad et al. Measurement of event-plane correlations in $\sqrt{s_{NN}} = 2.76$ TeV lead-lead collisions with the ATLAS detector. *Phys. Rev.*, C90(2):024905, 2014.
- [71] C. Nattrass and T. Todoroki. Event plane dependence of the flow modulated background in dihadron and jet-hadron correlations in heavy ion collisions. *Physical Review C*, 97(5), May 2018.
- [72] C. Nattrass. Re-examining the iconic dihadron correlation measurement demonstrating jet quenching. *Physical Review C*, 97(3), Mar 2018.
- [73] C. Nattrass et al. Disappearance of the Mach Cone in heavy ion collisions. *Phys. Rev.*, C94(1):011901, 2016.
- [74] Korinna Zapp, Gunnar Ingelman, Johan Rathsmann, Johanna Stachel, and Urs Achim Wiedemann. A monte carlo model for ‘jet quenching’. *The European Physical Journal C*, 60(4), mar 2009.

- [75] J. G. Milhano and K. C. Zapp. Origins of the di-jet asymmetry in heavy ion collisions. *Eur. Phys. J.*, C76(5):288, 2016.
- [76] K. C. Zapp. Geometrical aspects of jet quenching in JEWEL. *Phys.Lett.*, B735:157–163, 2014.
- [77] J. Noronha-Hostler. Solving the $R_{AA} \otimes v_2$ puzzle. *Journal of Physics: Conference Series*, 736(1), August 2016.
- [78] J. Noronha-Hostler et al. Event-by-event Hydrodynamics + Jet energy loss: A solution to the $R_{AA} \otimes v_2$ puzzle. *Phys. Rev. Lett.*, 116:252301, Jun 2016.
- [79] Andrew J. Larkoski, Jesse Thaler, and Wouter J. Waalewijn. Gaining (mutual) information about quark/gluon discrimination. *Journal of High Energy Physics*, 2014(11), nov 2014.
- [80] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. FastJet user manual. *The European Physical Journal C*, 72(3), mar 2012.
- [81] M. S. Abdallah et al. Differential measurements of jet substructure and partonic energy loss in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. *Phys. Rev. C*, 105:044906, Apr 2022.
- [82] J. Kevin Adkins. PhD thesis, University of Kentucky, 2019.
- [83] R. Brun et al. GEANT3. 9 1987.
- [84] Anders Andreassen, Patrick T. Komiske, Eric M. Metodiev, Benjamin Nachman, and Jesse Thaler. Omnifold: A method to simultaneously unfold all observables. *Phys. Rev. Lett.*, 124:182001, May 2020.
- [85] Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler. Energy flow polynomials: a complete linear basis for jet substructure. *Journal of High Energy Physics*, 2018(4), April 2018.

Appendix A

Relativistic Kinematics

A.1 Laboratory vs Center-of-mass frames

A typical collision is a two-to-many reaction like:

$$\underbrace{A}_{\text{projectile}} + \underbrace{B}_{\text{target}} \rightarrow C_1 + C_2 + C_3 + \dots, \quad (\text{A.1})$$

In *laboratory frame*, the target is at rest, so,

$$\mathbf{p}_B^{\text{lab}} = \vec{0} \implies p_B^{\text{lab}} = (m_B, 0, 0, 0), \quad (\text{A.2})$$

In *center-of-mass* frame, the sum of three-momentum vectors vanish in the initial and final states

$$\mathbf{p}_A^{\text{cm}} + \mathbf{p}_B^{\text{cm}} = \sum_i \mathbf{p}_{C,i}^{\text{cm}} = \vec{0}, \quad (\text{A.3})$$

Using Lorentz invariance of the Mandelstam variable $s \equiv (p_A + p_B)_\mu (p_A + p_B)^\mu$:

$$\begin{aligned} s &= (E_A^{\text{lab}} + m_B)^2 - |\mathbf{p}_A^{\text{lab}}|^2 = (E_A^{\text{cm}} + E_B^{\text{cm}})^2 \\ \implies s &= m_A^2 + m_B^2 + 2m_B E_A^{\text{lab}} = (E^{\text{cm}})^2 \end{aligned} \quad (\text{A.4})$$

As relativistic energies ($\mathcal{O}(10^{2-3}\text{GeV})$) are much higher than reactant masses $\mathcal{O}(1 - 10\text{GeV})$,

$$\sqrt{s} = E^{\text{cm}} \approx \sqrt{2m_B E_A^{\text{lab}}} \quad (\text{A.5})$$

As the center-of-mass energy that produces new particles increases only as $\sqrt{E^{\text{lab}}}$, making any gain in particle production more expensive at relativistic energies. Thus modern colliders usually operate in center-of-mass frames.

A.2 Coordinate system

Considering the non-linear addition of velocities in relativistic mechanics,

$$\beta = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}, \quad (\text{A.6})$$

Considering the transformation $y = \tanh^{-1} \beta$,

$$\begin{aligned} \tanh^{-1} \beta_1 + \tanh^{-1} \beta_2 &= \tanh^{-1} \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2} = \tanh^{-1} \beta \\ \implies y_1 + y_2 &= y \end{aligned} \quad (\text{A.7})$$

The addition is linear again. We call this transformed velocity *rapidity*, given by:

$$y = \tanh^{-1} \beta = \frac{1}{2} \ln \frac{1 + \beta}{1 - \beta} = \frac{1}{2} \ln \frac{E + p_z}{E - p_z} \quad (\text{A.8})$$

Rapidity can grow without bound, and at small velocities, $y \approx \beta$, making it a relativistic analogue of velocity. Since it transforms linearly under Lorentz boosts, rapidity distribution of particles retain their shapes. For relativistic energies, $E \approx p$, making A.8,

$$y \approx \frac{1}{2} \ln \frac{p + p_z}{p - p_z} = -\ln \left(\tan \frac{\theta}{2} \right) = \eta \quad (\text{A.9})$$

η is called the *pseudo-rapidity*, the massless limit of rapidity. As shown in A.2, pseudo-rapidity can be directly determined from particle production angle θ measured with respect

to the beam axis.

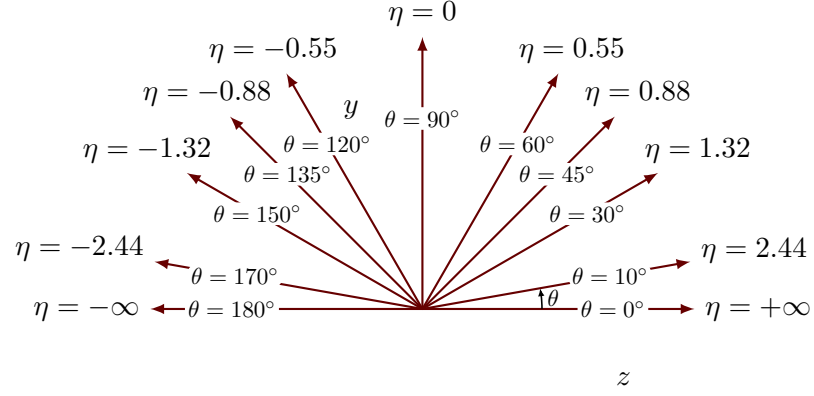


Figure A.1: η vs θ

Also considering *azimuthal angle* (φ) and *transverse-momentum* ($p_T = \sqrt{p_x^2 + p_y^2}$) of the particles, which are invariant under boosts along the beam-line by design, we can use (p_T, η, φ) as the coordinate system for the particles produced in the collision. Figure 2.6 illustrates this coordinate system for a cylindrical collider detector.